Mapper Graph Grid Search

This contains all my exhaustive mapper constructions and their evaluations

Smaller dimensions

I used the following columns:

- wave
- affective valence
- wemwbs total
- bangs factors (each with frustration and satisfaction)
  - autonomy
  - competence
  - relatedness
- gaming influence on lives (score for "no affect" is 20 as it is scaled 1-7, 1=greatly interfered, 7=greatly supported)
- total minutes played during last two weeks (wave duration)
- max binge minutes
- reaction time mean

PC1, PC2, Min-Max Scaling

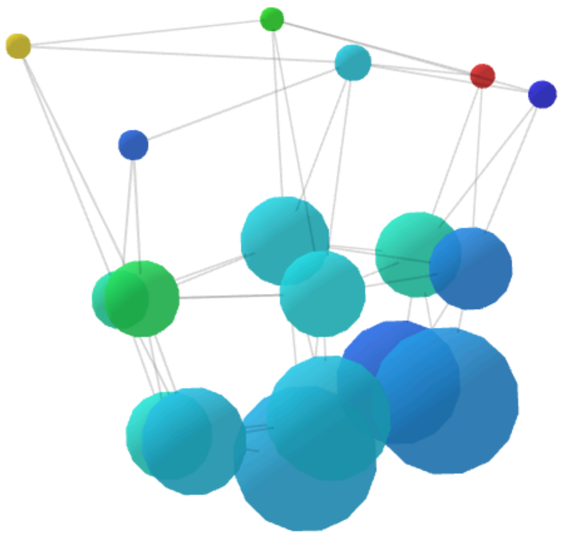
The highlighted graphs so far:

- 3 intervals show structure of 9 nodes, with a corner of more ADHD participants
- 5 intervals at 30% begin to show a USA style structure to the data and a very small ADHD branch
  - 50% intervals show a nicer structure
  - DBSCAN 0.5 and 3 show more detail into the main structure
- 10 intervals really begins to break down the main structure, including a little hole which is cool, and some isolated components
  - Switching to kmeans with 2 clusters actually splits the main nodes into two, which also highlights some higher proportions of ADHD when forced to split the data into 2.
  - 50% overlap shows more isolated components

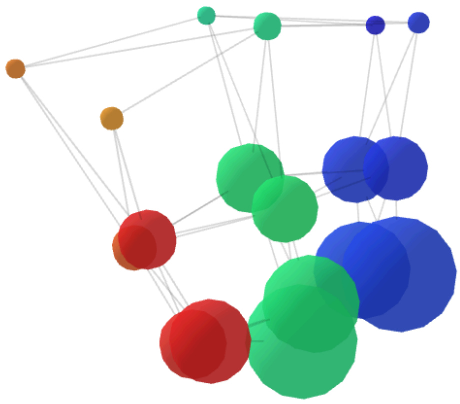
3 intervals, 30% overlap

I have learnt from this that minPts above 15 gives the same structure, if we have a high eps (0.2>).

Kmeans=2

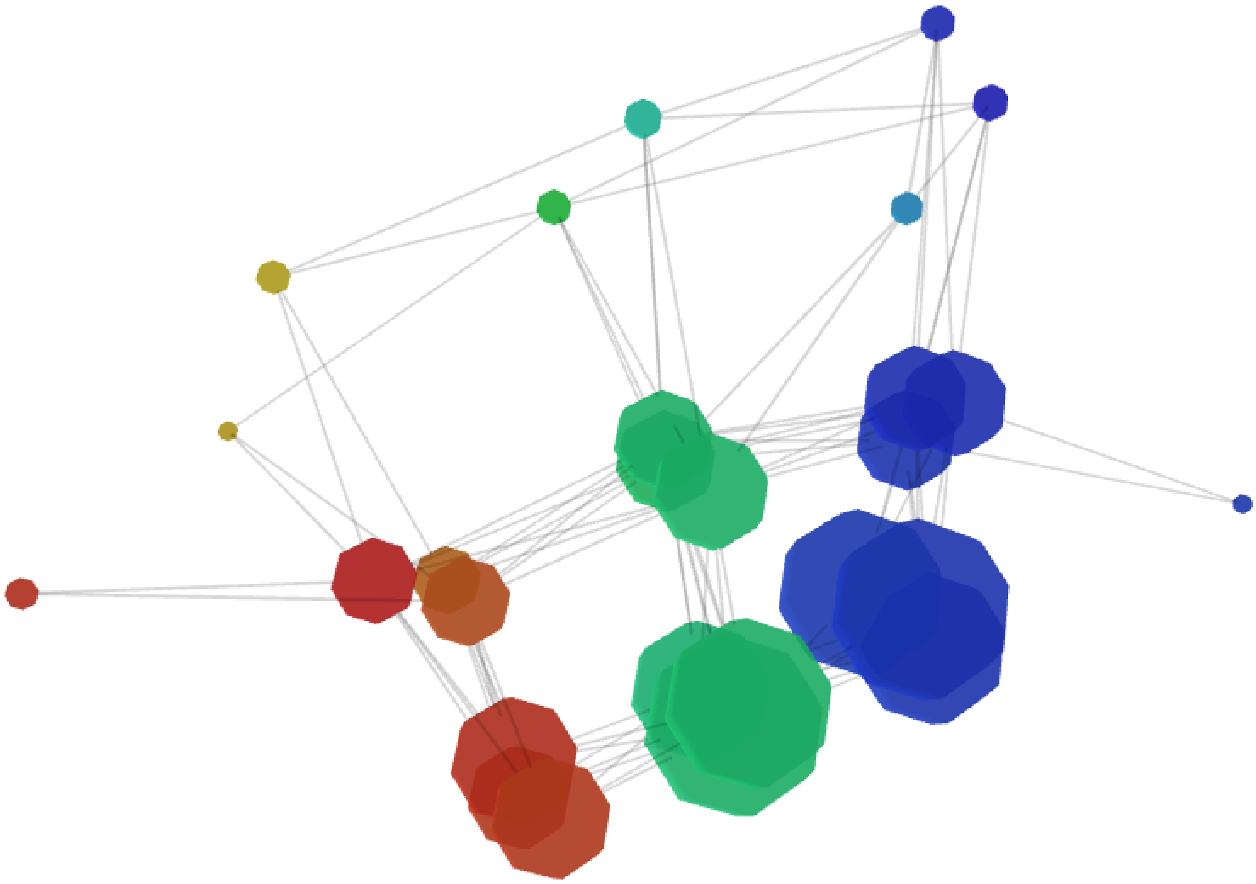


- Red shows 40% of participants are diagnosed with ADHD
- Bottom groupings show high wemwbs total, high bangs satisfaction, low frustration, above 20 average gaming influence
- left hand side plays a lot more minutes during waves (4000) compared to right (340)



Kmeans=3

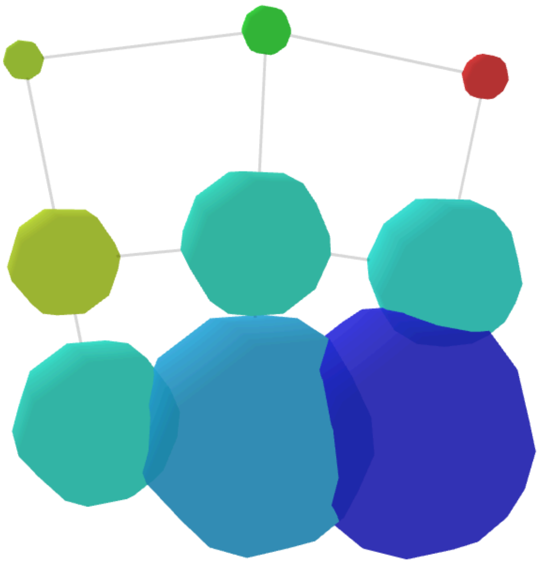
Similar structure, just more branching out nodes



- The extra nodes are just people who have similar telemetry, but have differing wellbeing
- Between 2-4 participants, either have none or one person with diagnosed ADHD

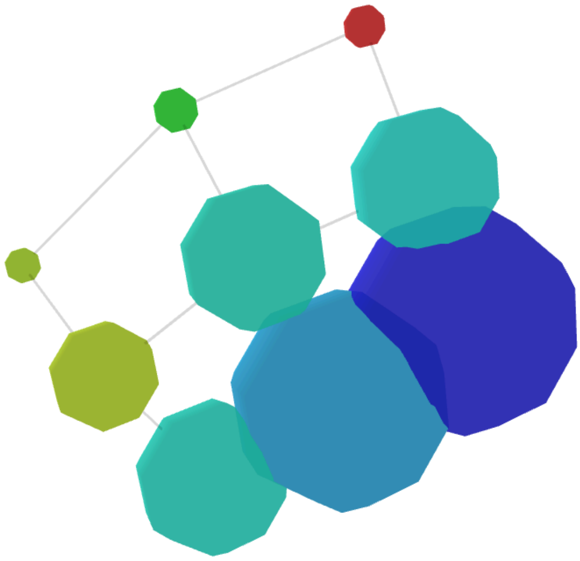
DBSCAN(eps=0.2, minPts=15)

No real change, perhaps because of minPts?



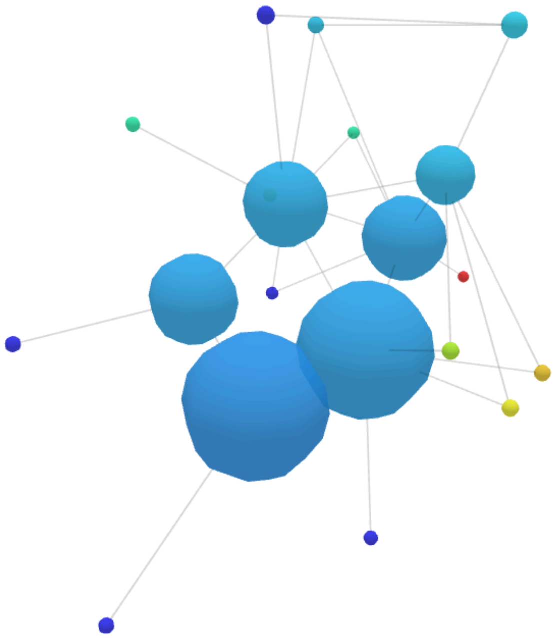
DBSCAN(eps=0.2, minPts=20)

No change again



DBSCAN(eps=0.5, minPts=3)

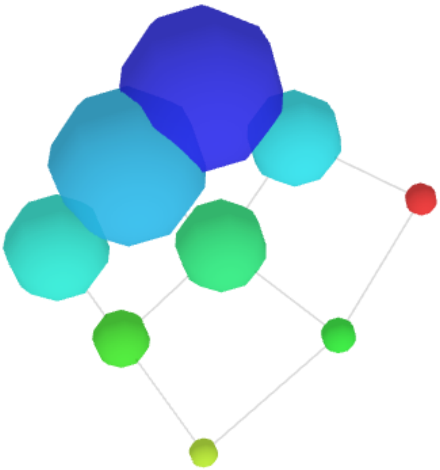
Interesting 3D shape, bad 2D:



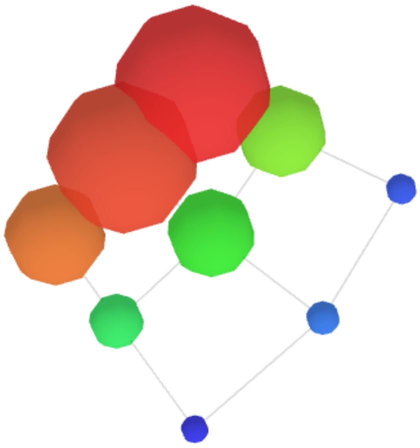
- Top left nodes (sizes between 3-14) show high levels of relatedness frustration and low wellbeing scores

DBSCAN(eps=0.15, minPts=15)

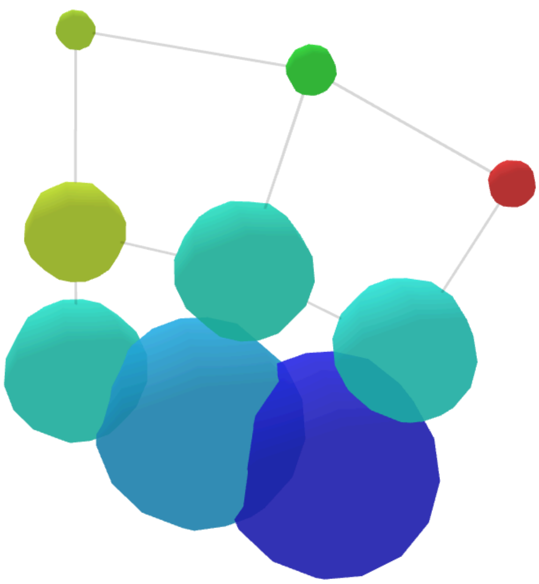
By ADHD proportion (red=0.25, blue=0.14)



Exhibiting lower wellbeing (wemwbs total, red=23, blue=13):

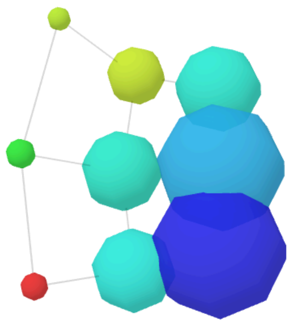


DBSCAN (eps=30, minPts=20)

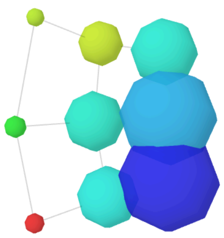


Massive simplification, large nodes have 2000 rows in. DBSCAN needs tweaking more

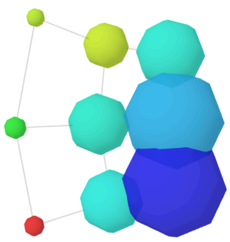
DBSCAN (0.1, 5)



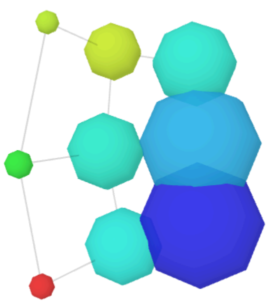
DBSCAN (0.1, 10)



DBSCAN (0.2, 5)



DBSCAN (0.2, 10)



3 Intervals, 50% overlap

DBSCAN (0.1, 5)

No changes

DBSCAN (0.1, 10)

No changes

DBSCAN (0.2, 5)

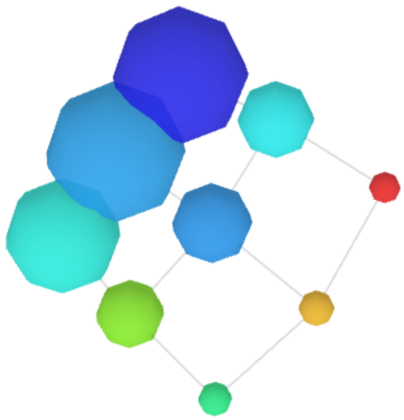
No changes

DBSCAN (0.2, 10)

No changes

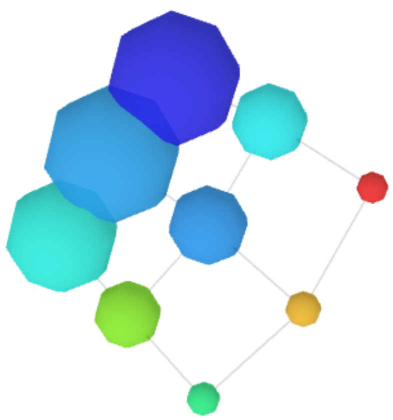
DBSCAN (0.2, 15)

Same structure, slightly different ADHD proportions (red=0.25, blue=0.16):



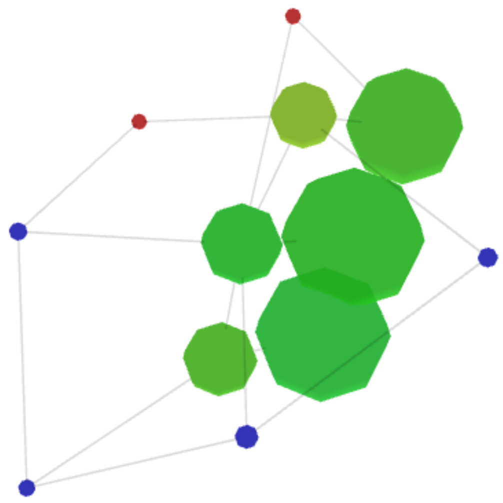
DBSCAN (0.2, 20)

No change:



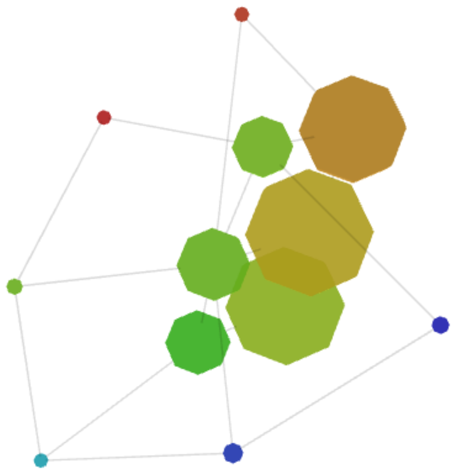
DBSCAN (0.5, 3)

Now have smaller end nodes (blue=0 now, red=0.33):



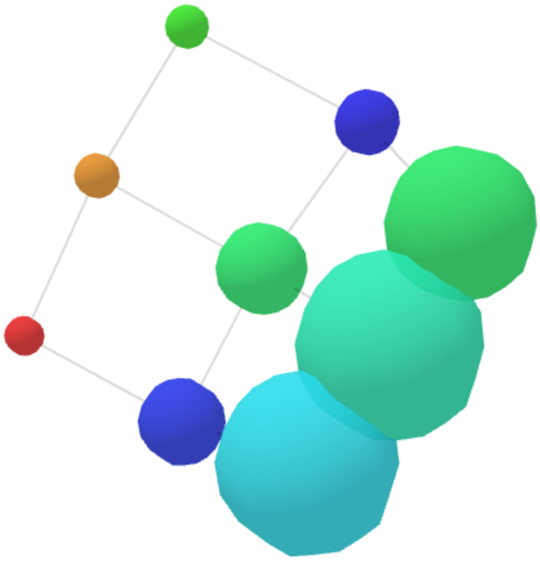
Bottom three exhibit worse life impact from gaming (blue=13.67), node sizes 3(dark blue), 6(lightblue):

lower relatedness satisfaction:

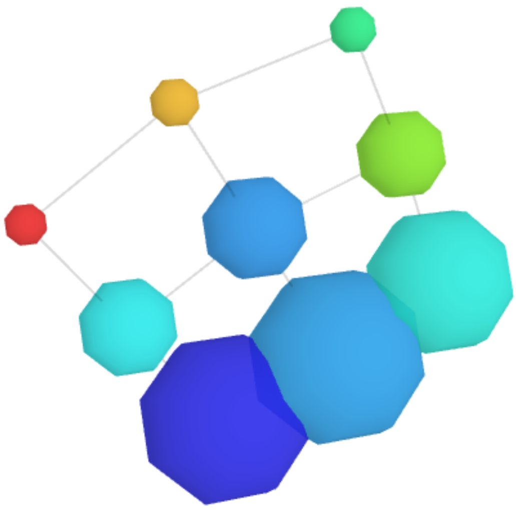


DBSCAN(0.5, 15)

Same a 0.5, I think the eps is too large.

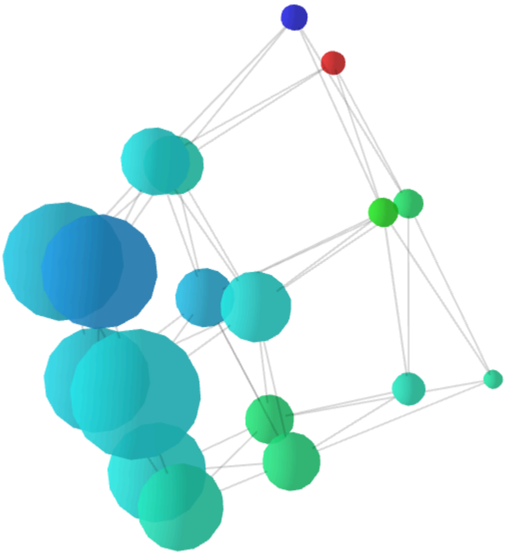


DBSCAN (30, 20)



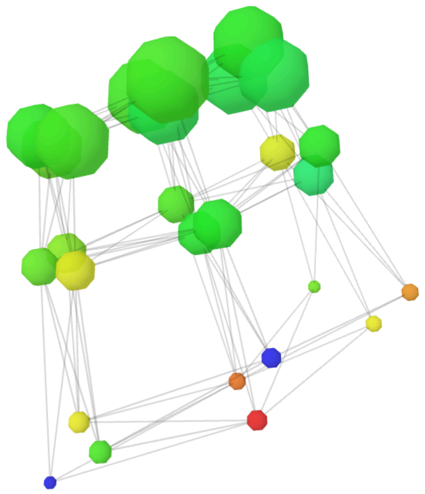
Kmeans (2)

Simply splitting the large DBSCAN into two nodes each:



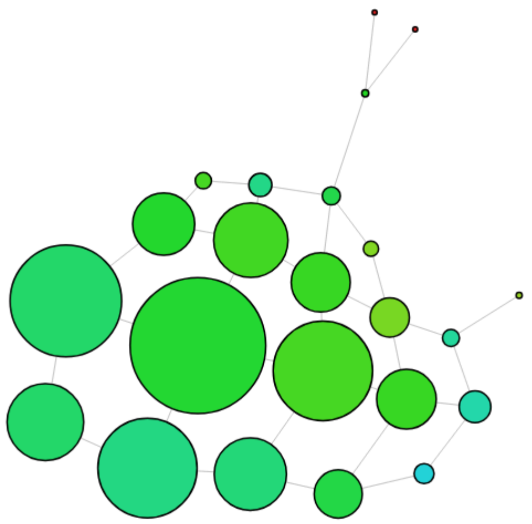
Kmeans(3)

Same problem as before:

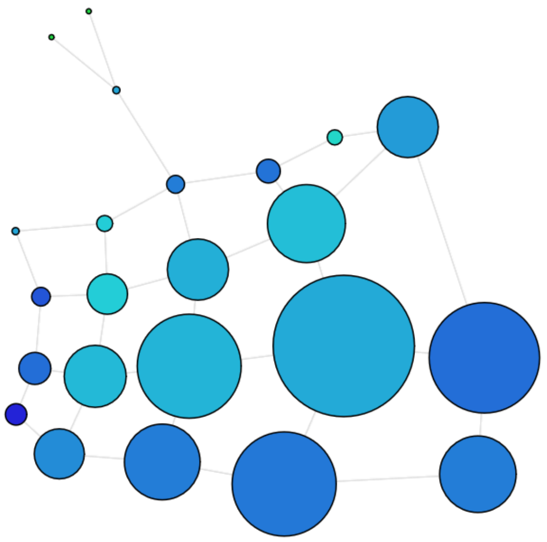


5 intervals, 10% overlap

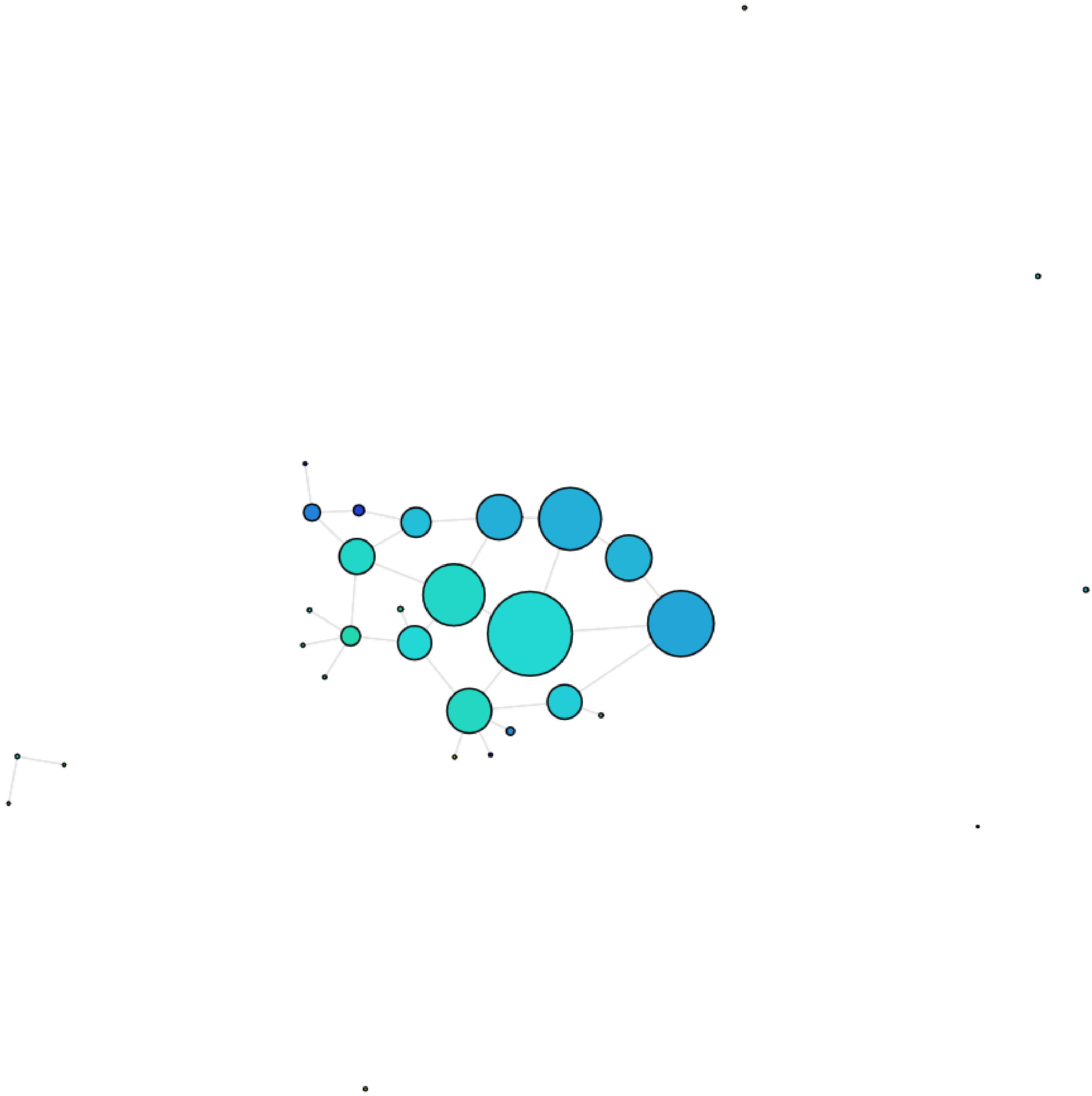
DBSCAN(0.05, 10)



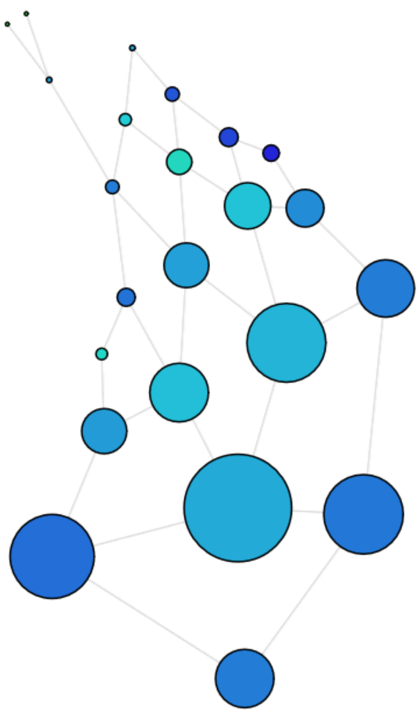
DBSCAN (0.1, 5):



DBSCAN (0.5, 3):

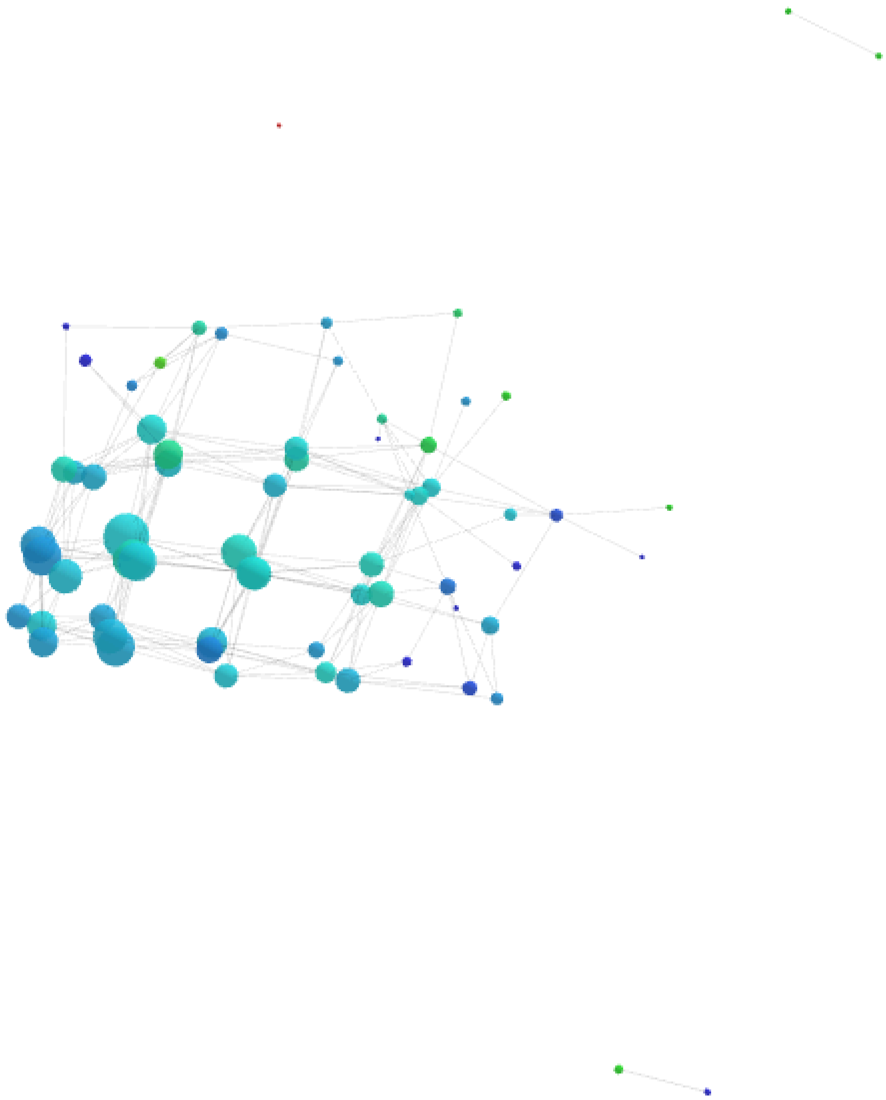


DBSCAN (0.6, 26):



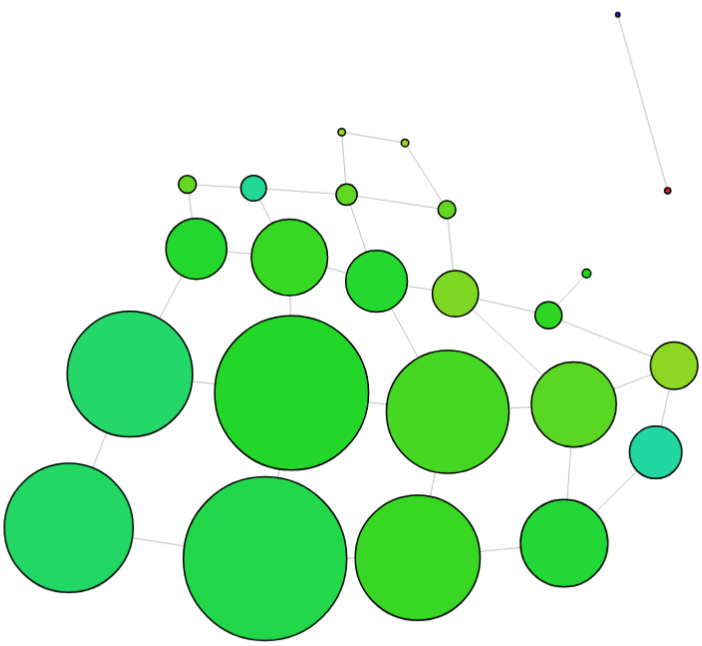
kmeans(2):

kmeans(3):



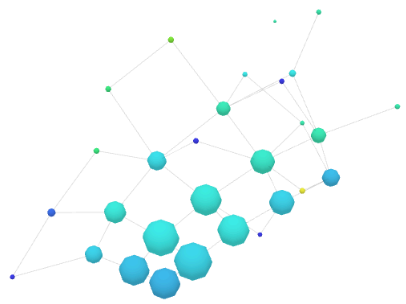
5 Intervals, 20% overlap

DBSCAN(0.05, 10):

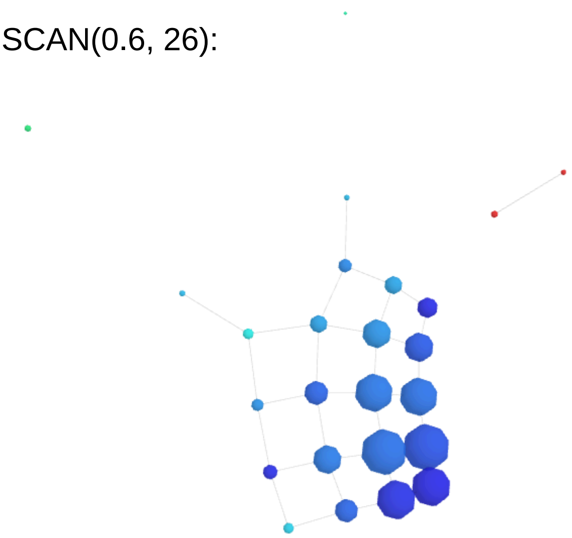


DBSCAN (0.1, 5):

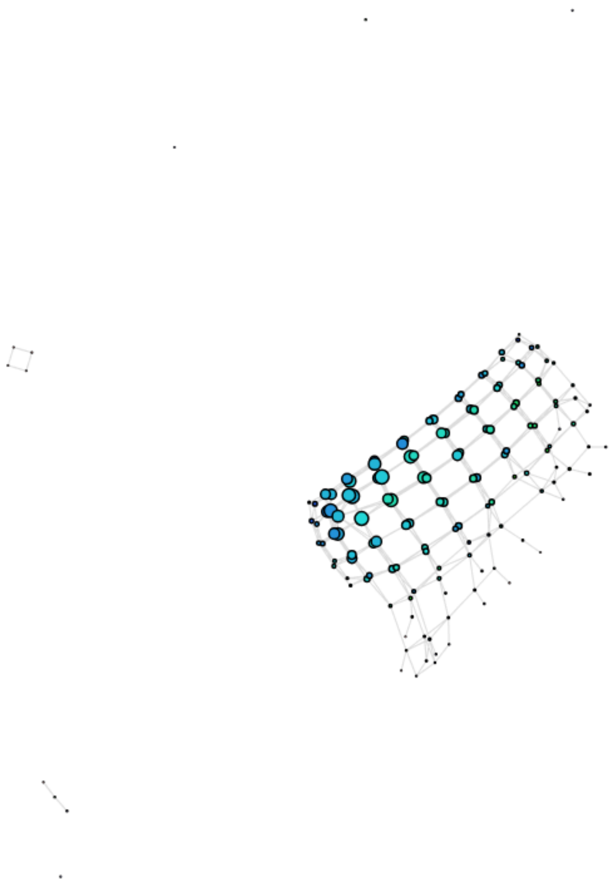
DBSCAN(0.5, 3):



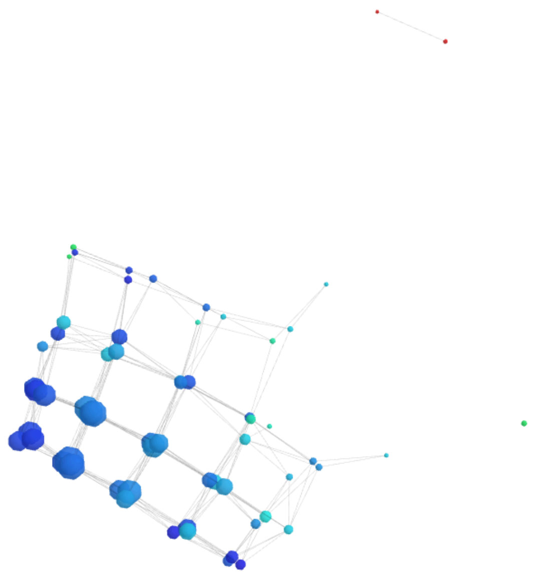
DBSCAN(0.6, 26):



kmeans(2):



kmeans(3):



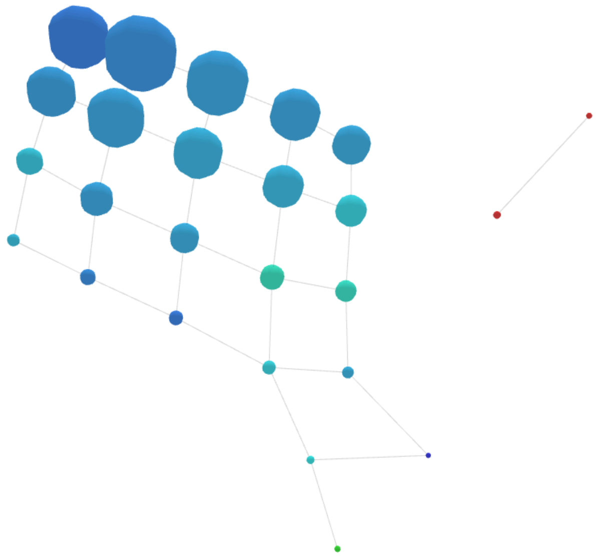
5 Intervals, 30% overlap

I am hoping we break the barrier of the 9 core nodes

DBSCAN (0.1, 5)

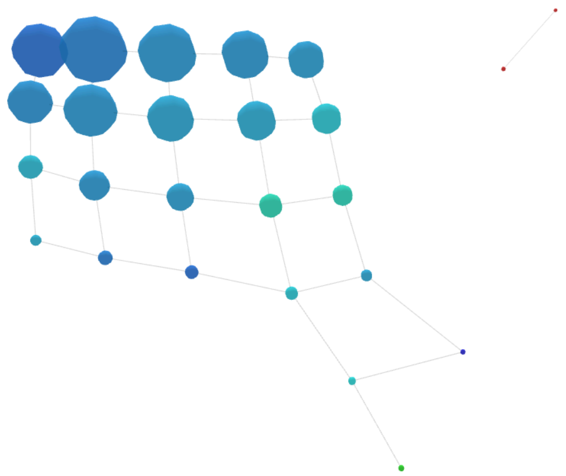
Structure improved, we can see now the red node has become an isolated component, they are just ADHD diagnosed patients with node sizes 2 and 1. They are nodes 6 and 1. This is just one person of different waves who has low wellbeing, quicker reaction times, and performs low on the bangs scales.

Coloured by ADHD proportion



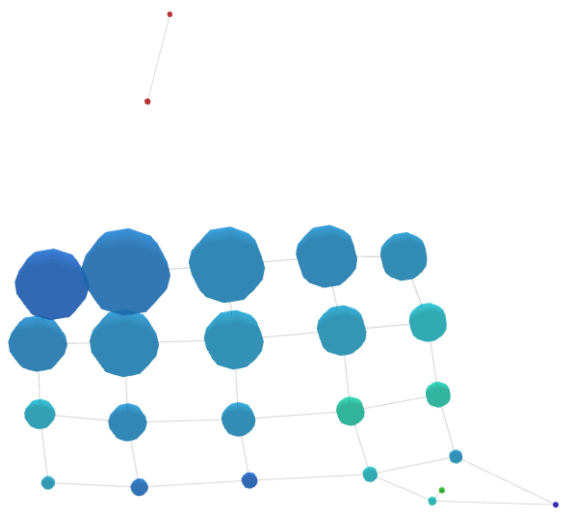
DBSCAN (0.1, 10)

Same as before



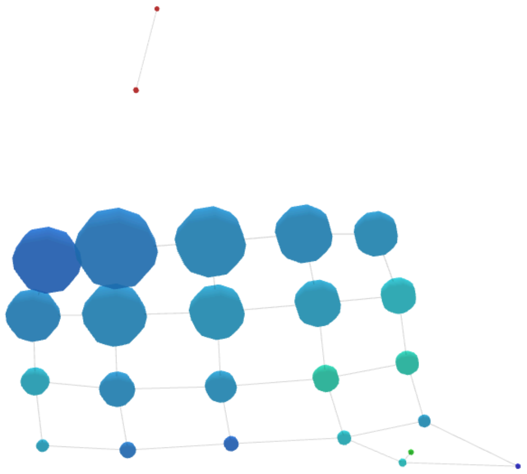
DBSCAN (0.2, 5)

Same again



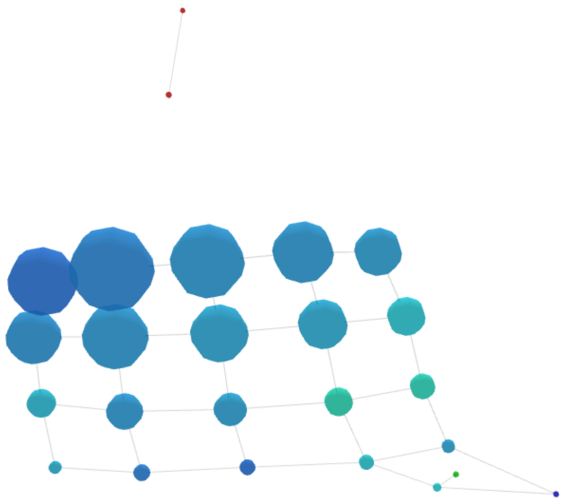
DBSCAN (0.2, 10)

Same



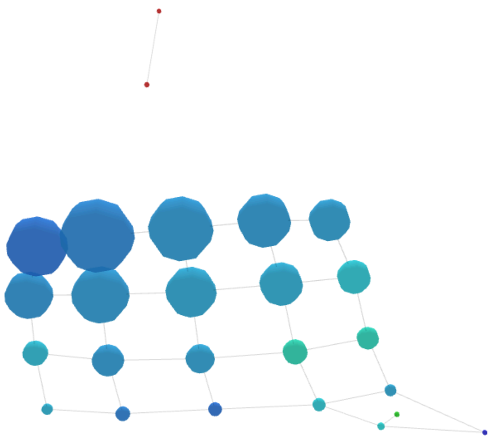
DBSCAN (0.2, 15)

Same



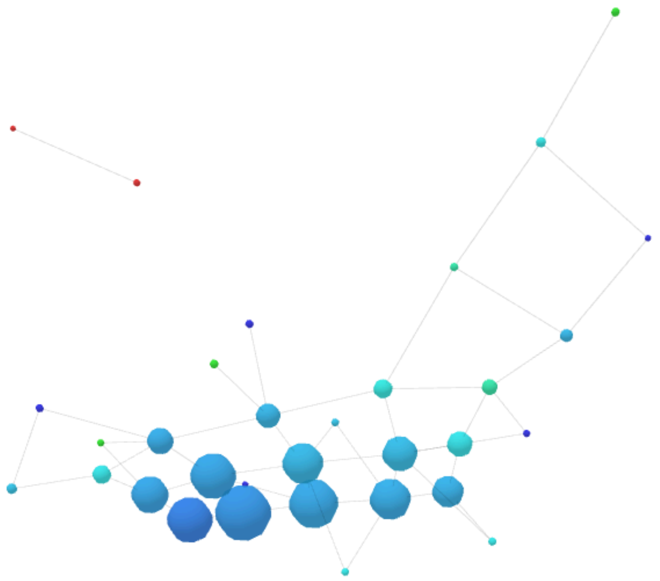
DBSCAN (0.2, 20)

Same

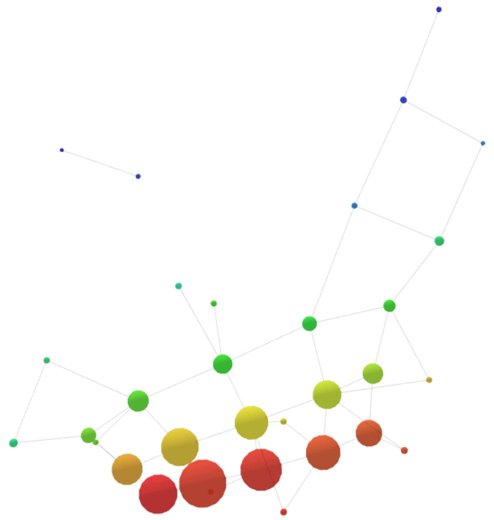


DBSCAN (0.5, 3)

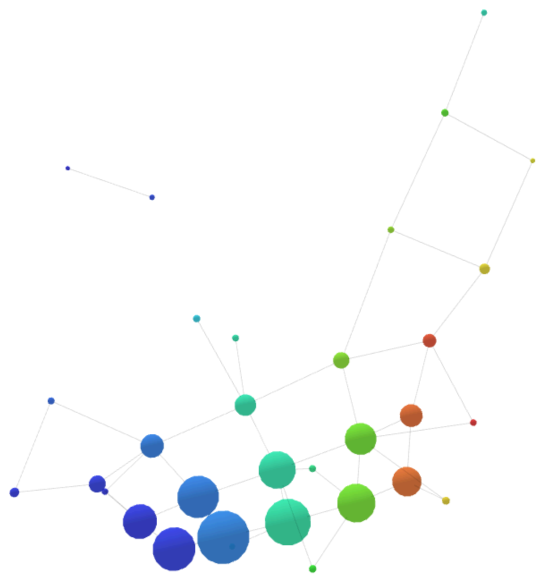
We keep the isolated component, but the main is spread out more



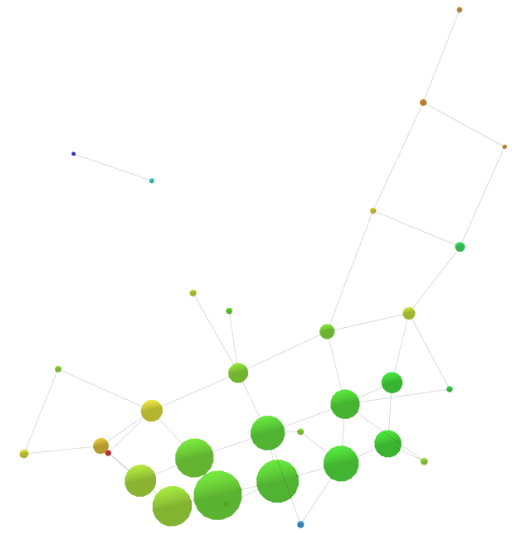
We see little nodes along the edge have more people with ADHD in them. These have different participants in them which is interesting.
Top flare has poorer mental well-being, and experiences more frustation with videogames:



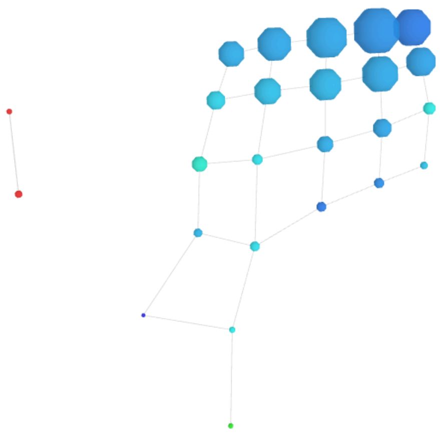
Structure left to right is by total wave playtime:



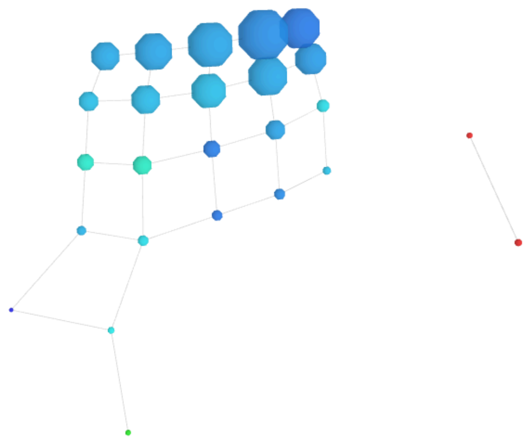
It has nothing to do with the wave:



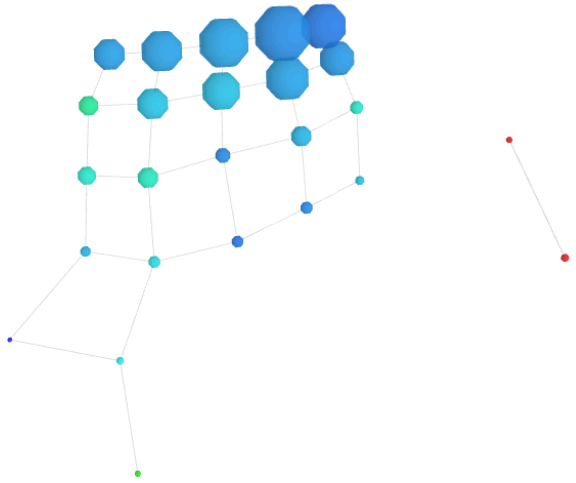
DBSCAN (0.5, 10)



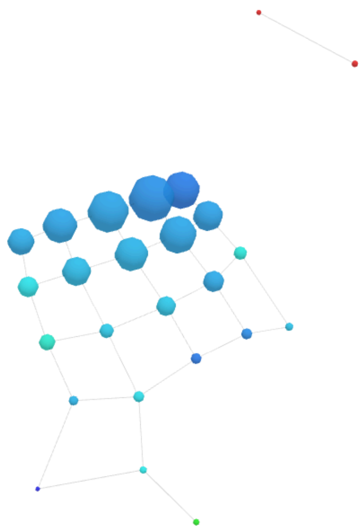
DBSCAN (0.5, 15)



DBSCAN (0.5, 20)

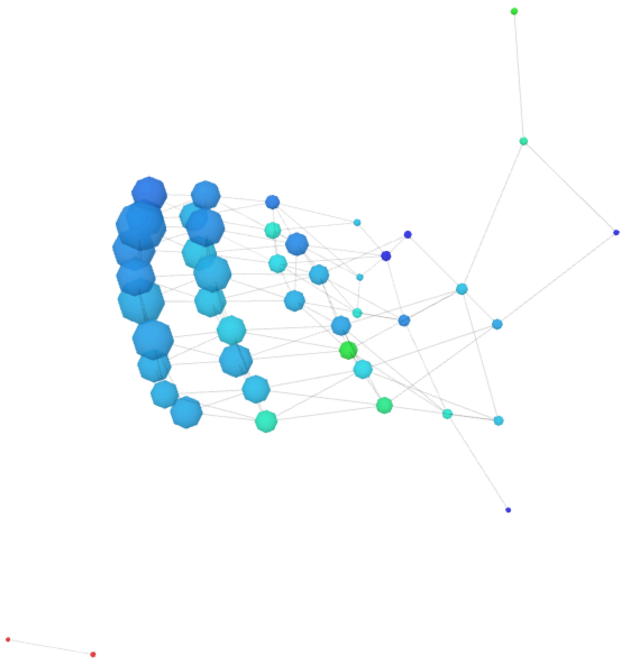


DBSCAN (0.6, 26)



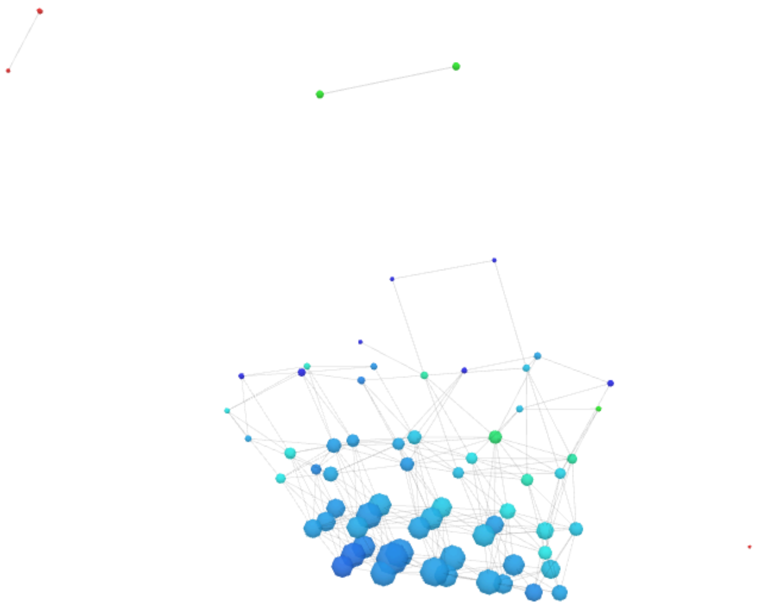
Kmeans (2)

Same structure of DBSCAN, but nodes are now split into 2:



Kmeans (3)

Similar, but we get another isolated component:

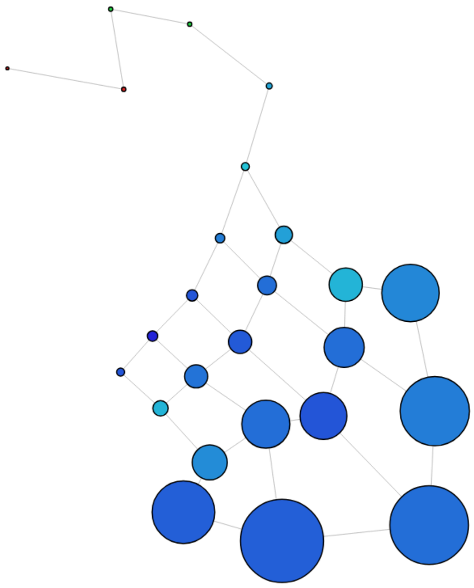


This is the flare from before, but has lost its connection after being forced into 3 clusters.

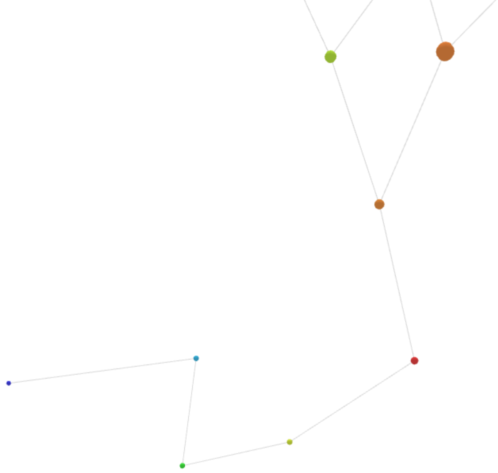
5 Intervals, 50% overlap

DBSCAN (0.1, 5)

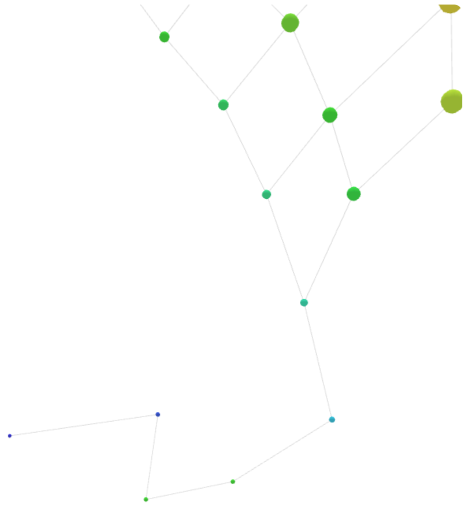
With more overlap, the isolated ADHD component has now joined the flare.



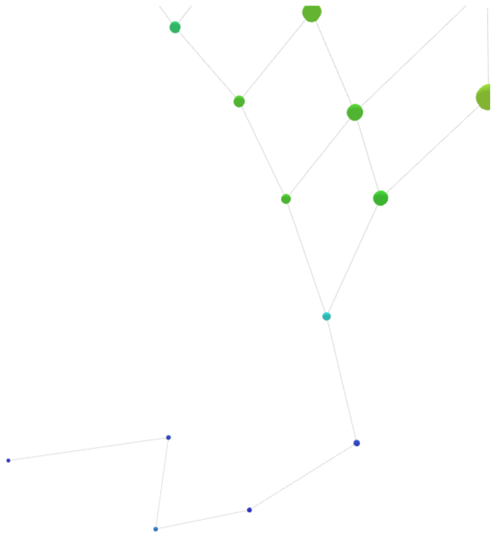
The max binge minutes does transition at the flare:



The affective valence also transitions:



Wemwbs total:



DBSCAN (0.1, 10)

No change, clearly too sparse

DBSCAN (0.2, 5)

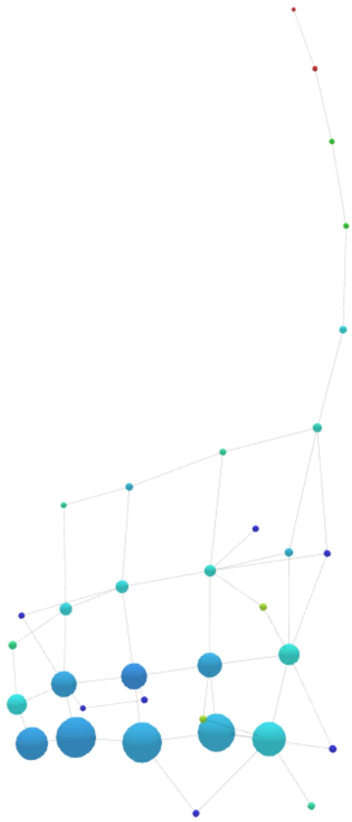
No change

DBSCAN (0.2, 10)

No change

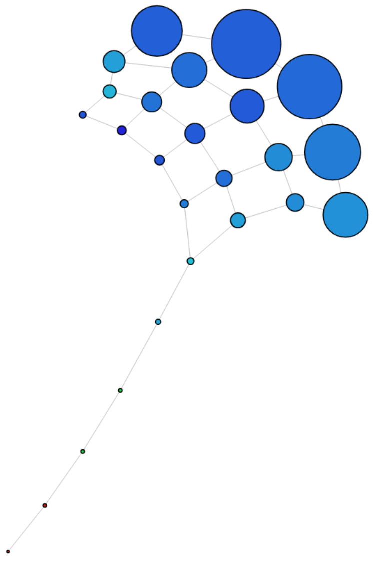
DBSCAN (0.5, 3)

We get the core structure again (including the flare), but we split the dense nodes up a bit more



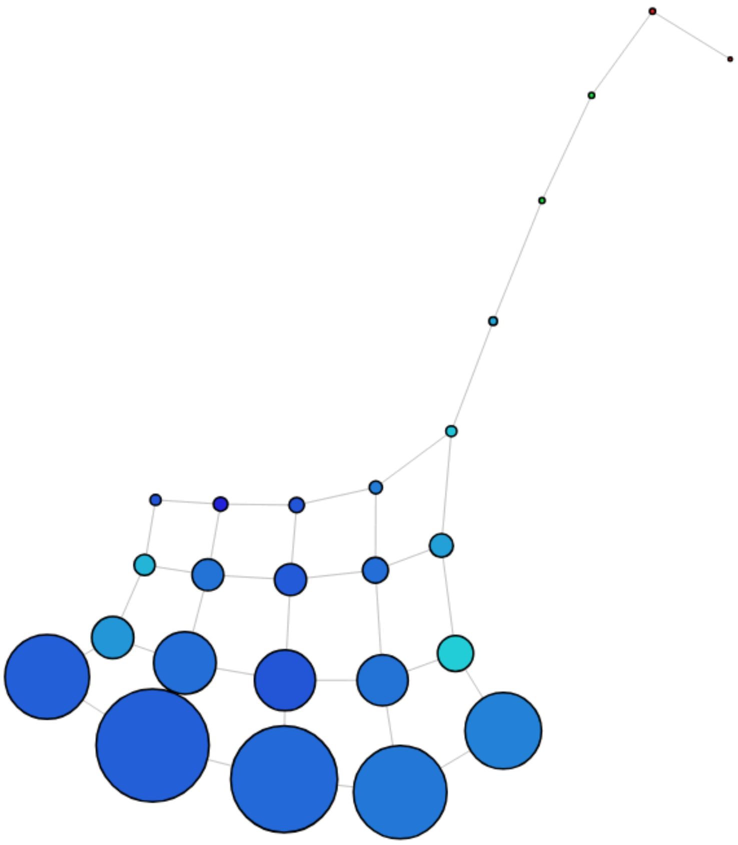
DBSCAN (0.5, 10)

Requiring a higher density loses the smaller nodes:



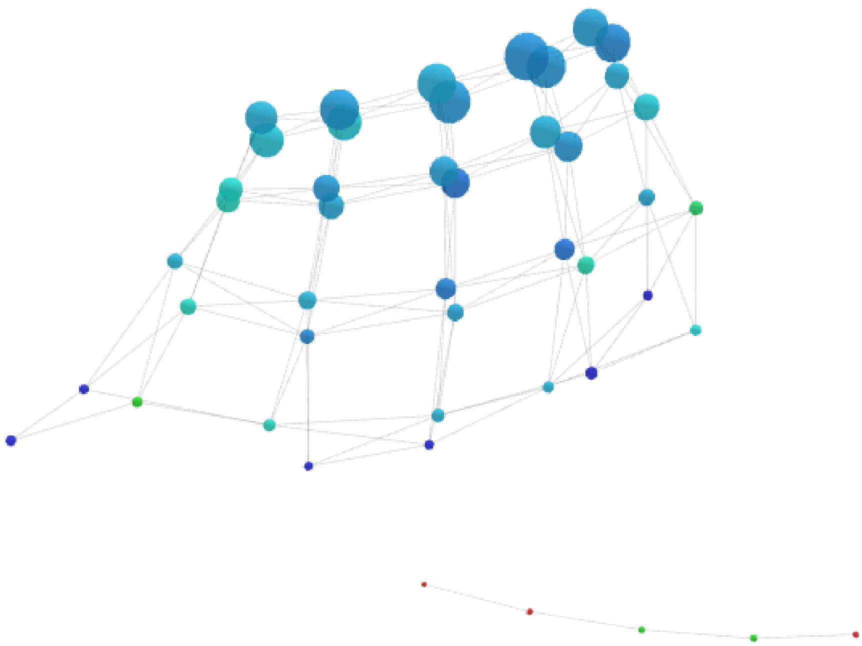
DBSCAN (0.6, 26)

Increasing the radius and the number of points doesn't change anything:



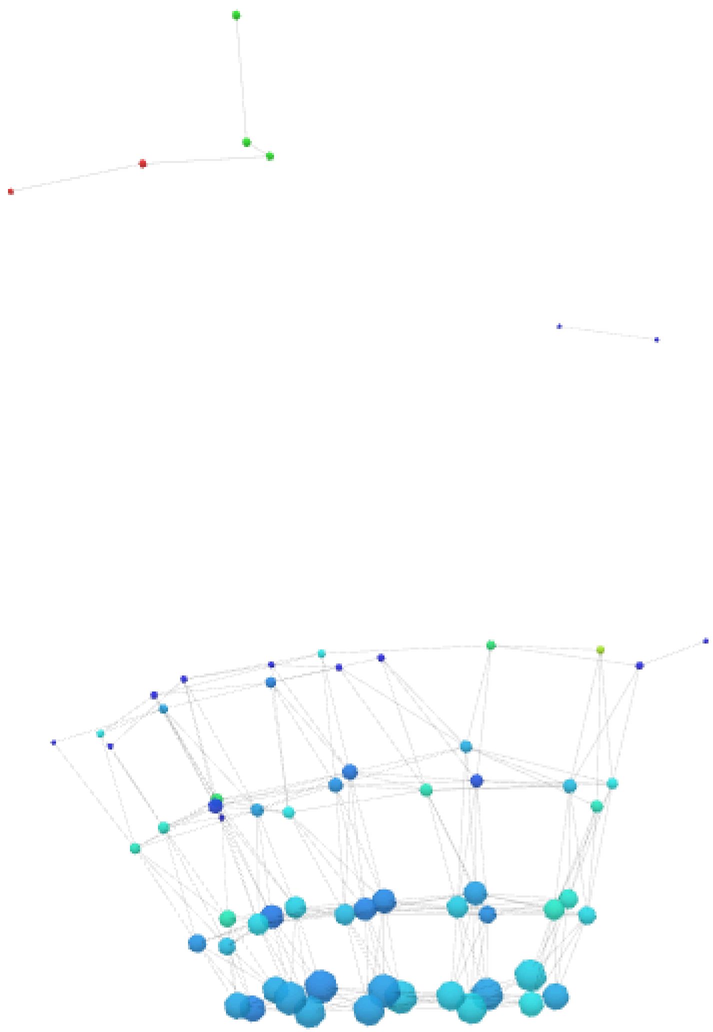
Kmeans (2)

We see the flare becomes separate:



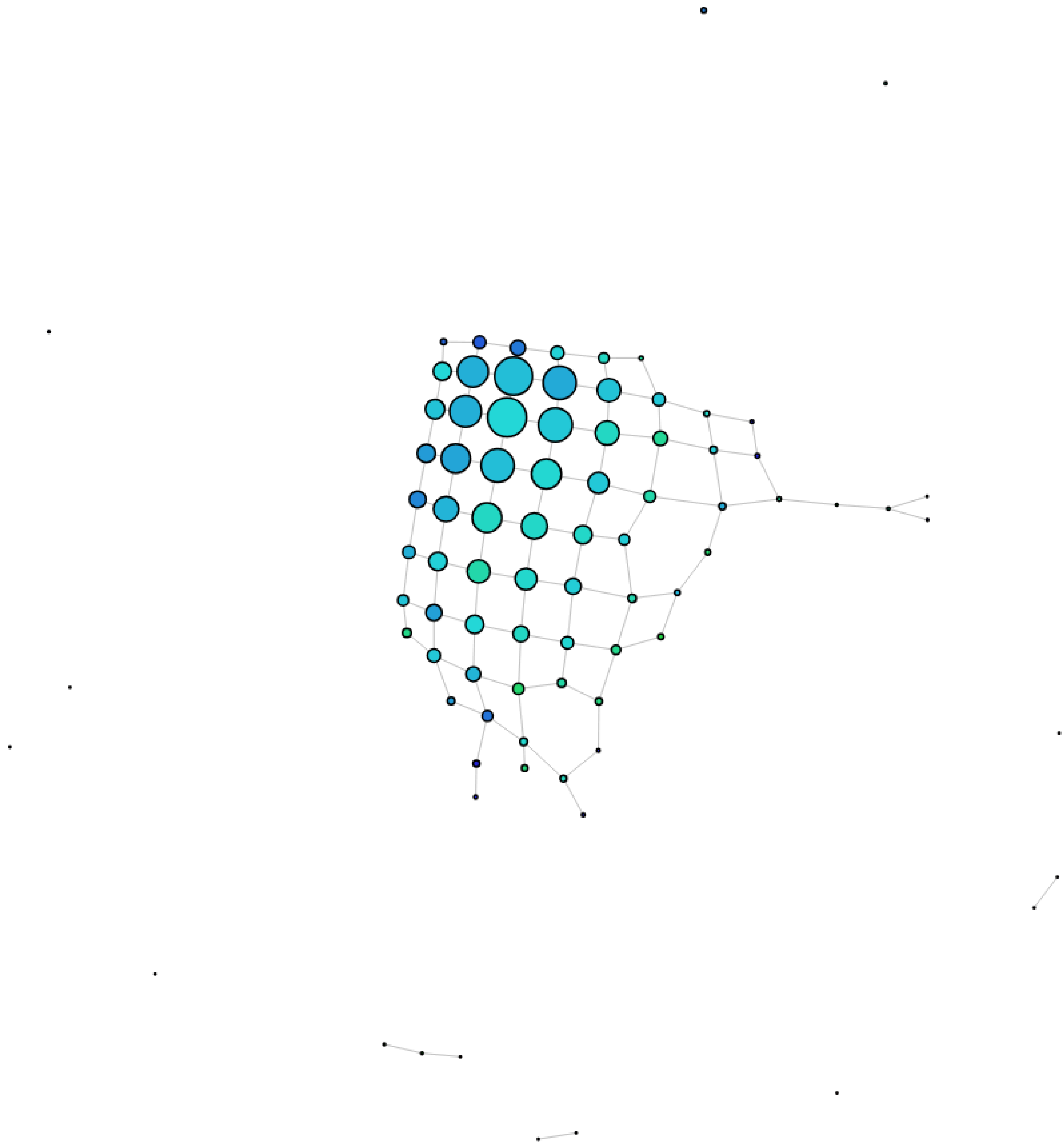
Kmeans (3)

Forcing to cluster into 3 adds another isolated component:



10 Intervals, 10% overlap

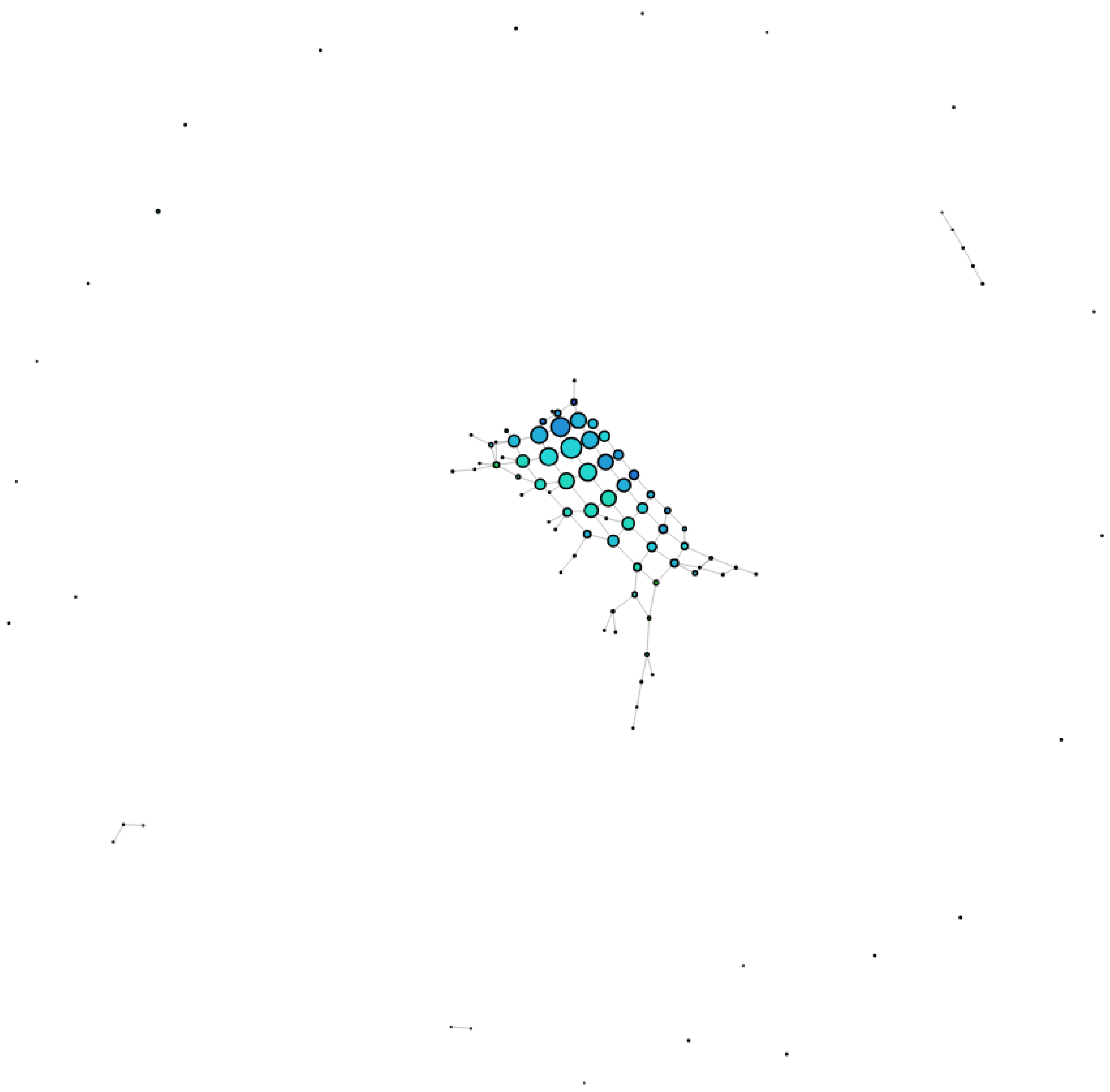
DBSCAN(0.05, 10):



DBSCAN(0.1, 5):

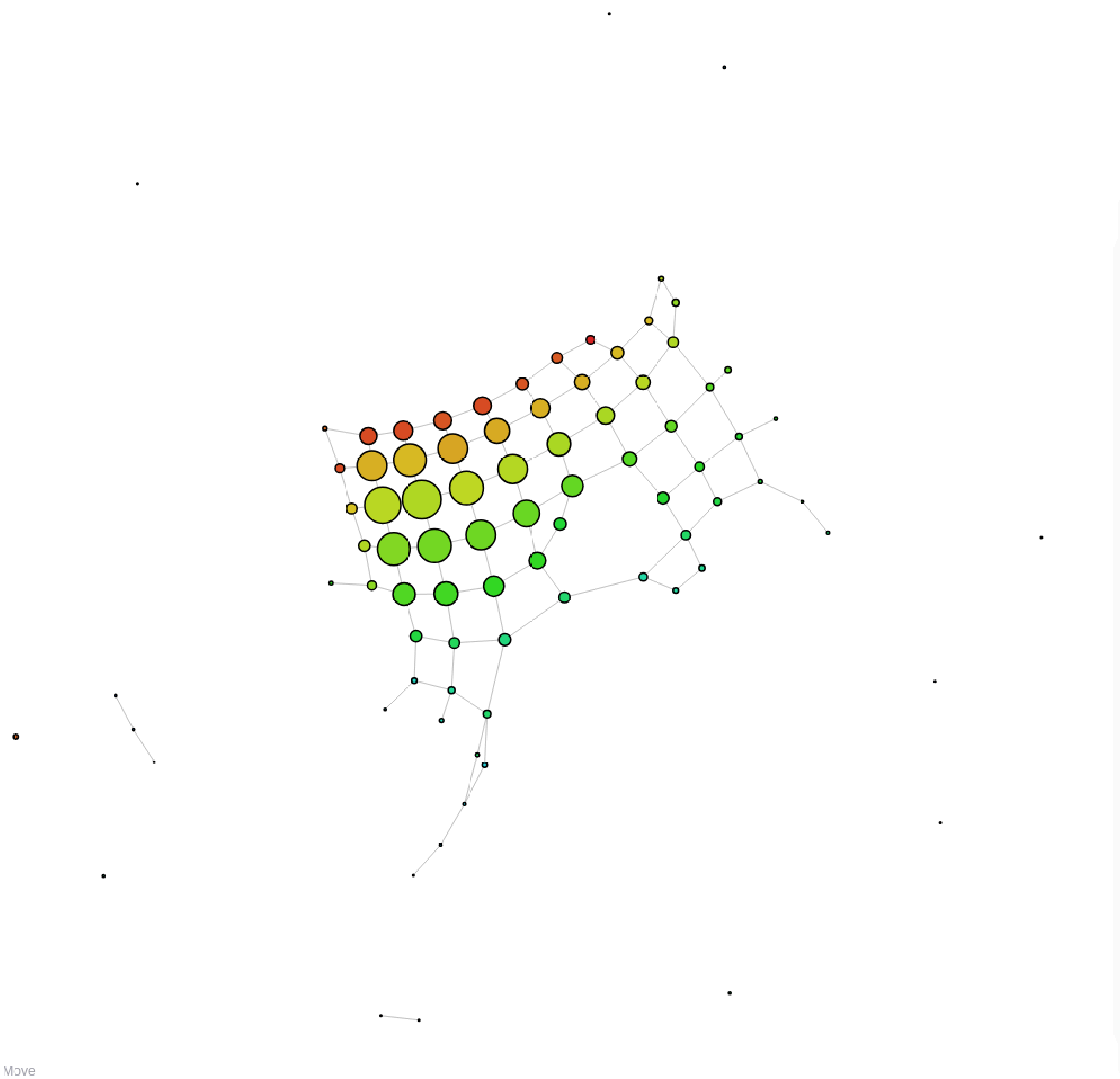
Move

DBSCAN(0.5, 3):



Move

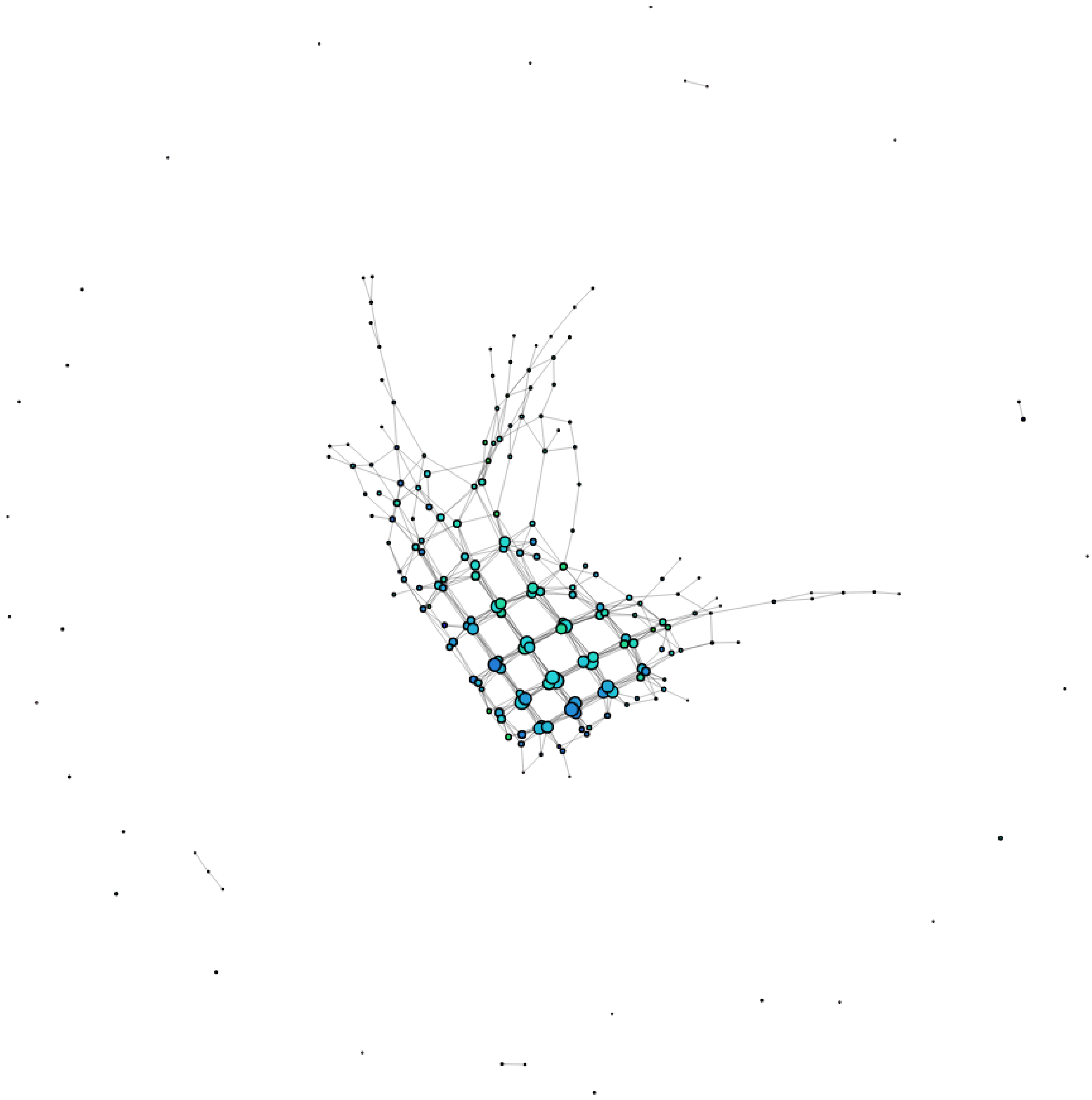
DBSCAN(0.6, 26):



Kmeans(2):

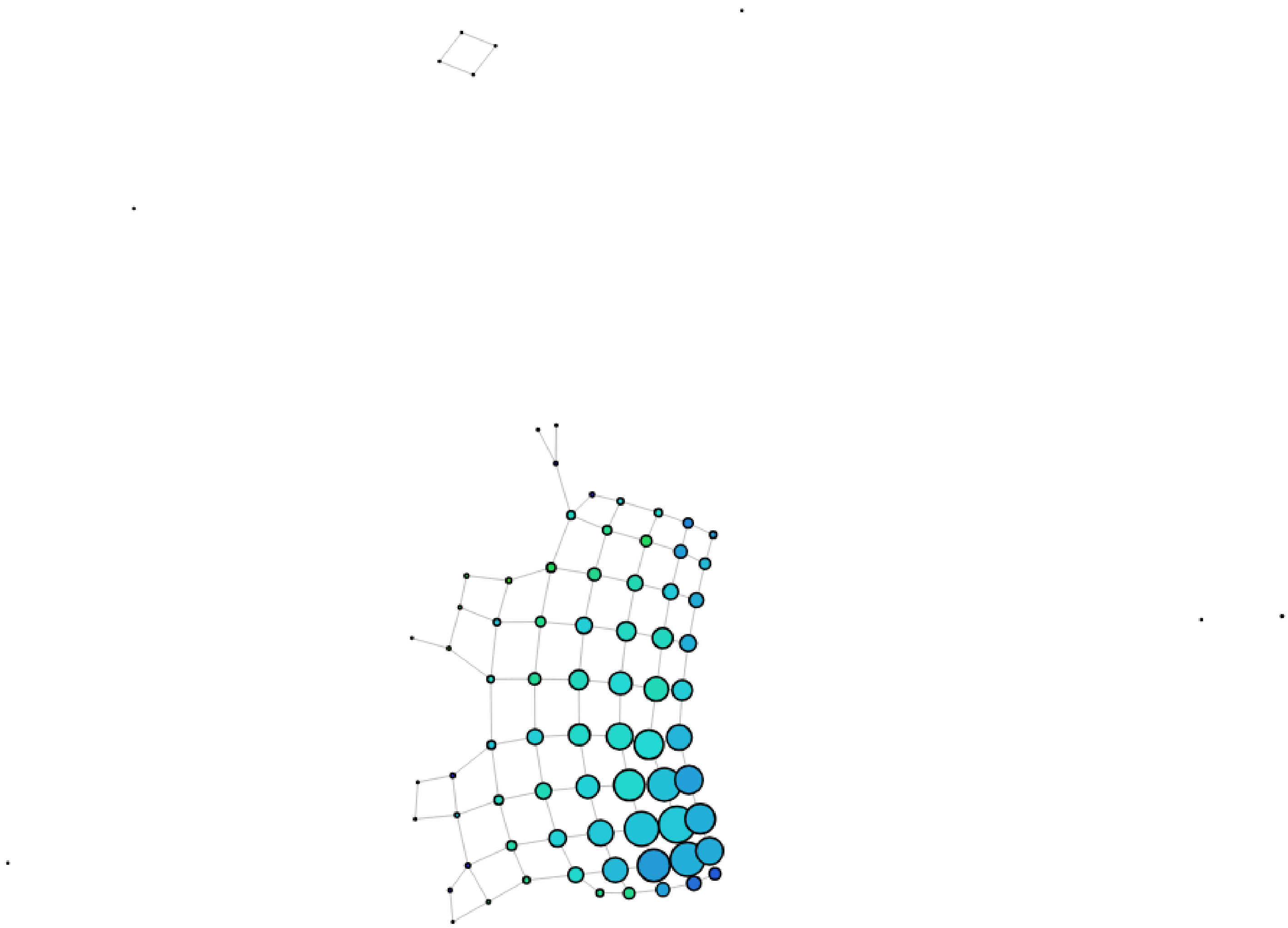
Move

Kmeans(3):

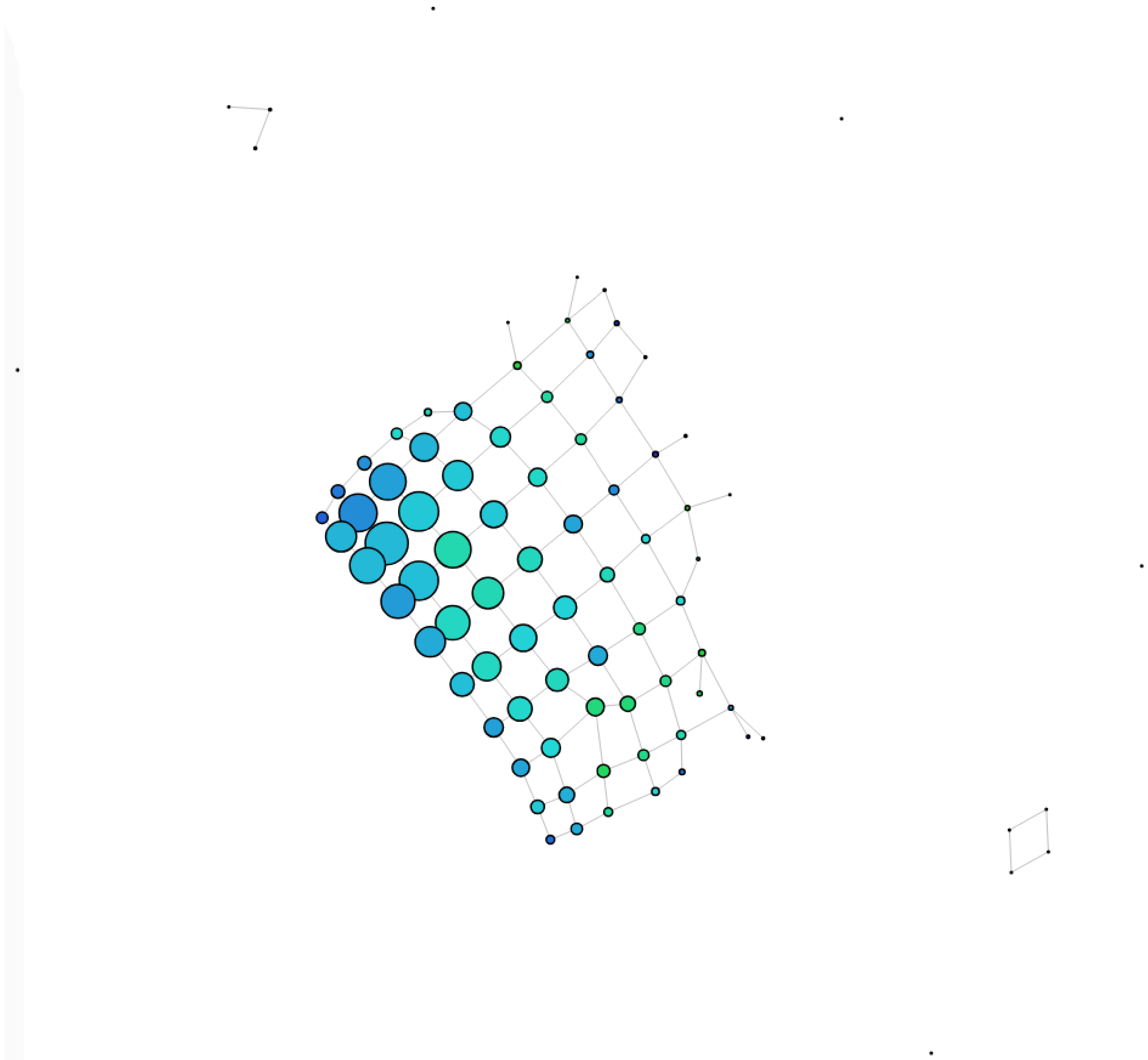


10 Intervals, 20% overlap

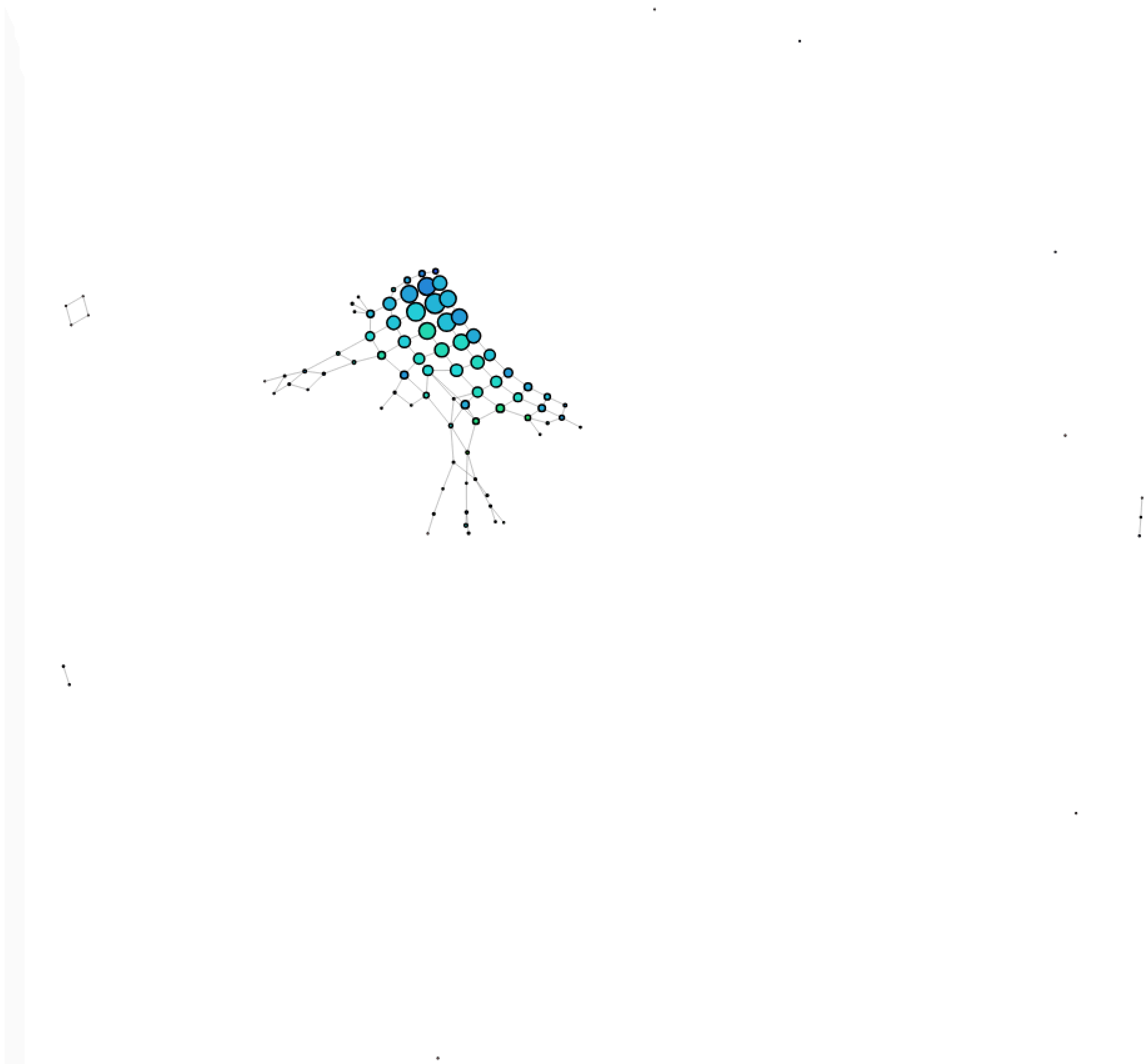
DBSCAN(0.05, 10):



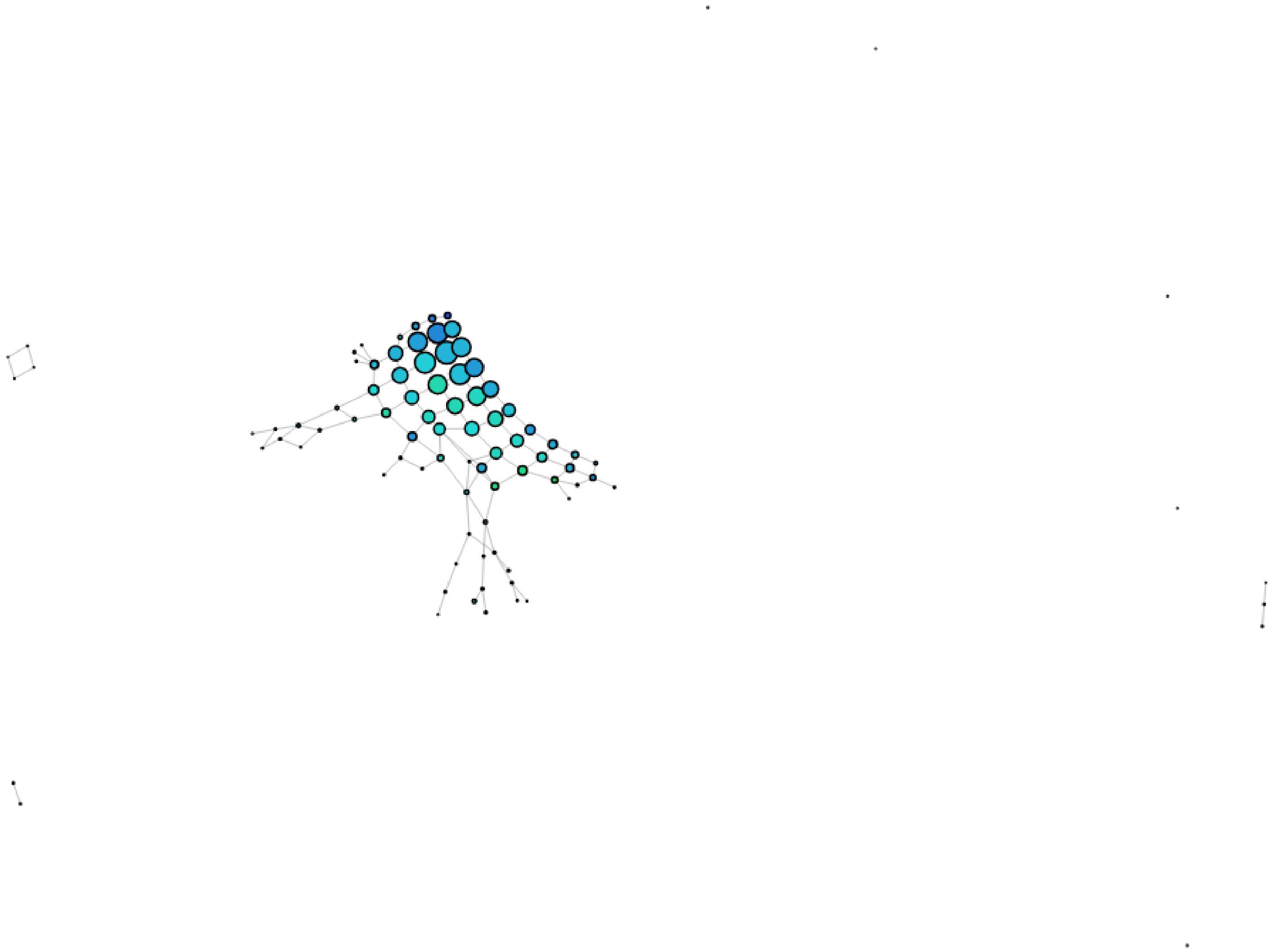
DBSCAN(0.1, 5):



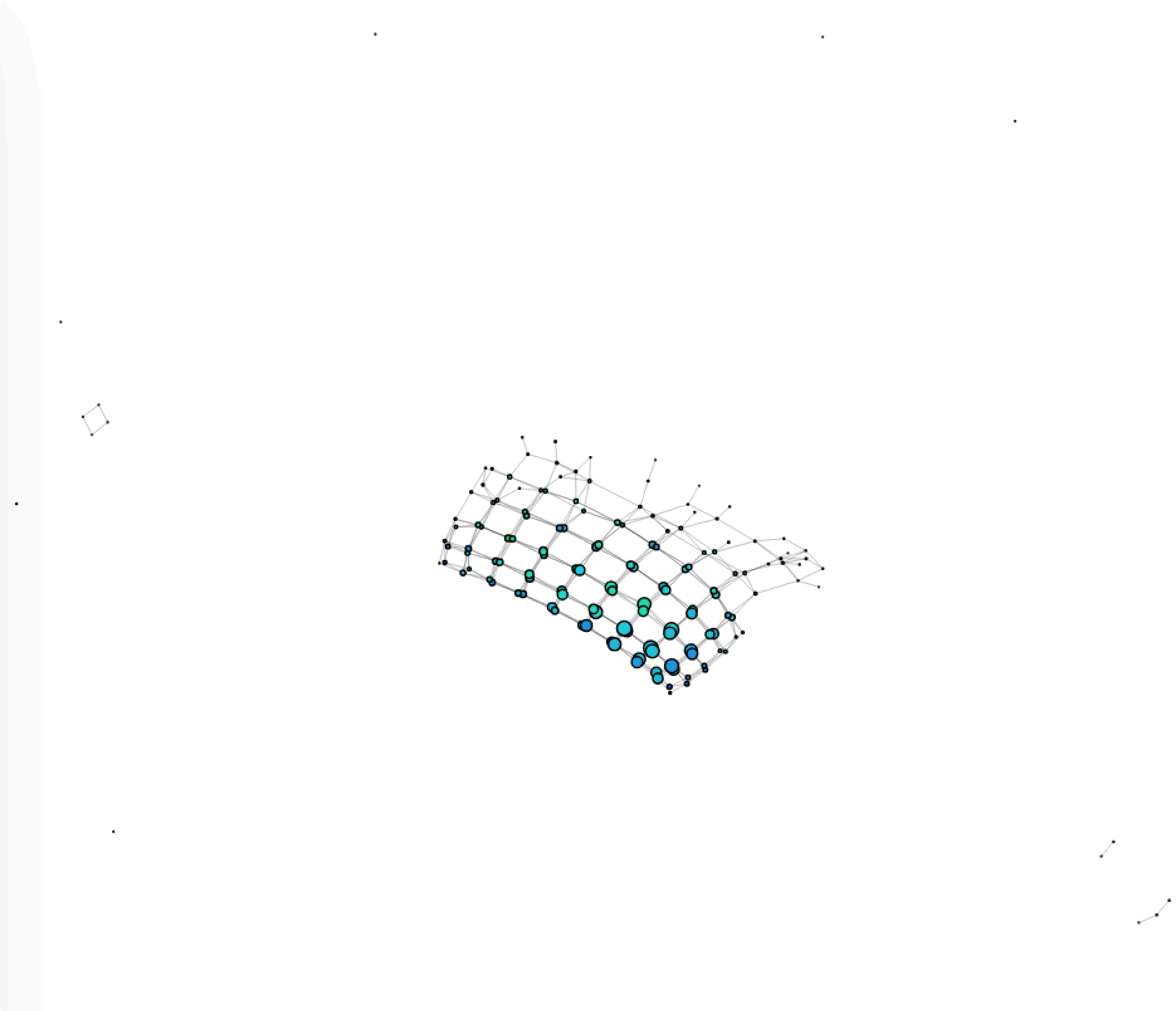
DBSCAN(0.5, 3):



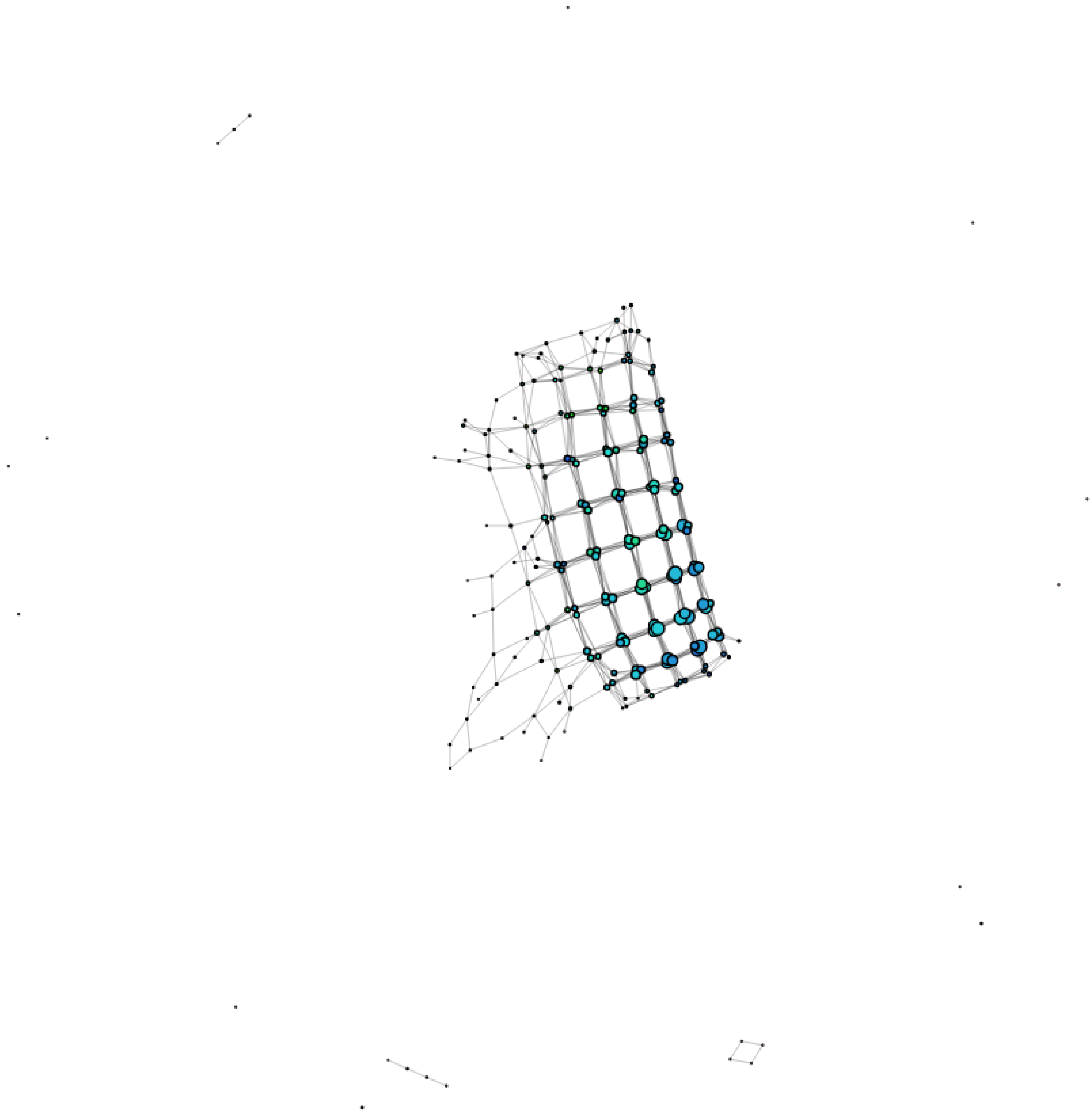
DBSCAN(0.6, 26):



Kmeans(2):

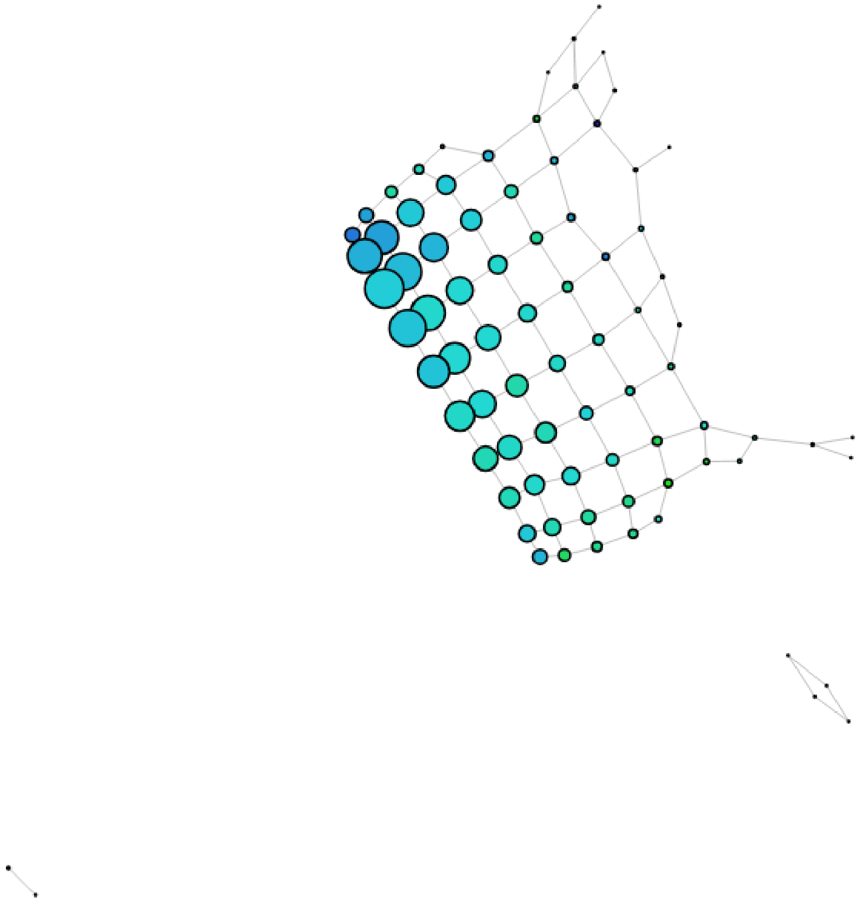


Kmeans(3):



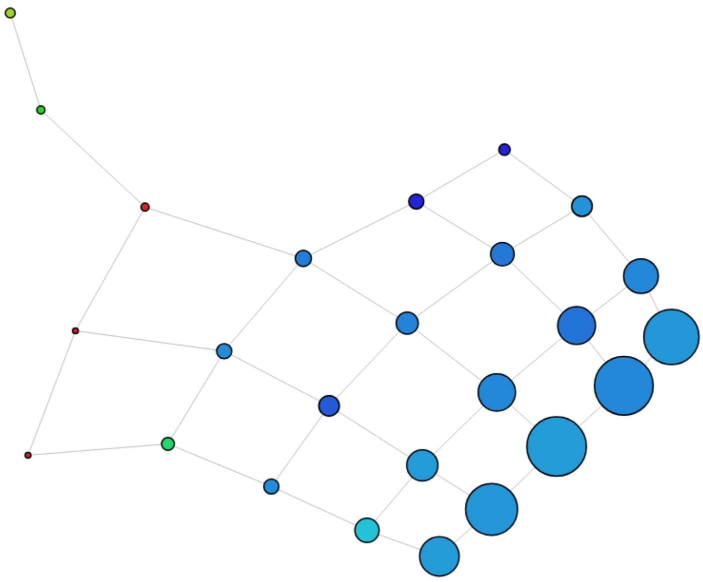
10 Intervals, 30% overlap

10 intervals, 30% overlap, DBSCAN(0.05, 10)



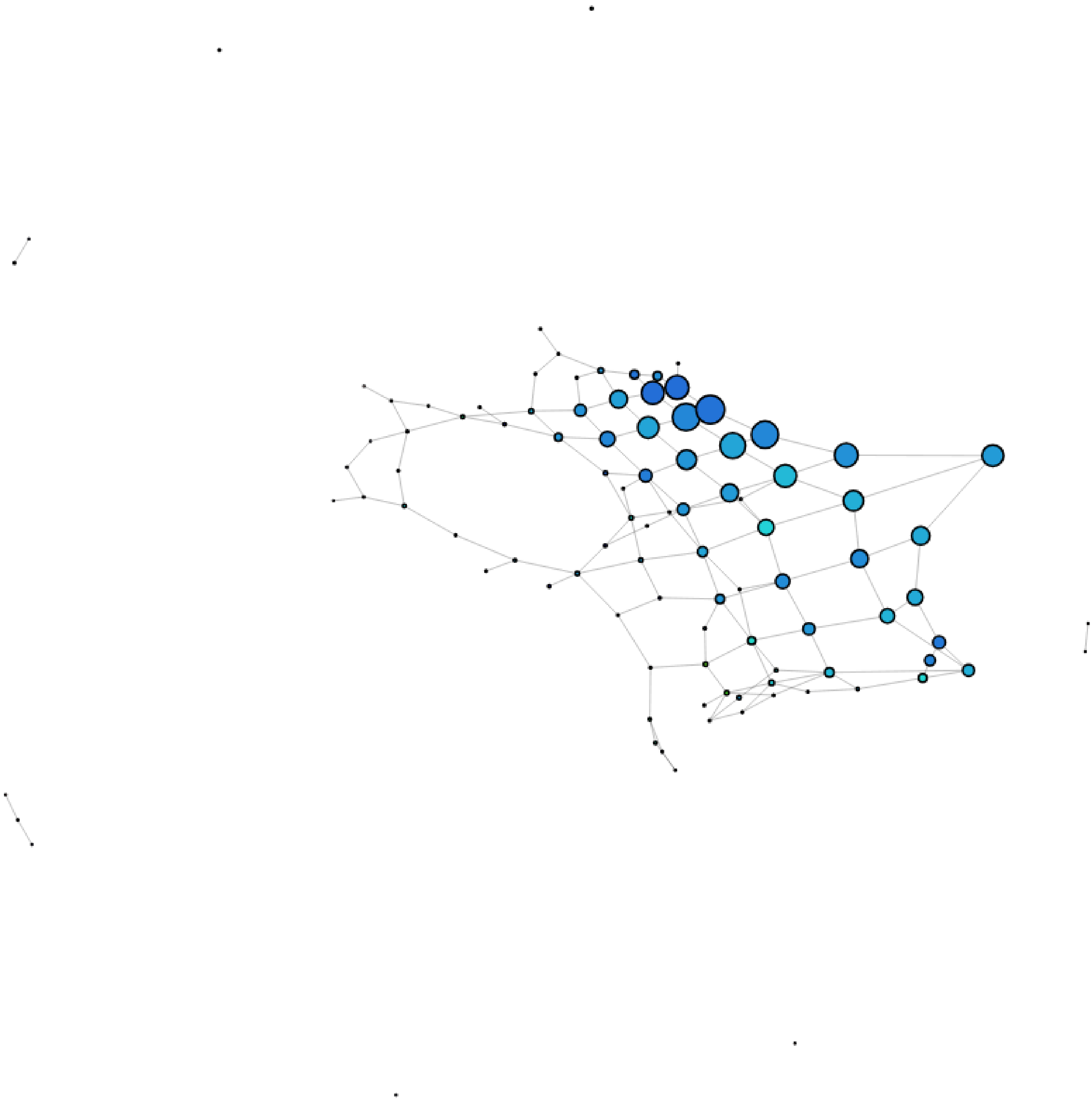
DBSCAN (0.1, 5)

More overlap joins the ADHD participants into the graph, possibly because there were more



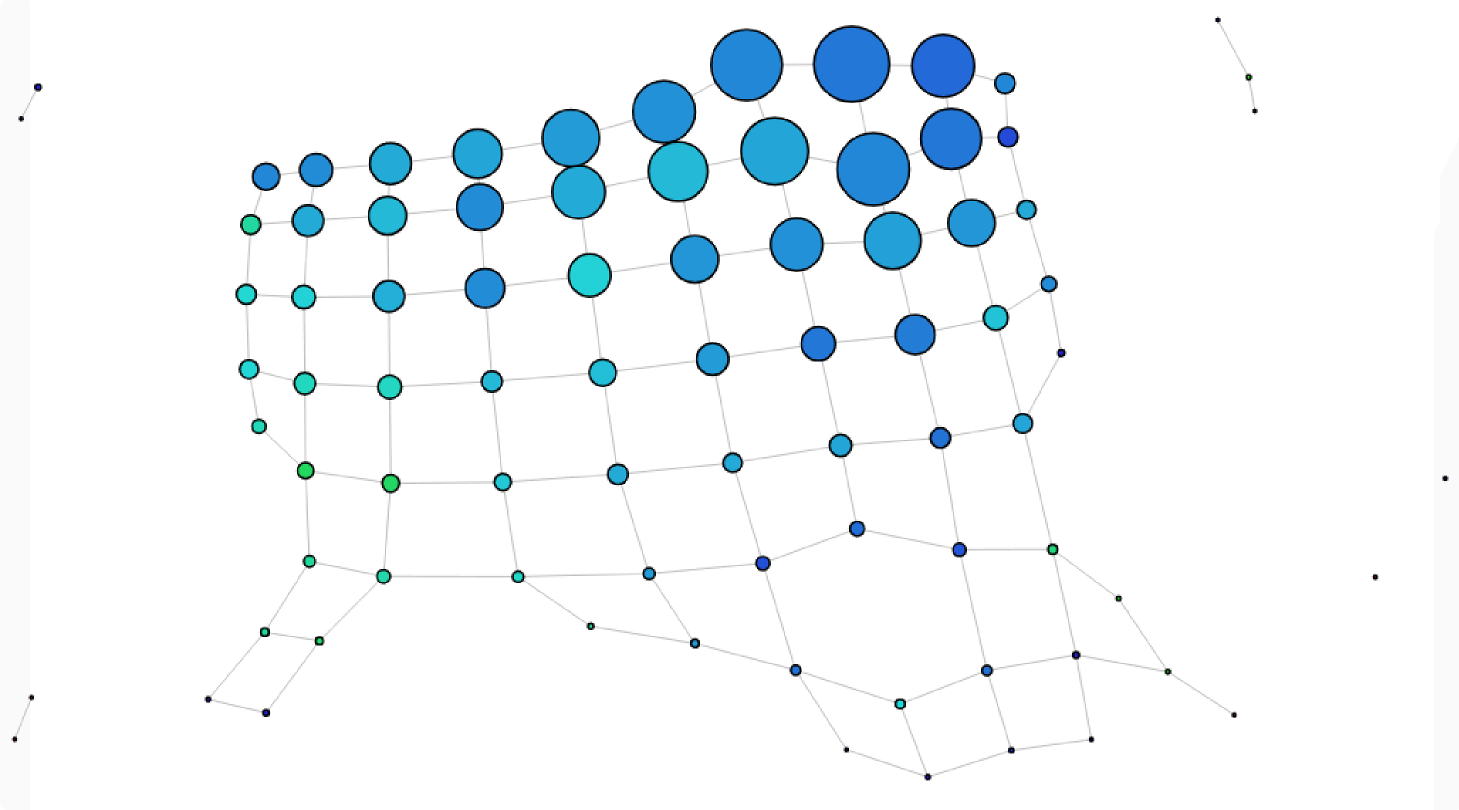
DBSCAN (0.5, 3)

Starting to capture some interesting properties:



DBSCAN (0.6, 26)

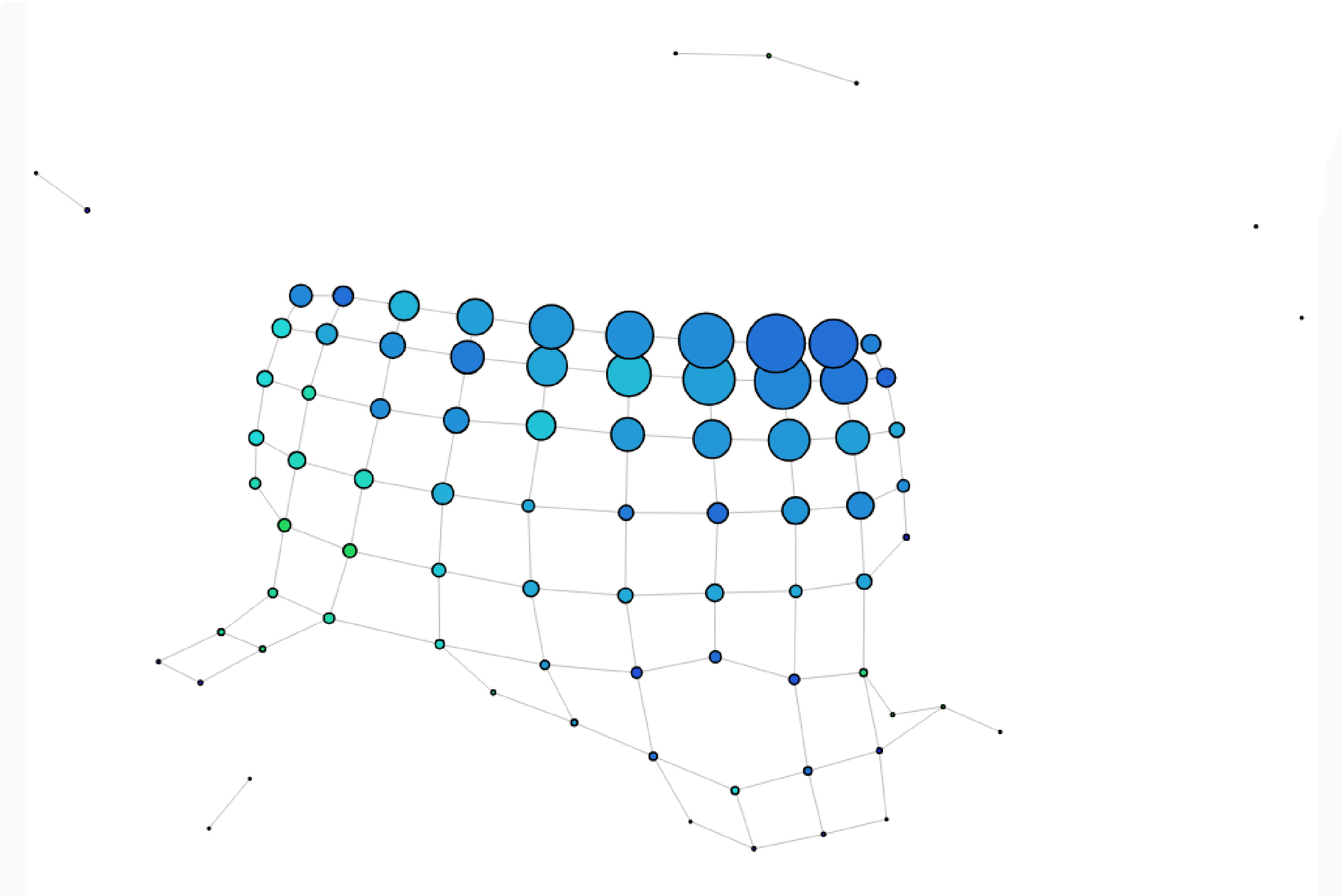
By increasing the thresholds (larger radius, but significantly larger number of points required for a core point) we get a more generalised version of the previous graph:



- wemwbs moves down the graph

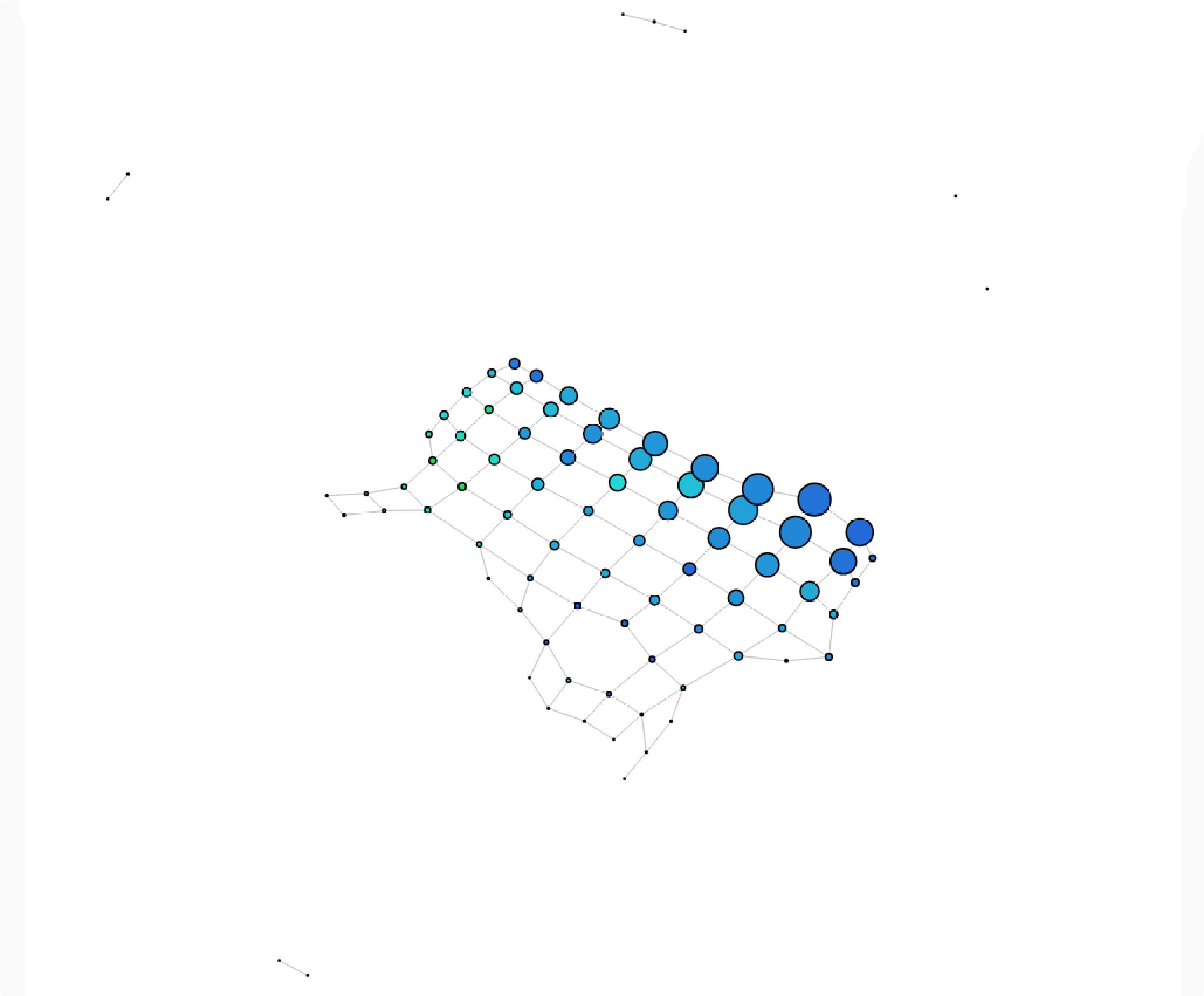
DBSCAN (0.5, 20)

No real change to previously:



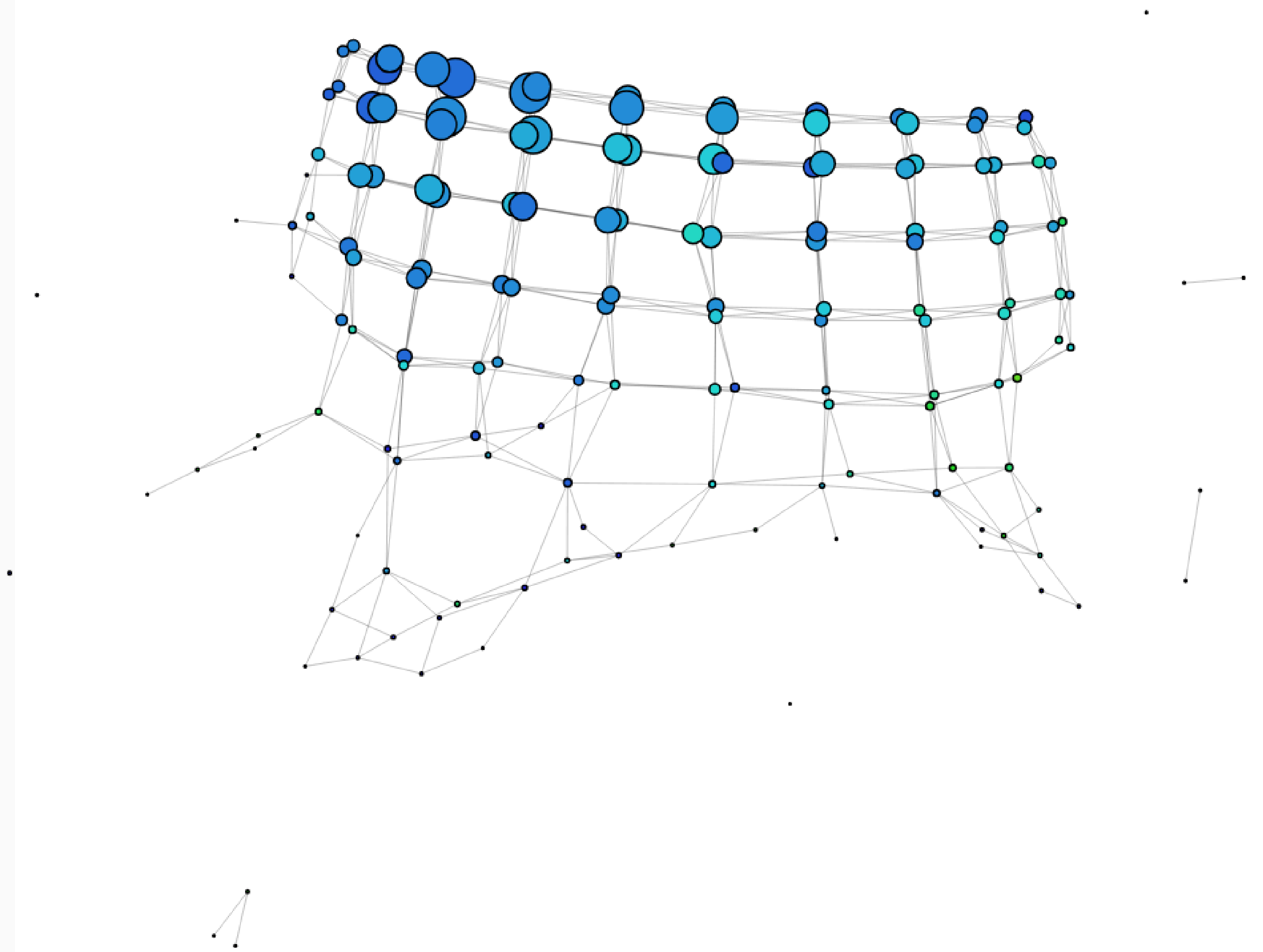
DBSCAN (0.5, 15)

And again



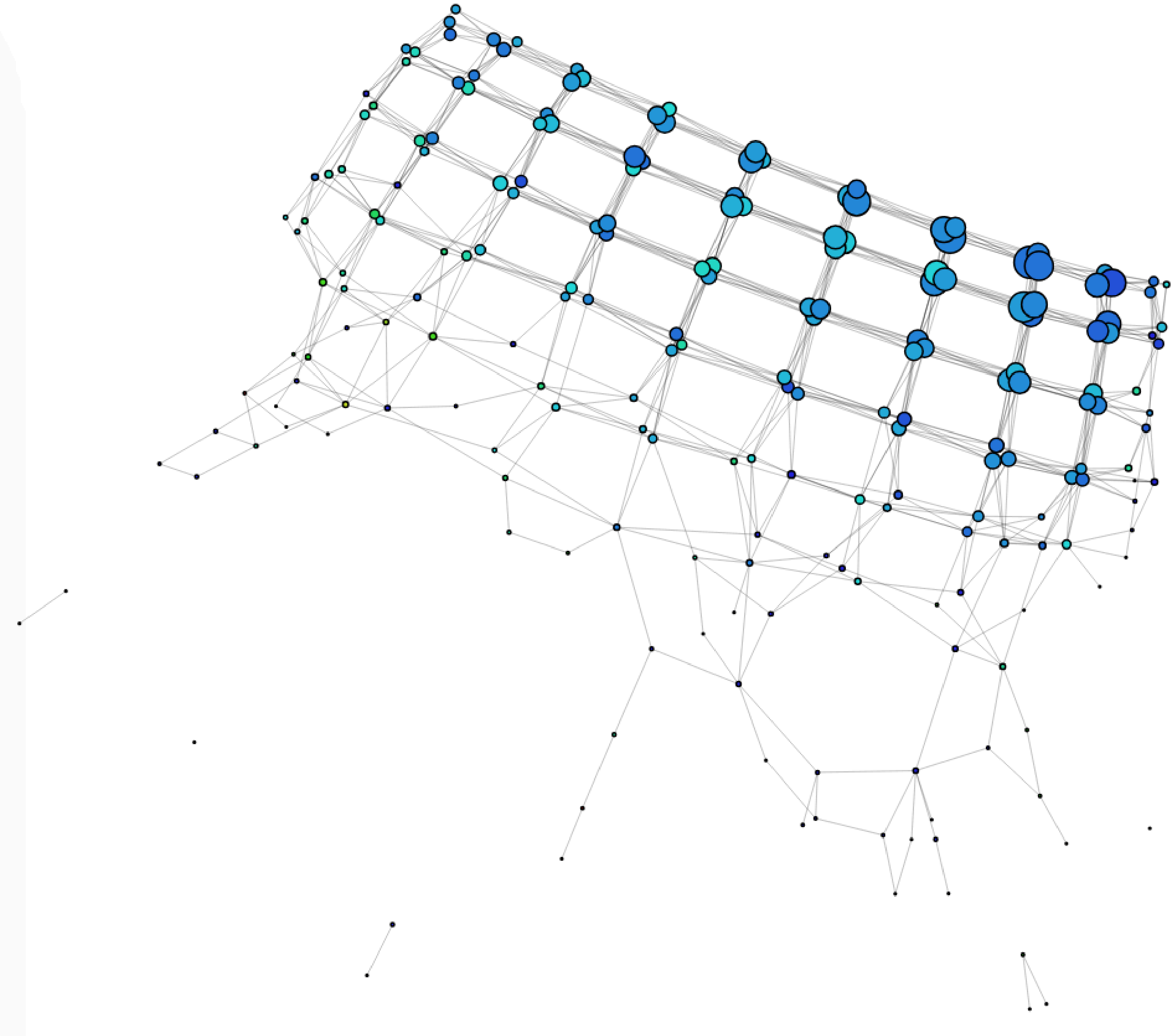
Kmeans (2)

We get the same structure again, however if we look closely at the nodes that are forced to cluster together, we can see that a few have very different proportions of ADHD participants.



Kmeans(3)

Not much of a change in proportions



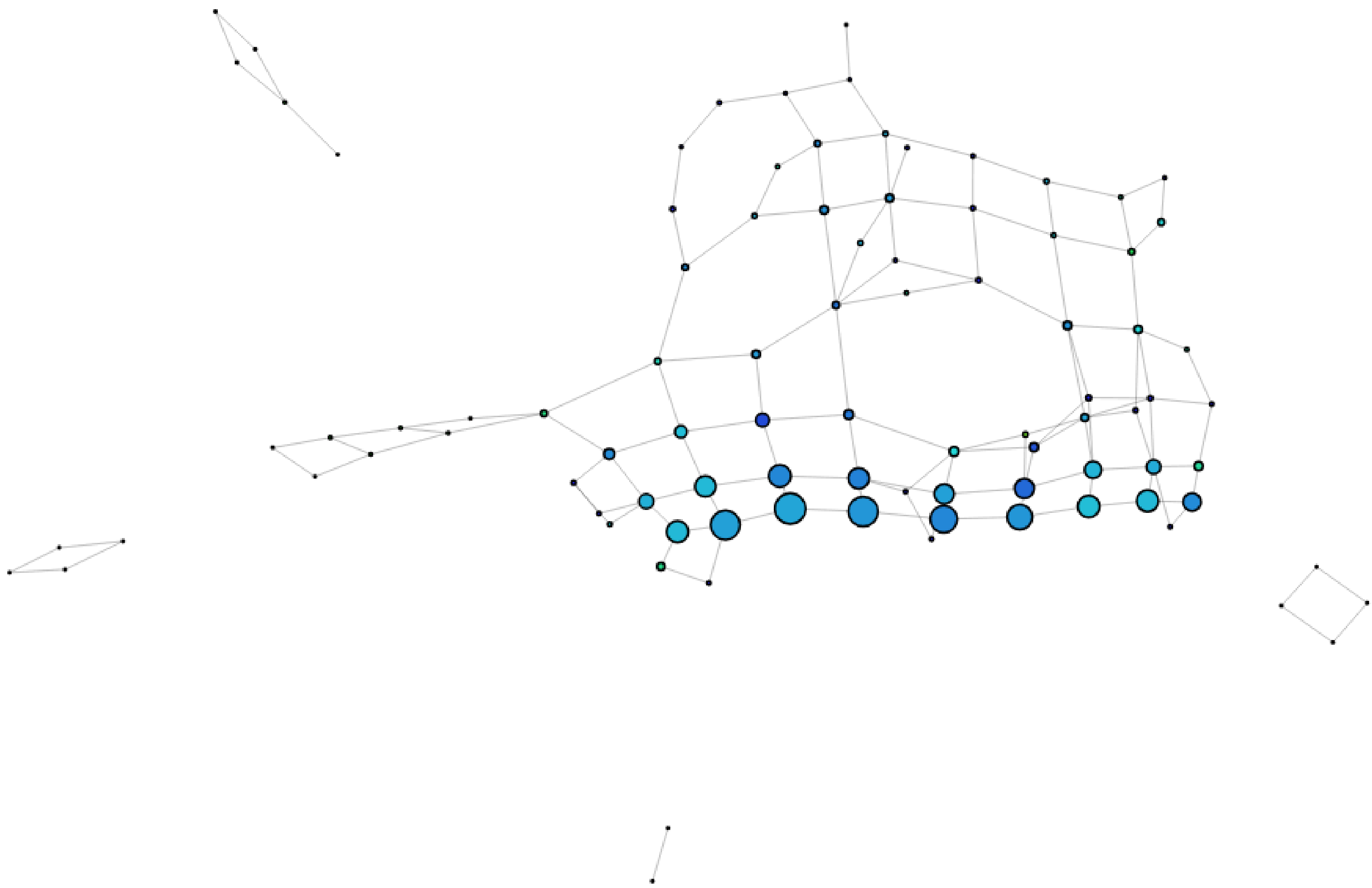
10 Intervals, 50% overlap

DBSCAN (0.1, 5)

We can see a structure similar to 30% overlap, but we have one individual as two separate square components (different wave numbers) as they have very low wellbeing scores.

DBSCAN (0.5, 3)

Allowing nodes to be formed in the main structure by reducing the min pts and increasing the radius shows a little more detail into the main structure, whilst keeping the isolated components.

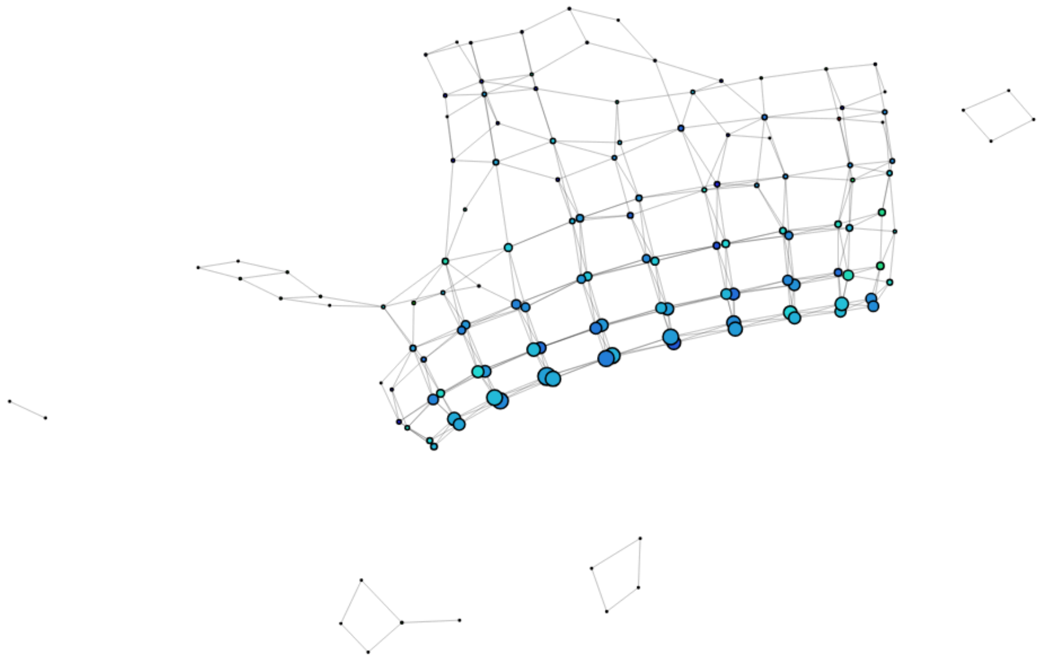


DBSCAN (0.6, 0.26)

The hole is now gone, and we see a little more generalised version again

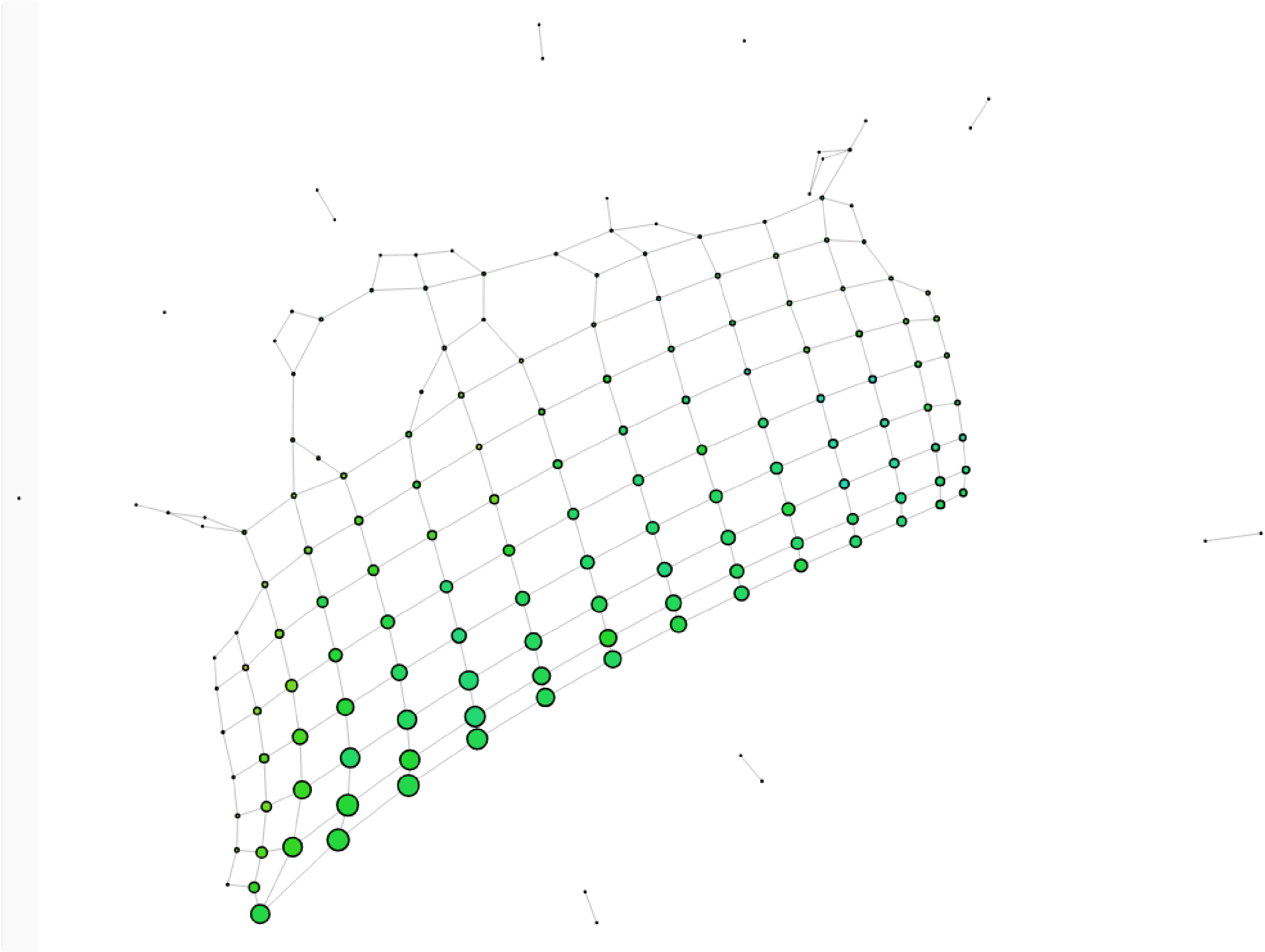
Kmeans (2)

This looks quite good, we see the same structure, same isolated components, but more variety in ADHD proportions. It is clear that by forcing the data to be clustered into two rather than density, it is calculating that there are some disparities in the data, and that these actually get more of the ADHD participants.



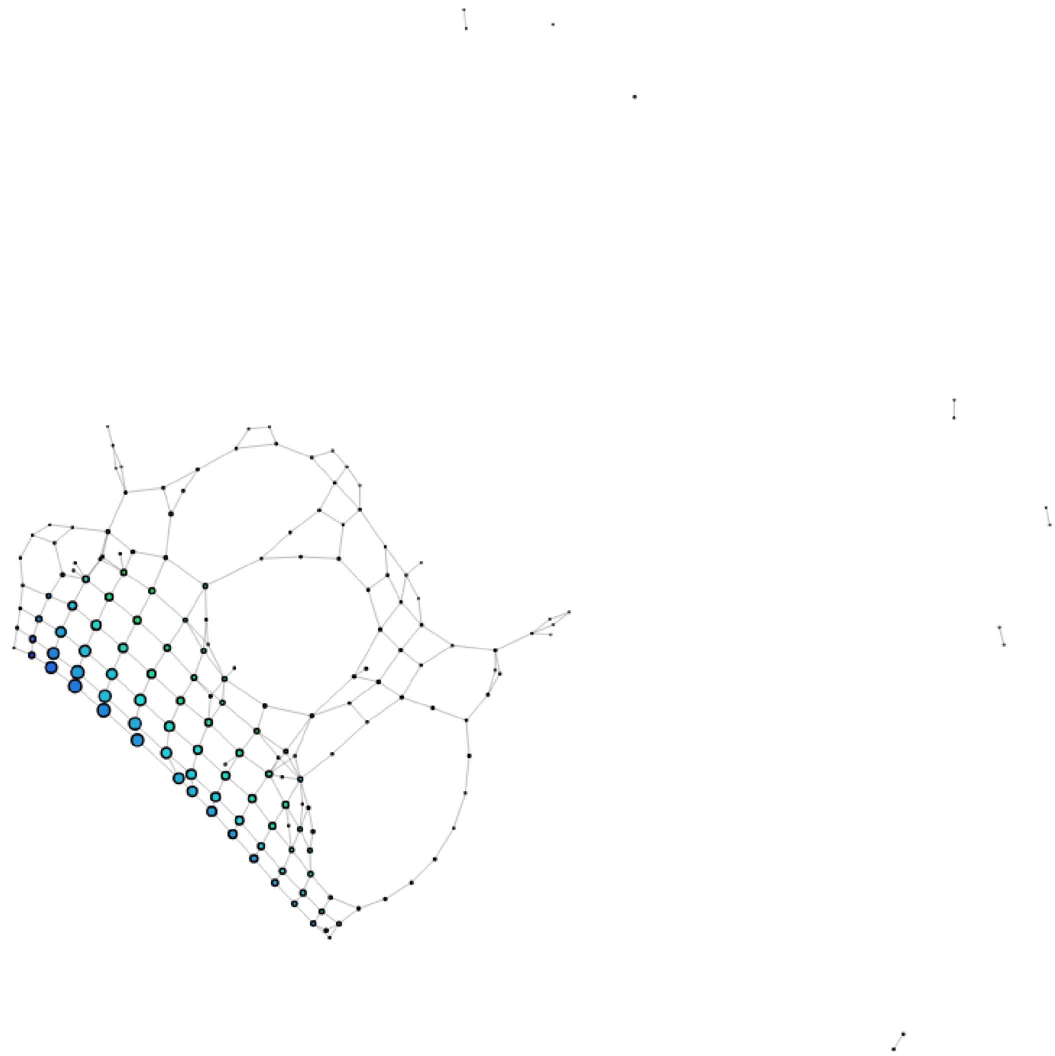
15 Intervals, 30% overlap

DBSCAN (0.1, 5)



DBSCAN (0.5, 3)

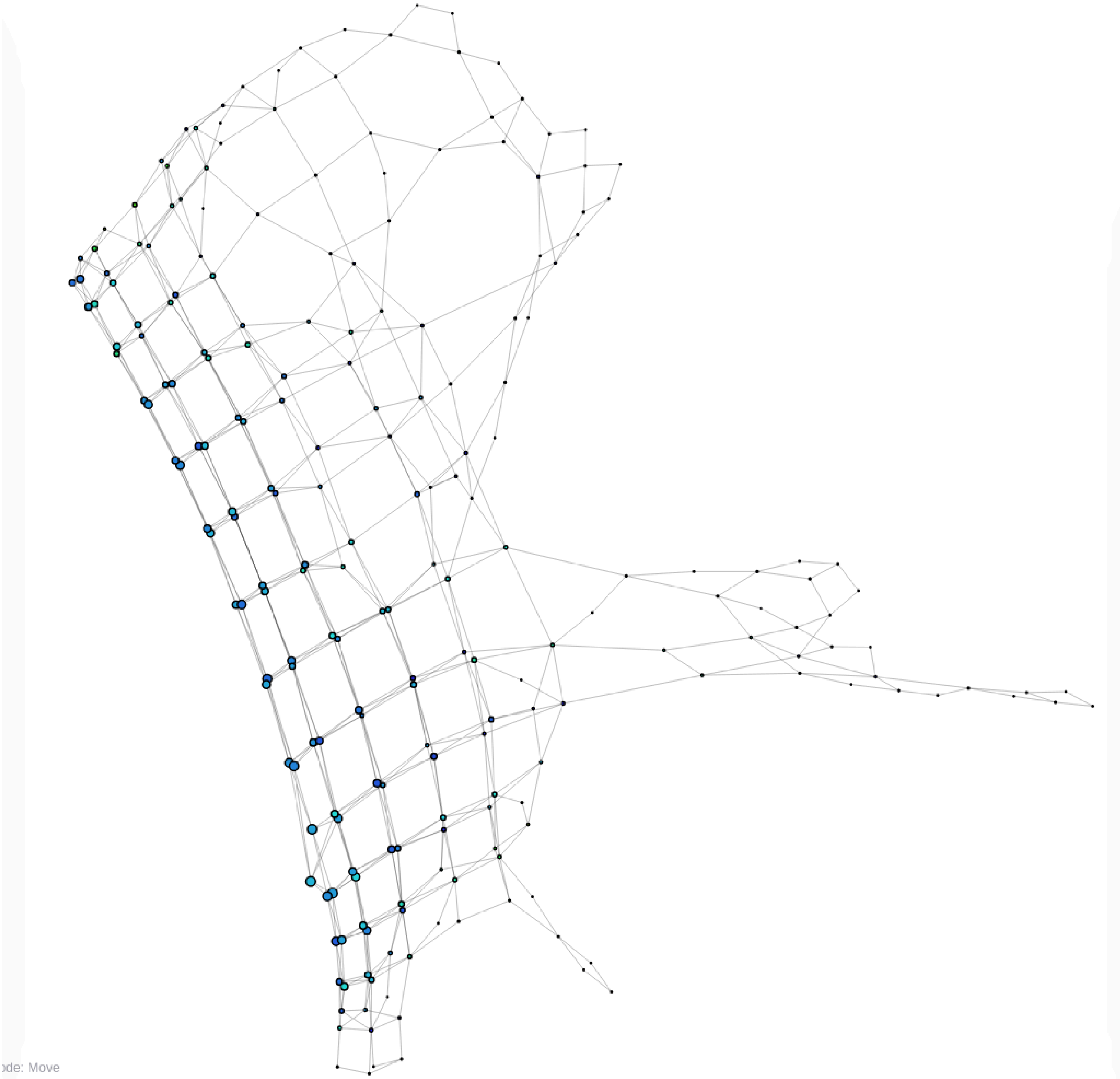
We get an an awesome structure:



- Similar rel frus, reaction times, comp frus, affective valence
- Top circle has slightly higher value of gaming on their lives than the other two, and auto frus
- top has lower rel satisfaction. slightly lower wemwbs
- bottom has higher comp satisfaction, slightly more adhd between middle and bottom, a lot higher binge minutes
- Top circle plays less

Kmeans (2)

Still has isolated components

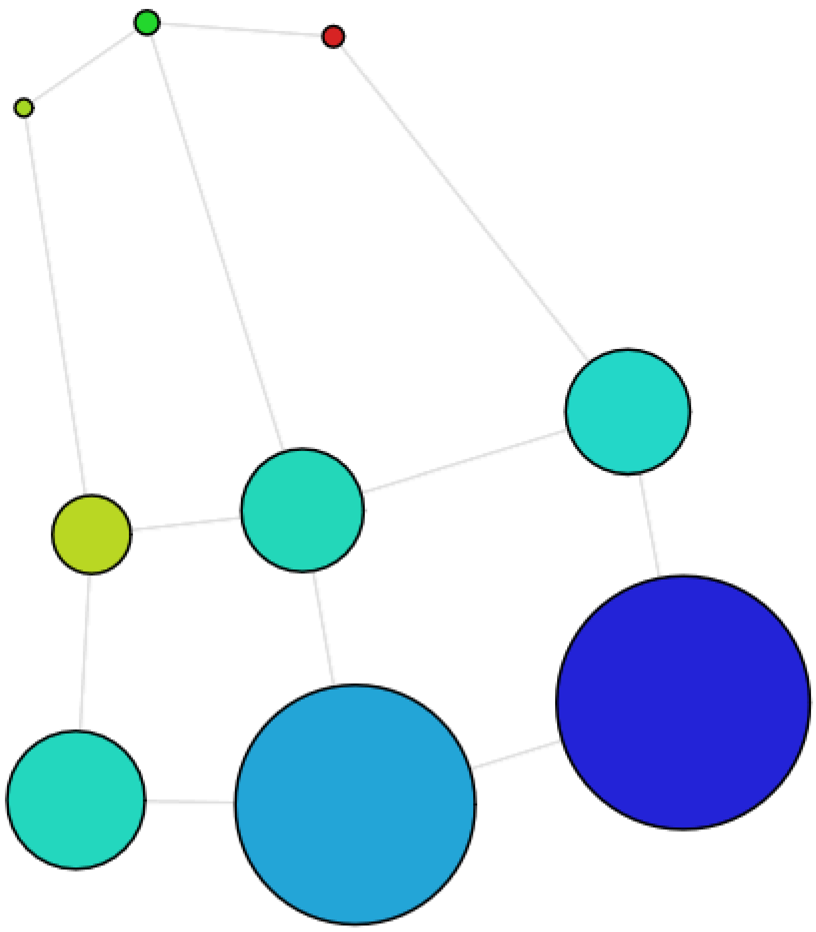


- No circular structure
- Personally feel we have stretched the graph out too far here

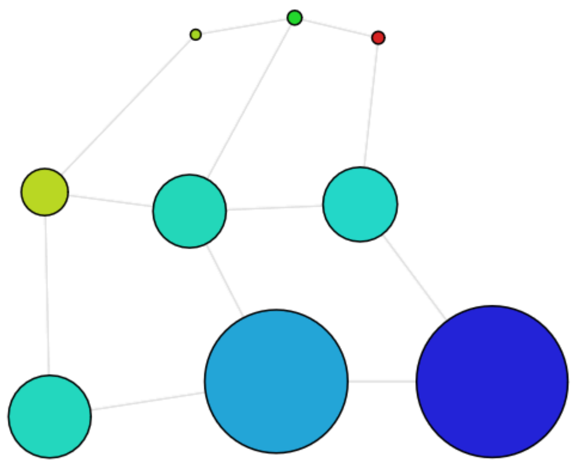
PC1, PC2, Z-scaling

3 intervals, 30% overlap

DBSCAN (0.1, 5)



DBSCAN (0.6, 26)

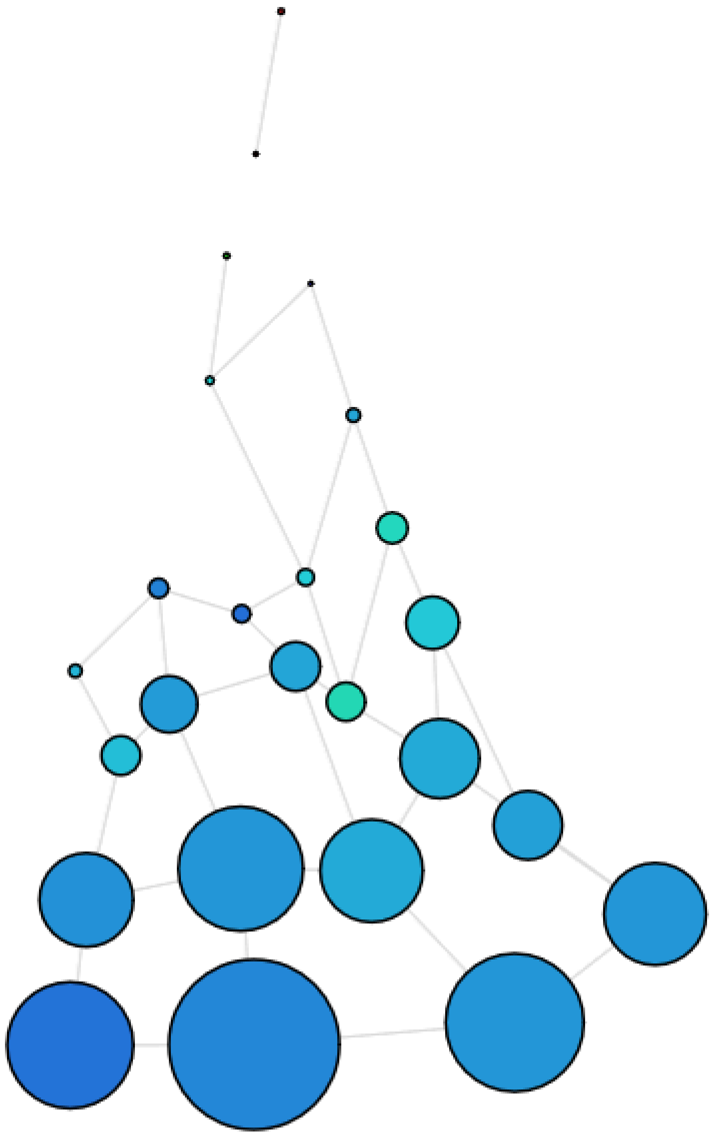


3 intervals, 50% overlap

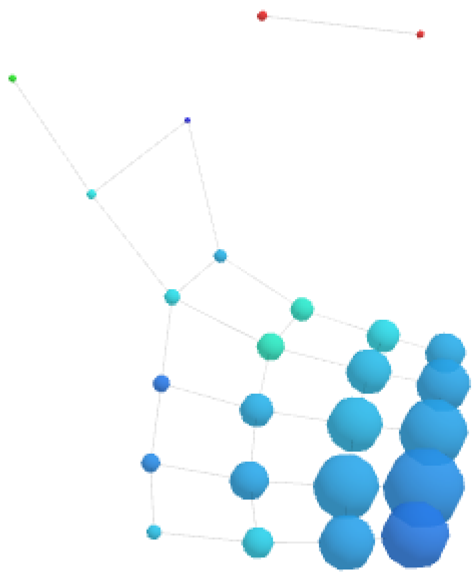
No change in structure compared to 30% overlap

5 intervals, 30% overlap

Same structure as min max scaling, including the isolated component:
DBSCAN (0.1, 5):

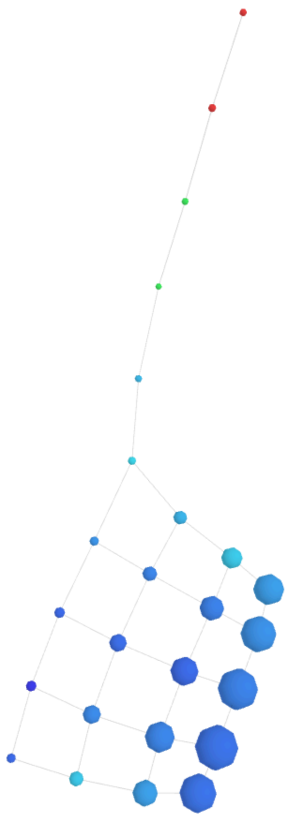


DBSCAN (0.6, 26):

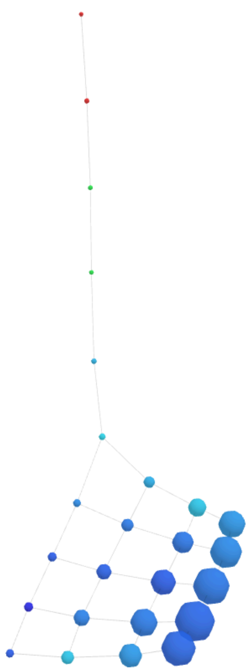


5 Intervals 50% overlap

DBSCAN (0.1, 5):

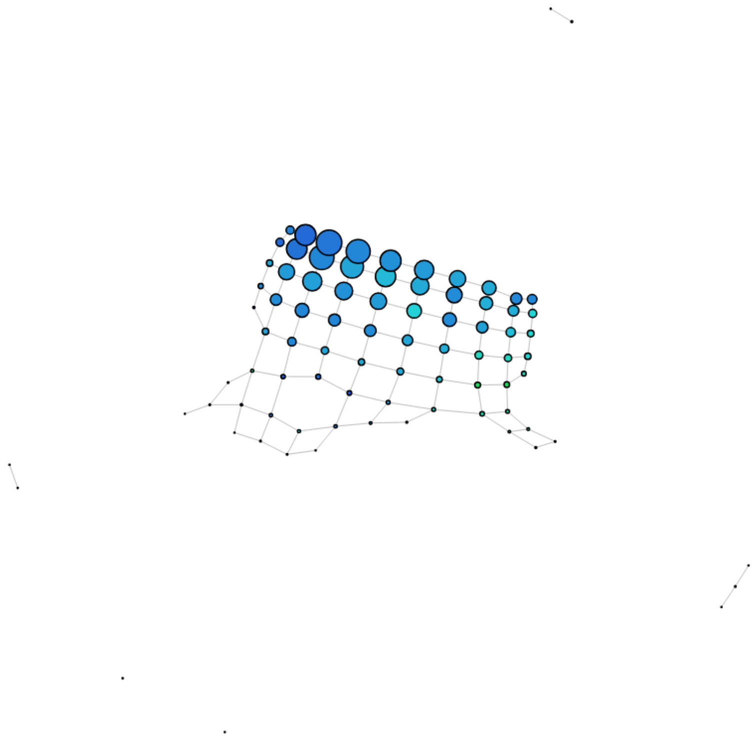


DBSCAN (0.6, 26):

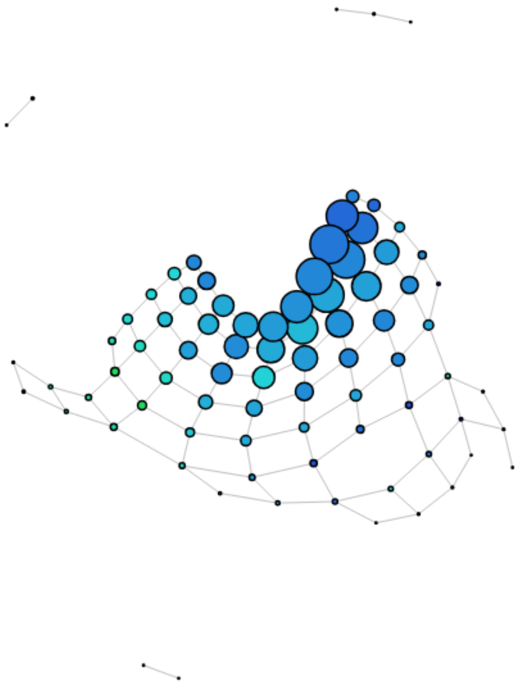


10 intervals 30% overlap

DBSCAN (0.1, 5):



DBSCAN (0.6, 26):



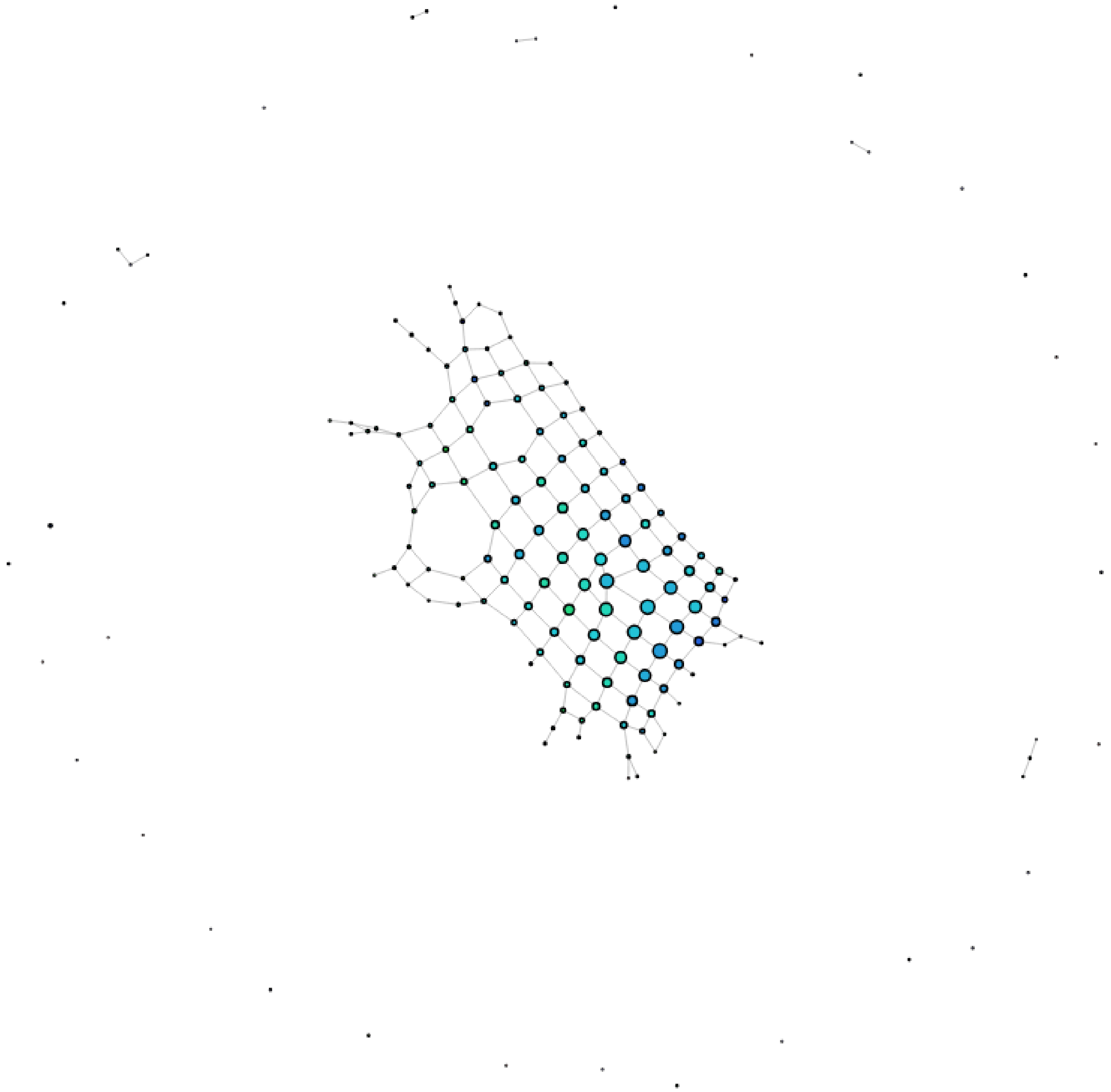
10 Intervals, 50% overlap

DBSCAN (0.1, 5):

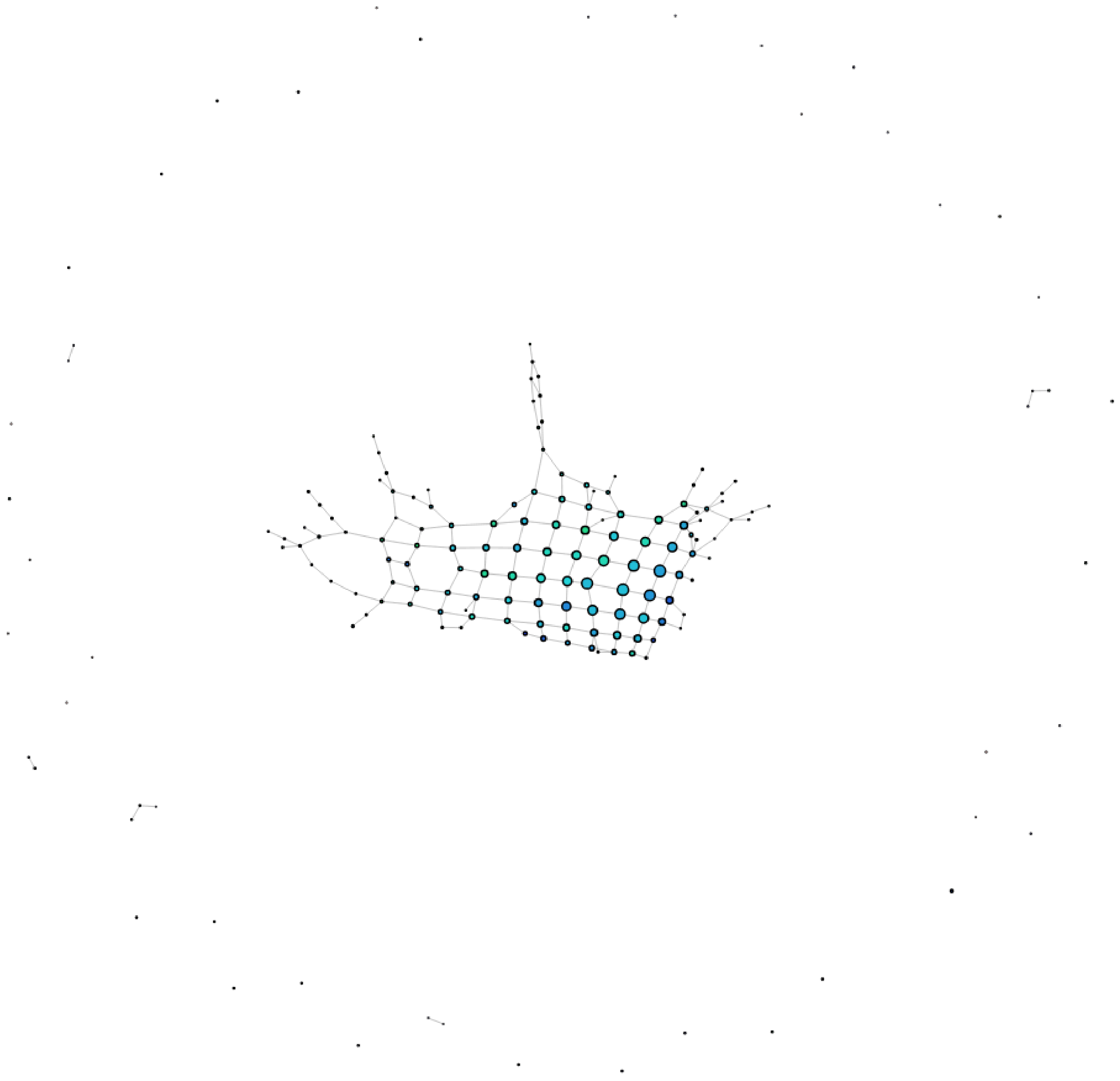
DBSCAN (0.6, 26):

15 Intervals, 10% overlap

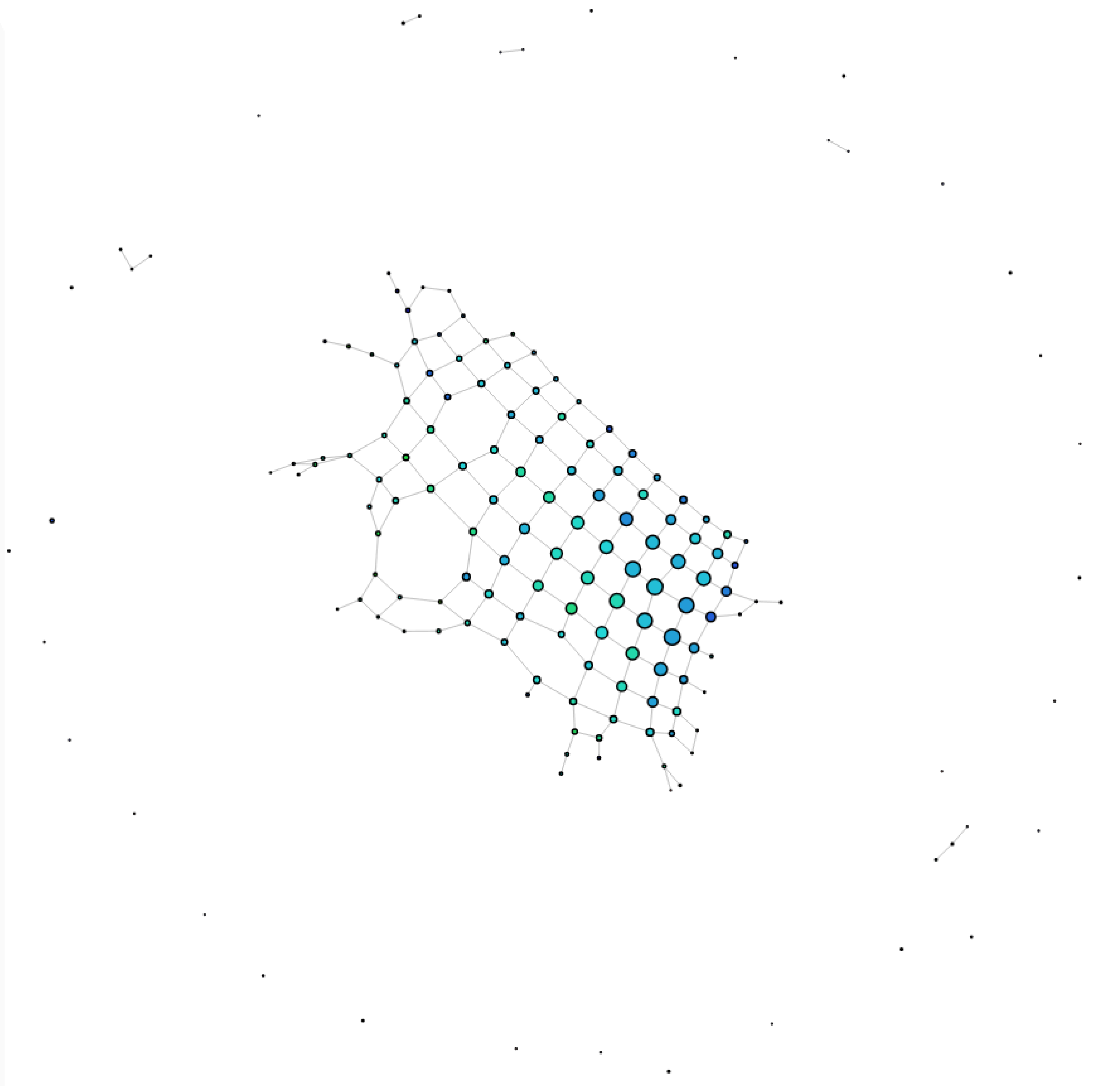
DBSCAN(0.1, 5):



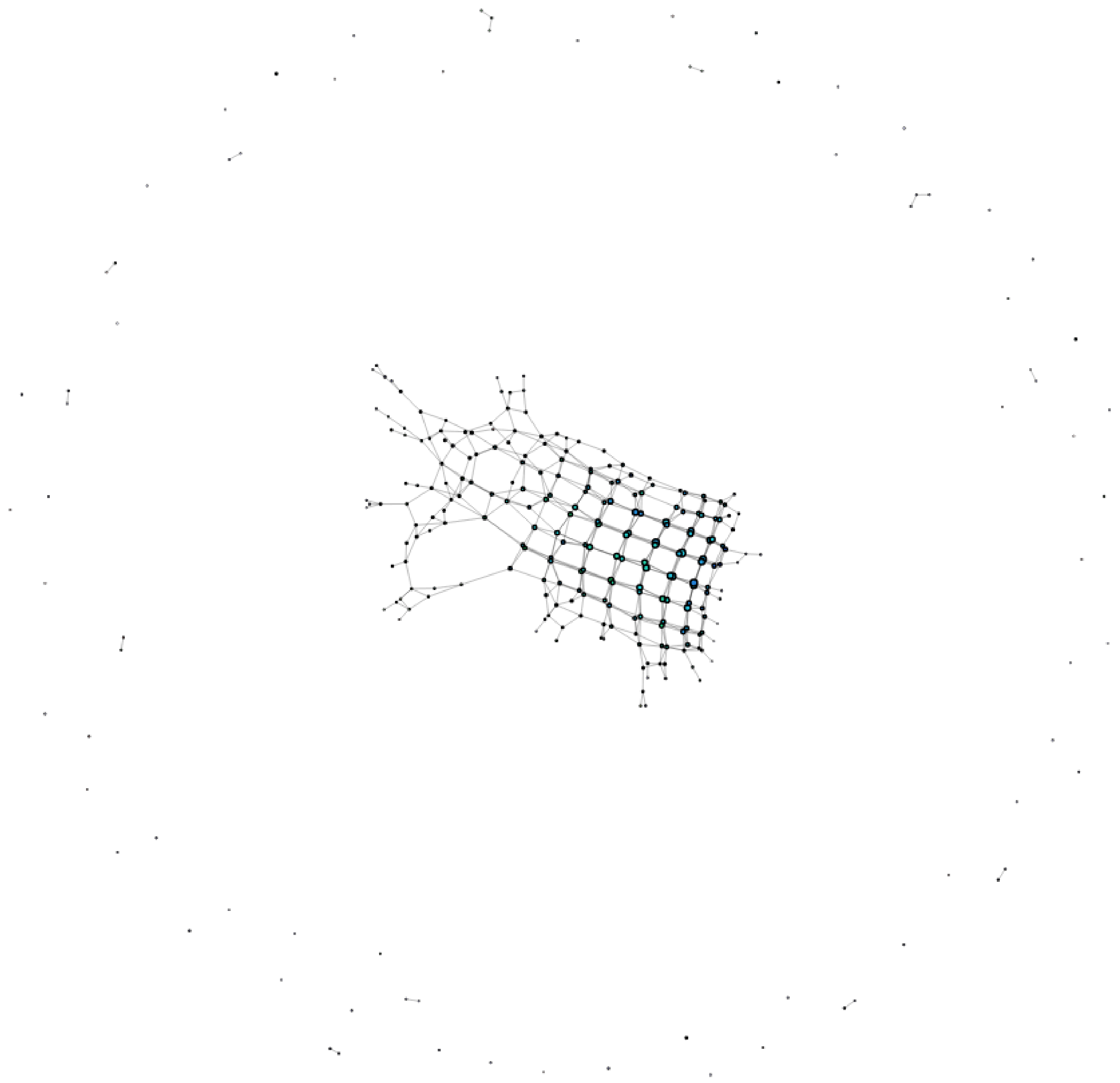
DBSCAN(0.5, 3):



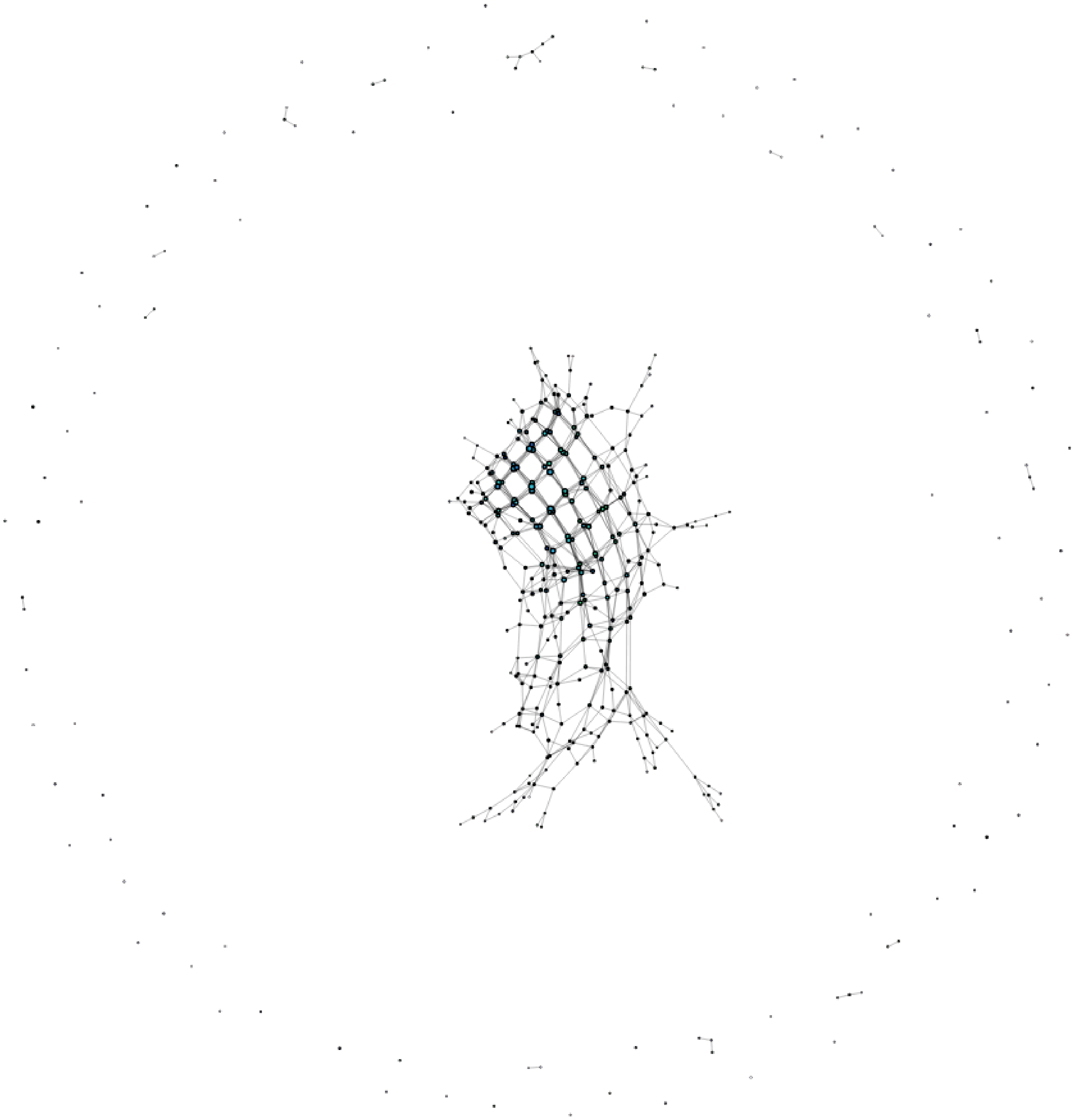
DBSCAN(0.6, 26):



Kmeans(2):

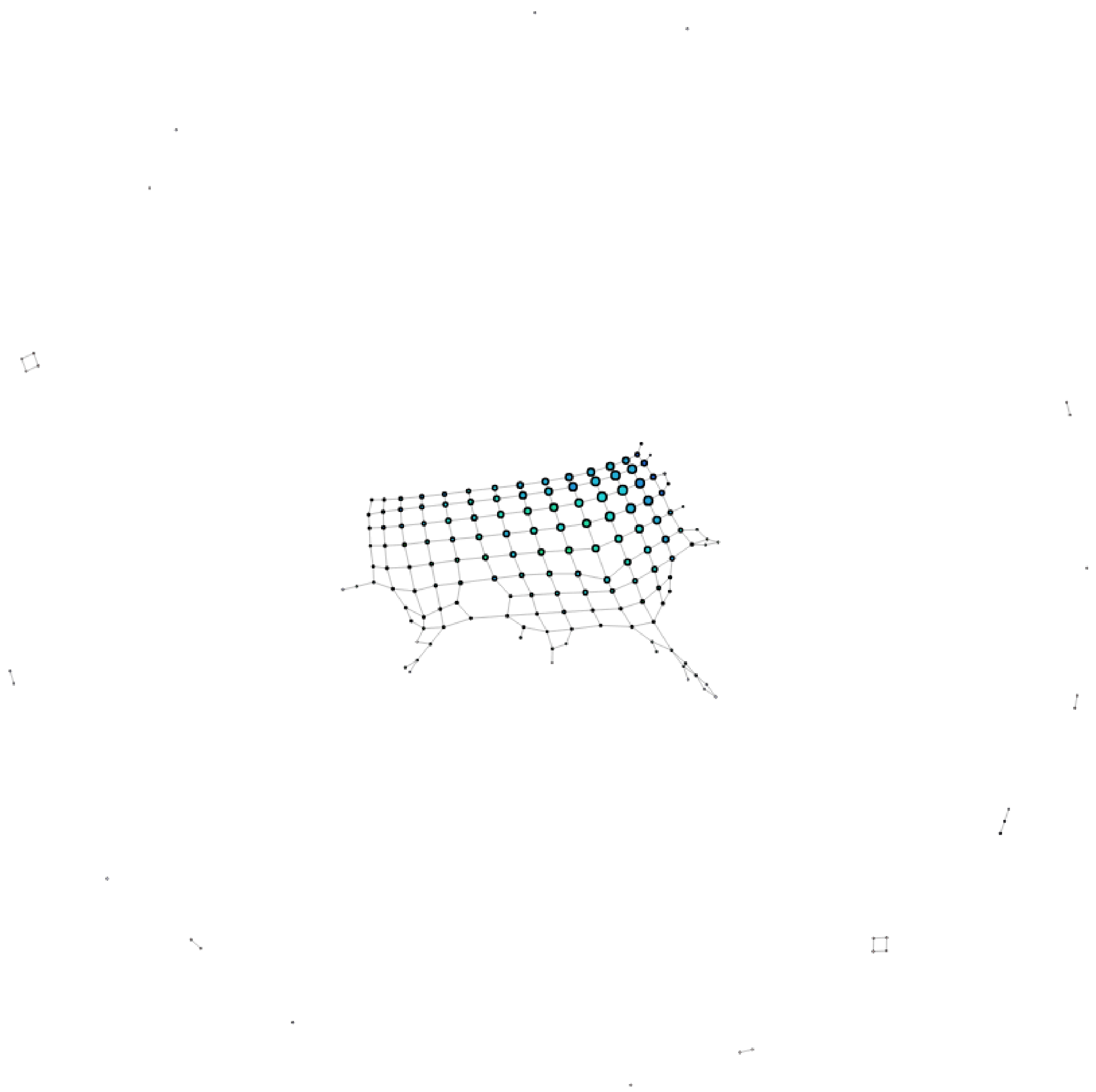


Kmeans(3):

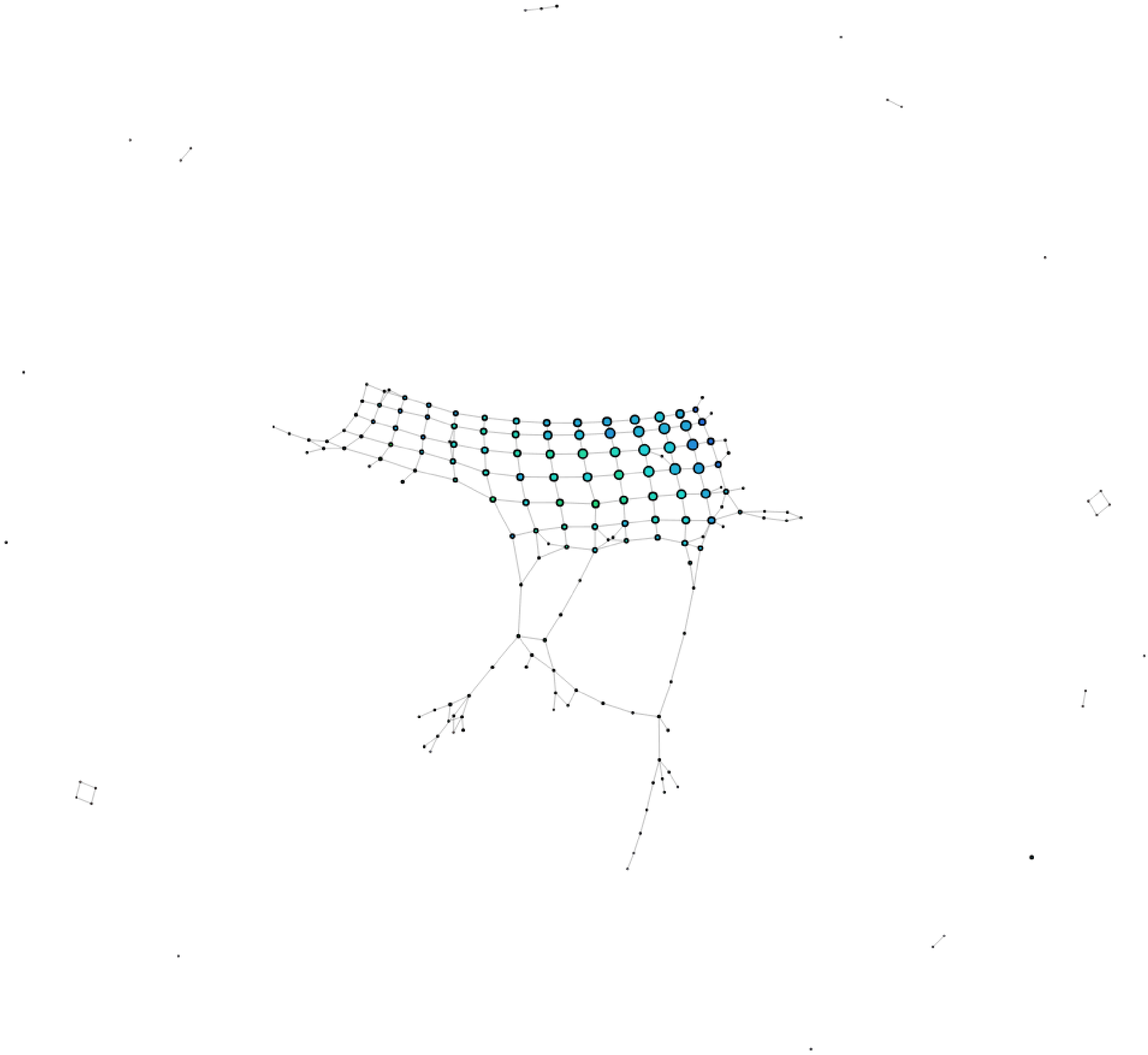


15 Intervals, 20% overlap

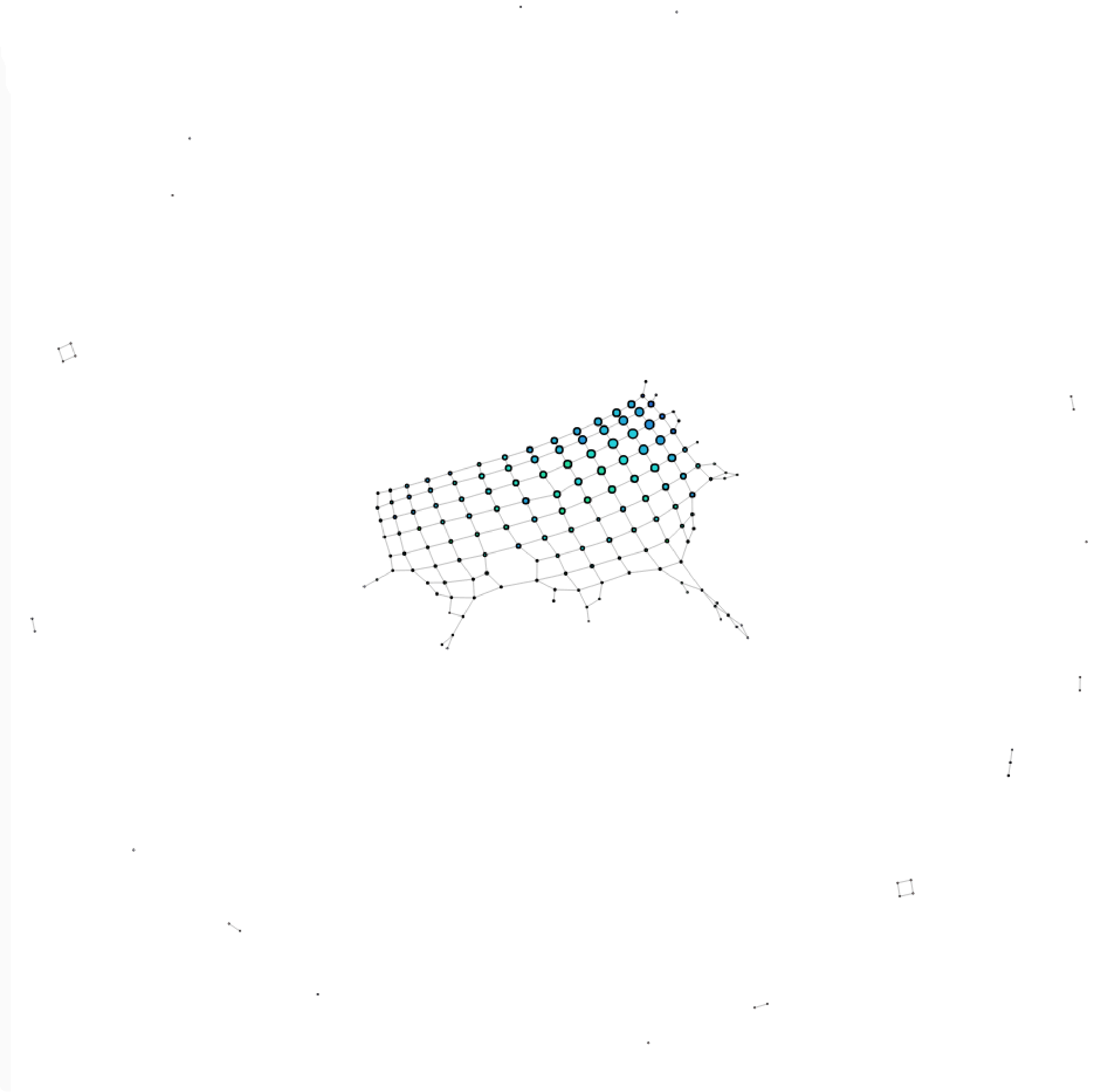
DBSCAN(0.1, 5):



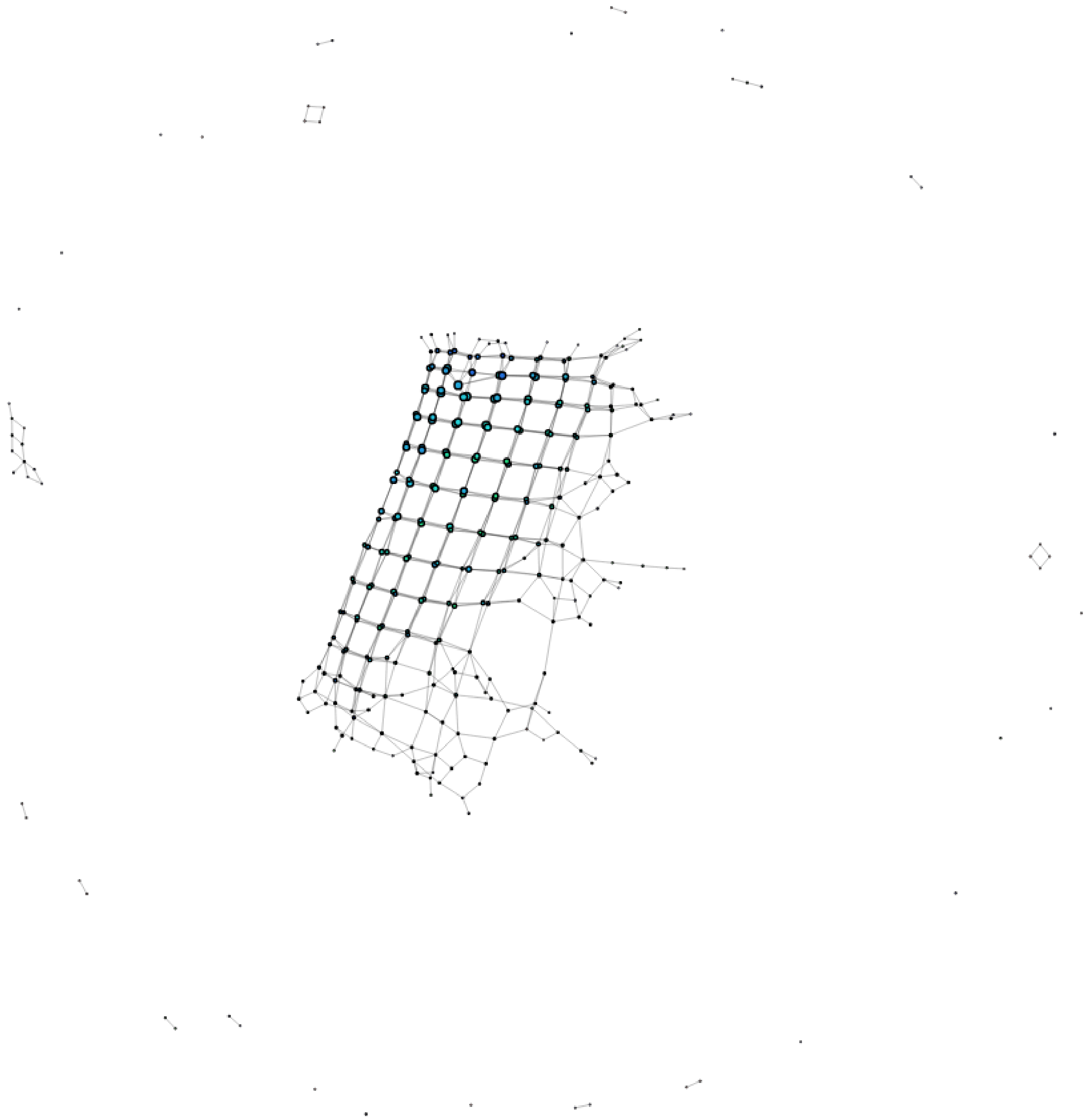
DBSCAN(0.5, 3):



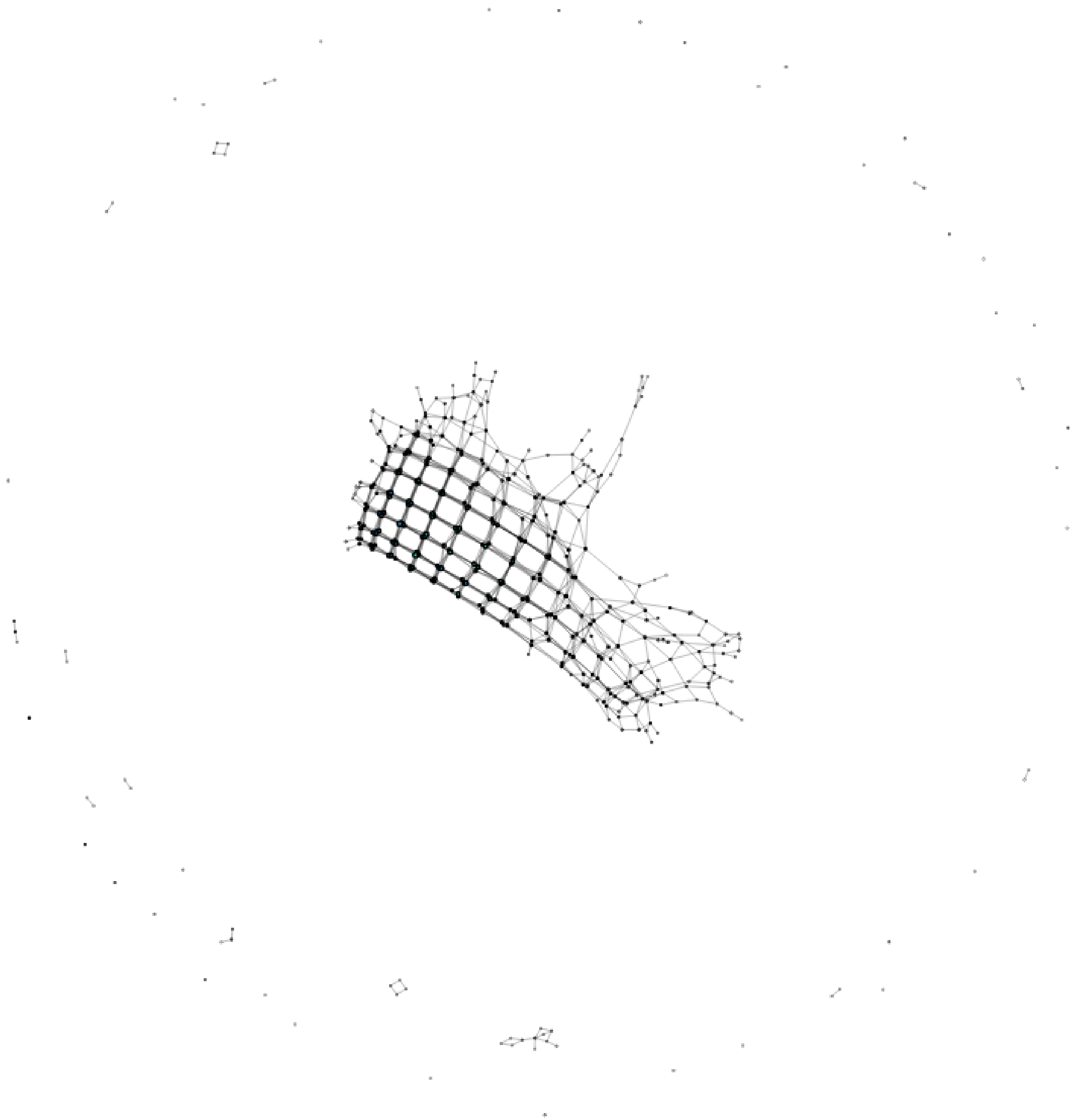
DBSCAN(0.6, 26):



Kmeans(2):

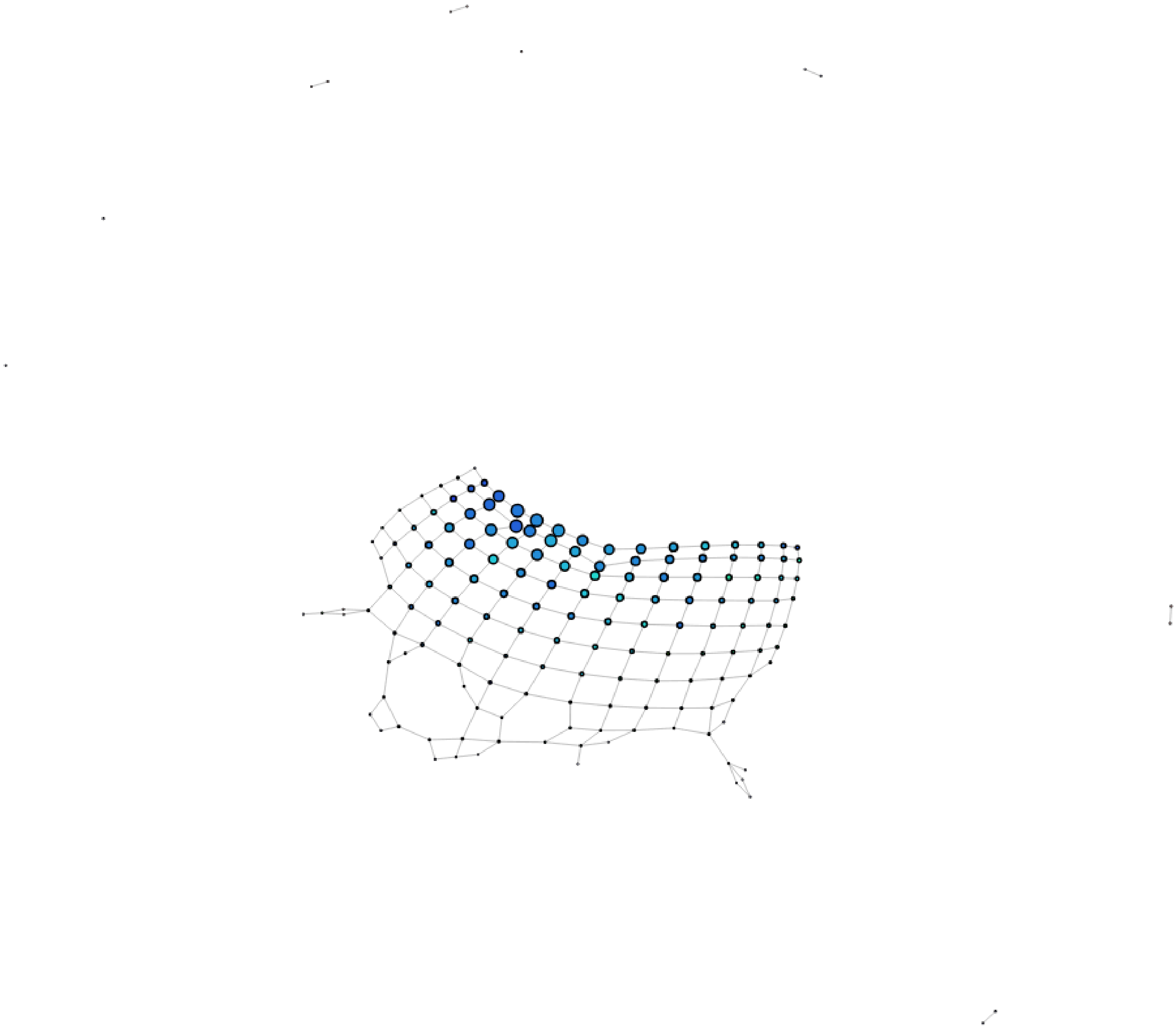


Kmeans(3):



15 intervals, 30% overlap

DBSCAN(0.1, 5):

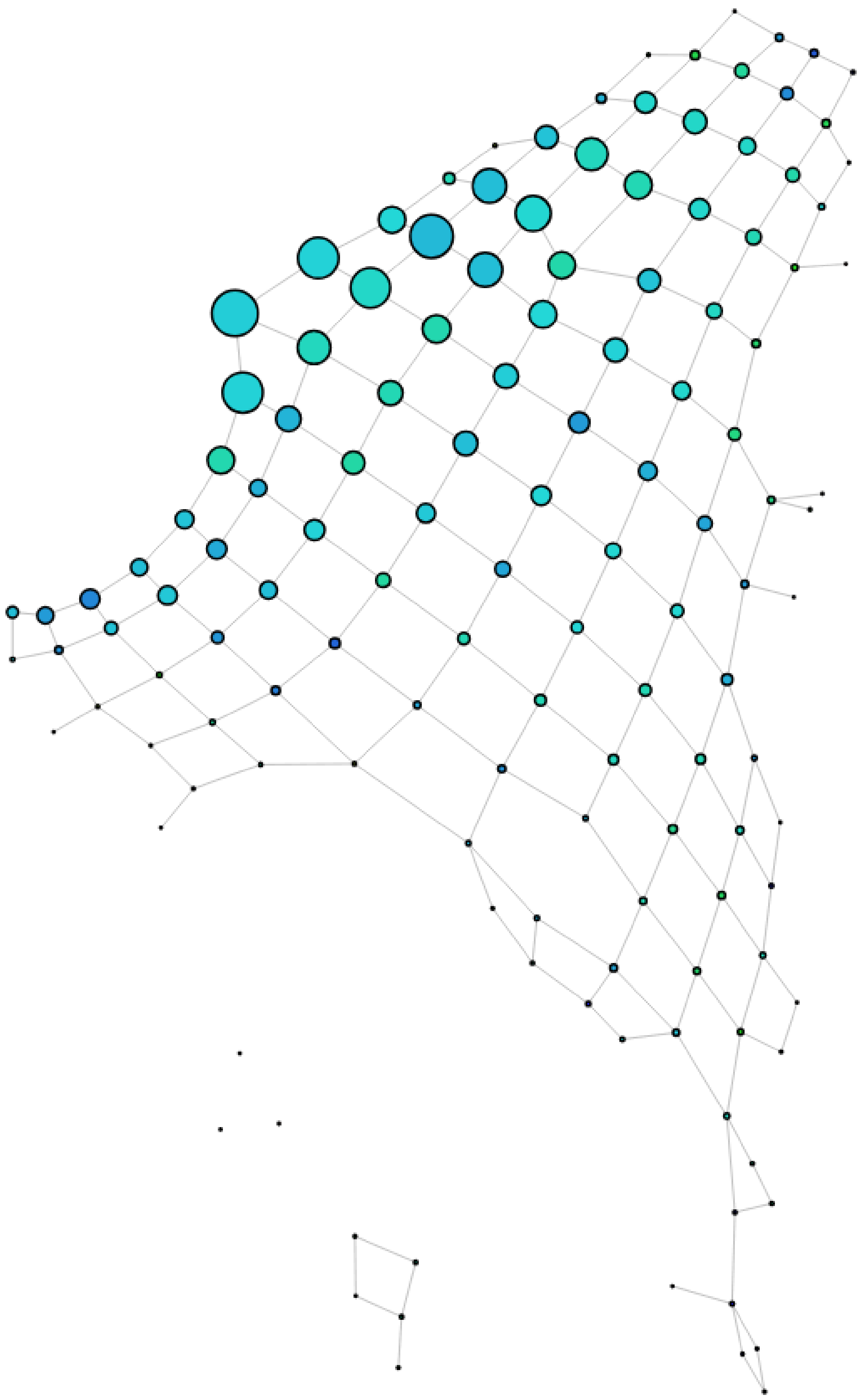


PC1 and Eccentricity

Here, we aim to form a structure that splits more with variance, and by a measure of centrality. We aim to capture a bit more of the outlying participants. We also performed a logarithm of the total wave minutes and binge watching. This actually may be counter intuitive as we want to measure outliers, but removing the skewed distribution.

15 Intervals, 30% overlap, DBSCAN(0.1, 5)

Aiming to create clusters of more dense data (within a radius of 0.1 and has at least 5 points):



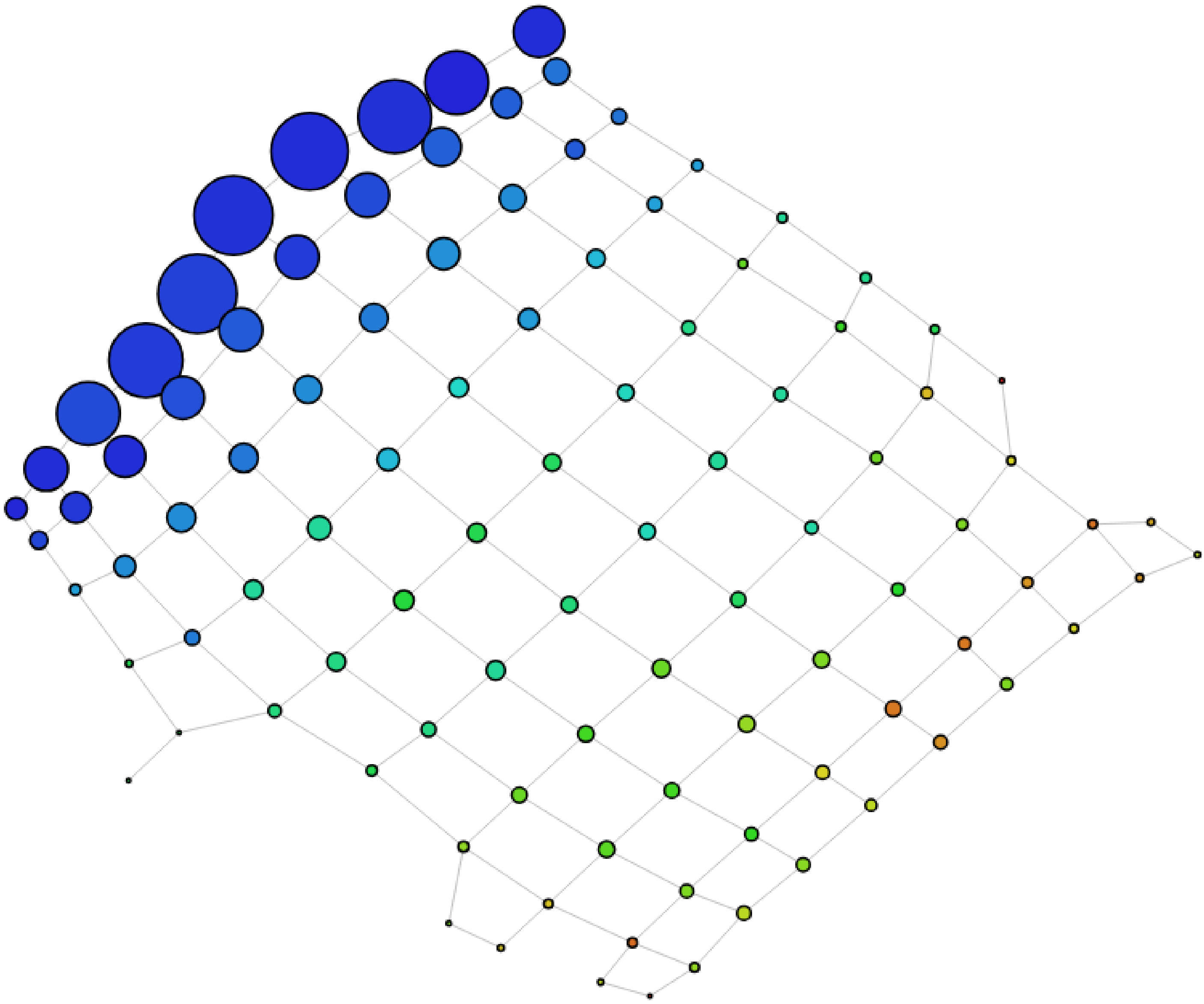
This is interesting because we have a little tail, that captures

XGBoost and Eccentricity

This is to force the network to create a structure that forces out the ADHD participants from the dense cluster of data. Using the eccentricity also forces geometric outliers out into the open.

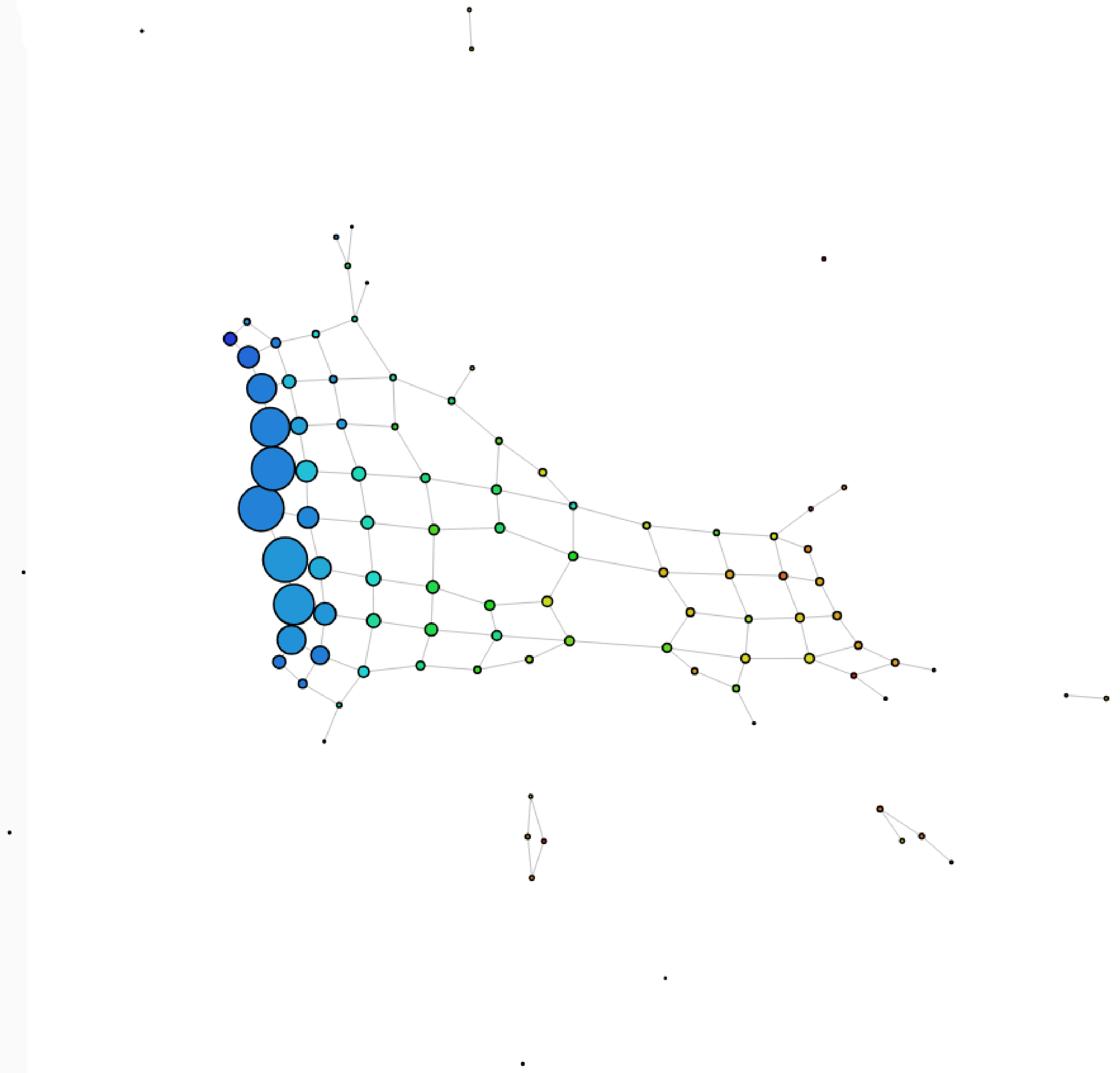
10 Intervals, 30% overlap, DBSCAN(0.05, 10)

Requiring 10 nodes in the neighbourhood of radius 0.05, we are creating clusters that are more dense, trying to spread the dense overall data out more. Coloured by proportion of participants with any form of ADHD:



10 Intervals, 20% overlap, DBSCAN(0.05, 10)

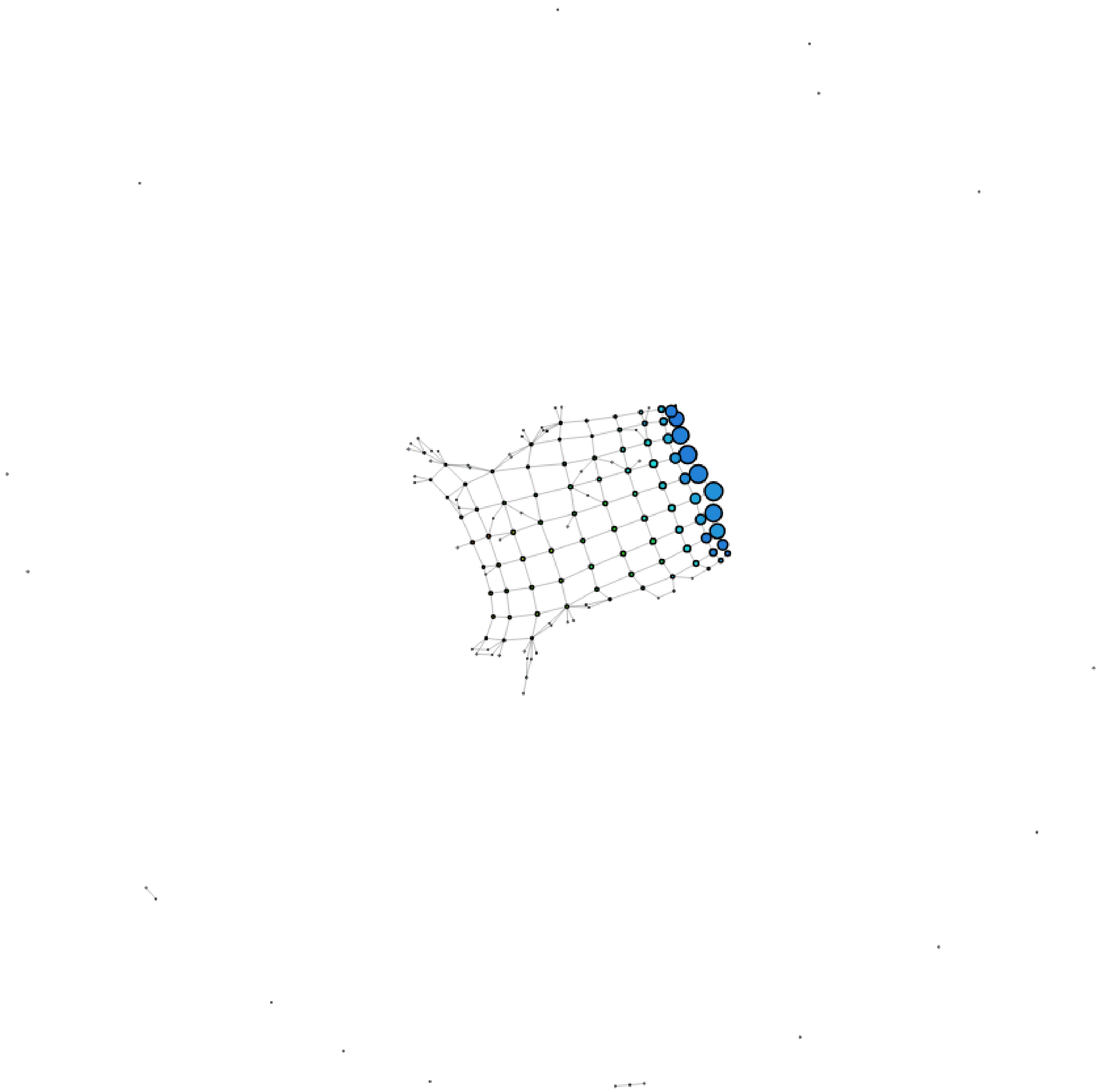
Reducing the overlap gets us a little more structure as 10 Intervals, 30% overlap, DBSCAN(0.05, 10) got us just a plan landscape.



Note that increasing the number of neighourhood required to 5 did not change the structure.

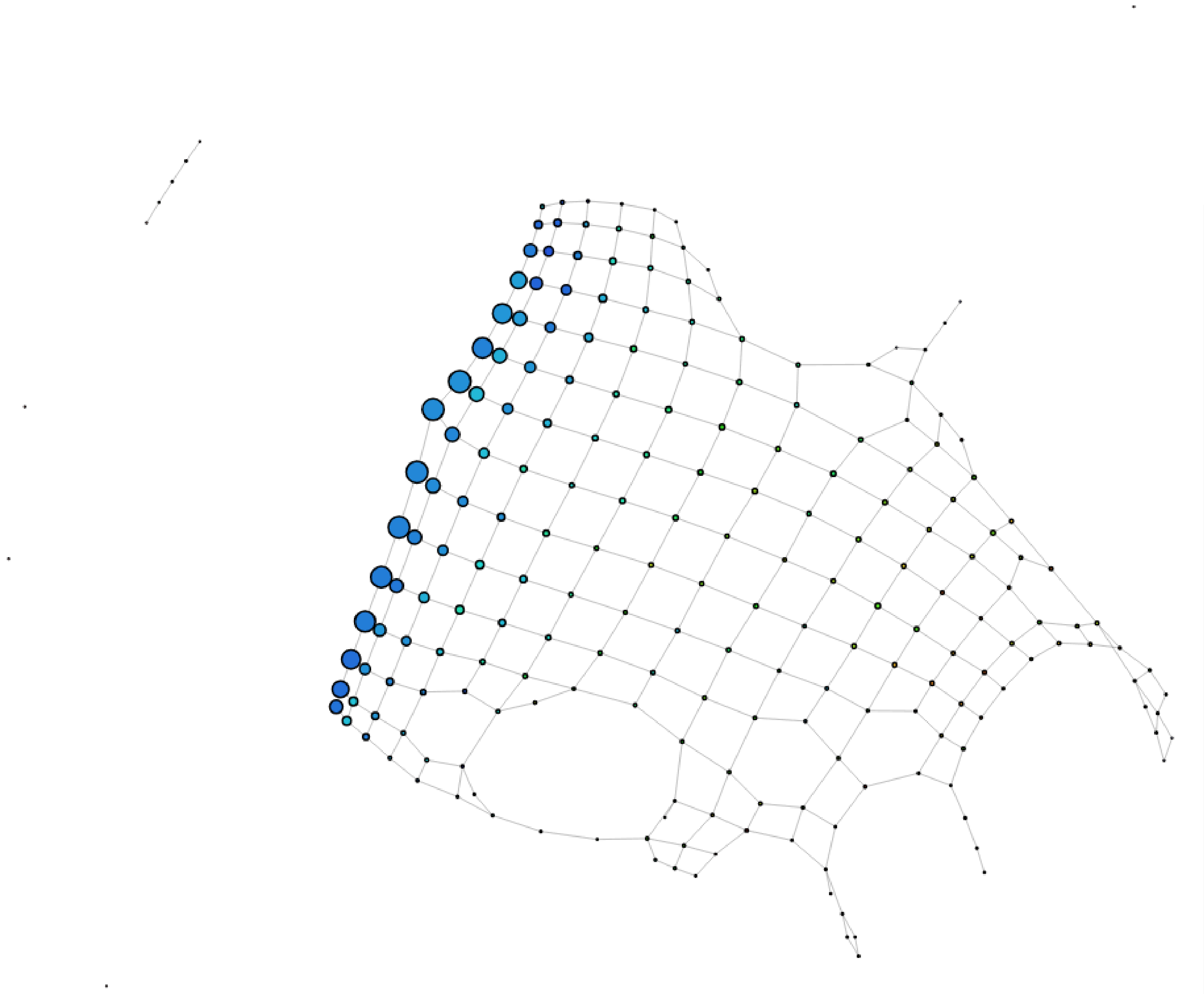
10 Intervals, 30% overlap HACA (single, 10 bins)

Too sparse in inter connections



15 Intervals, 30% overlap, DBSCAN(0.05, 10)

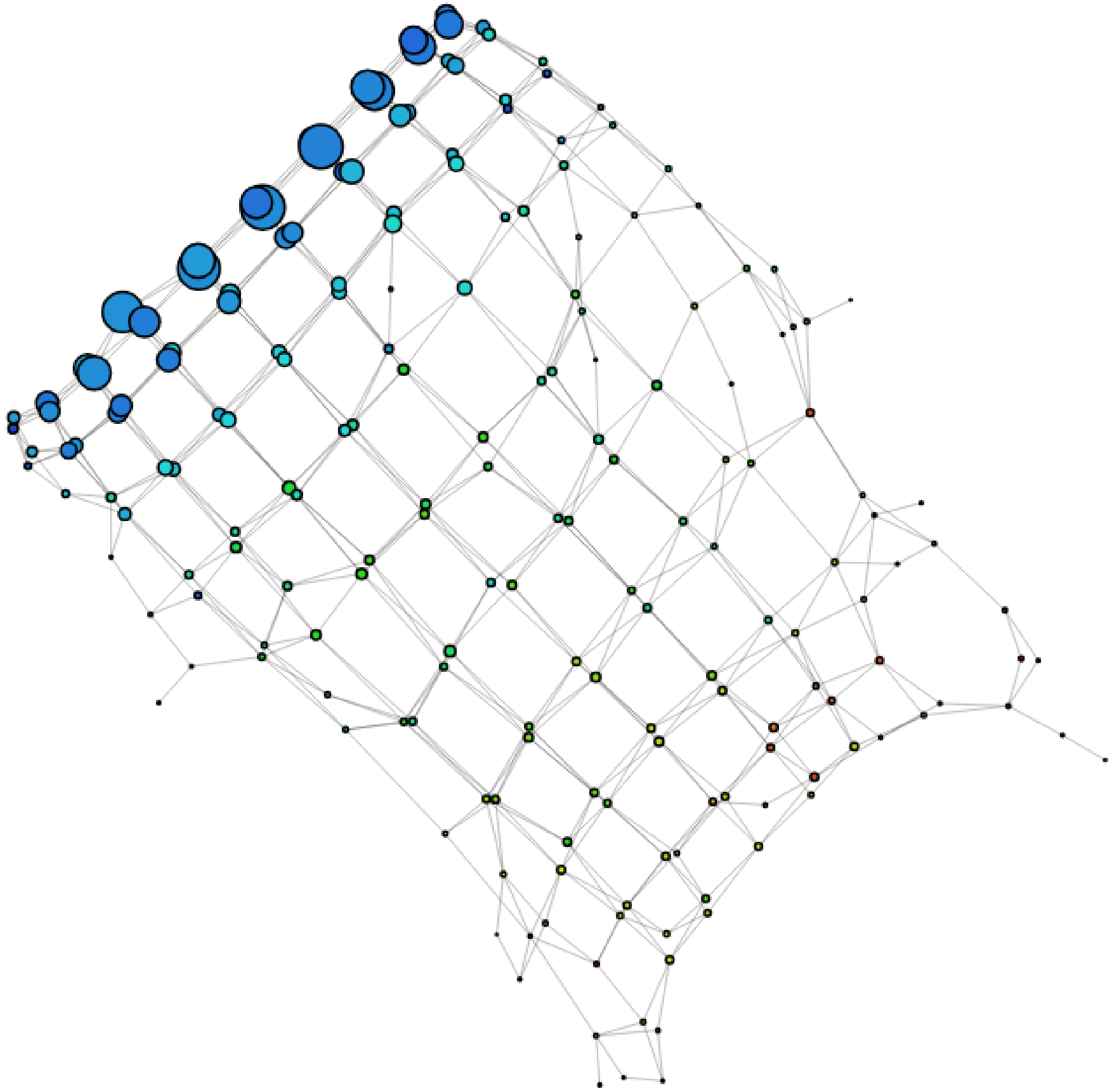
Increasing intervals does not help:



We are merely showing more noise, not insights

10 Intervals, 30% overlap, kmeans(2)

By using XGBoost, which already forces the graph to separate by ADHD, kmeans is useless here (we already have covered data by their likelihood of ADHD)

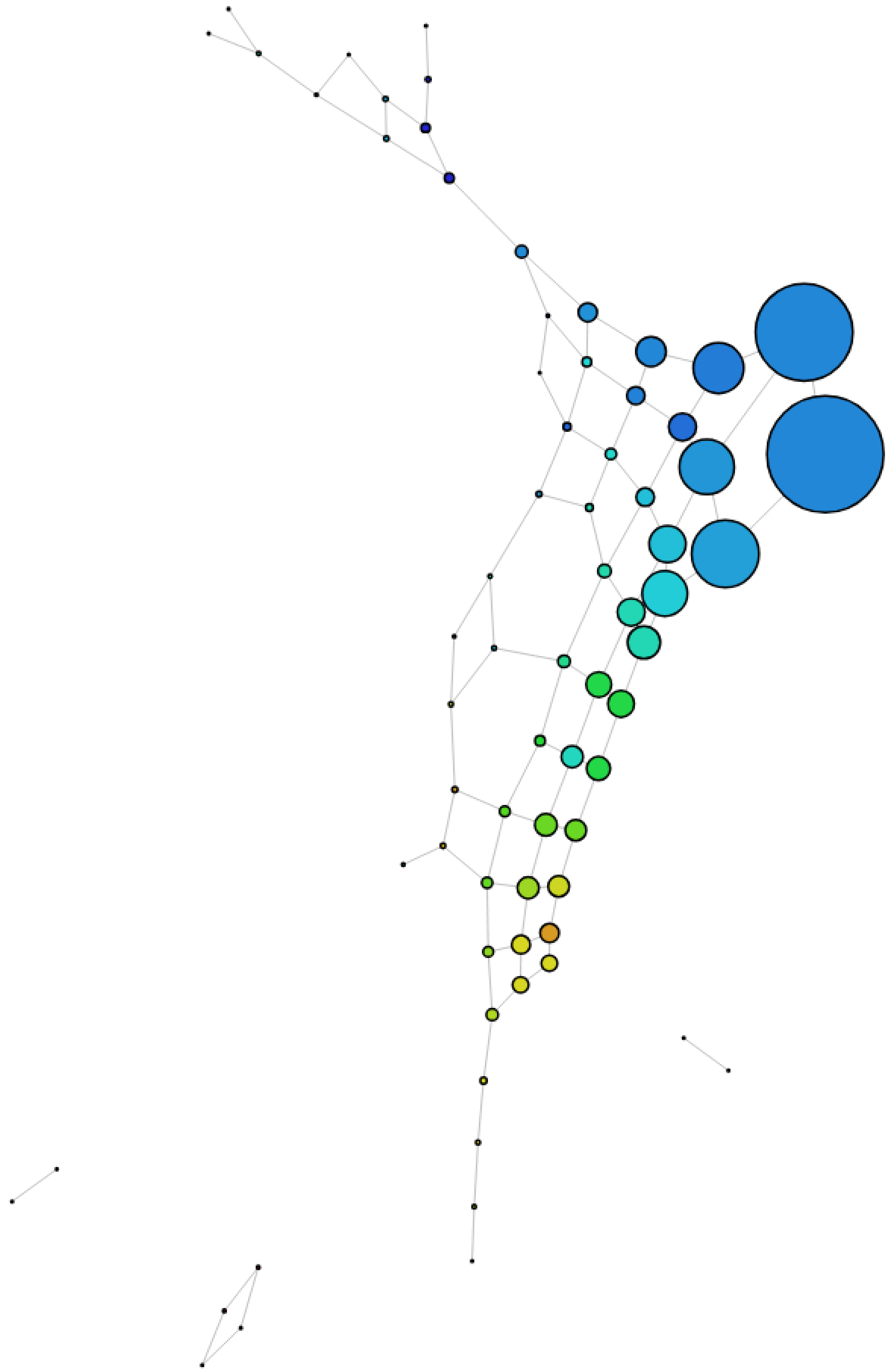


XGBoost and Mean Distance

Rather than using a measure of geometric centrality, using the sum of a row's residuals finds a measure of statistical centrality.

10 Intervals 30% overlap, DBSCAN(0.05, 10)

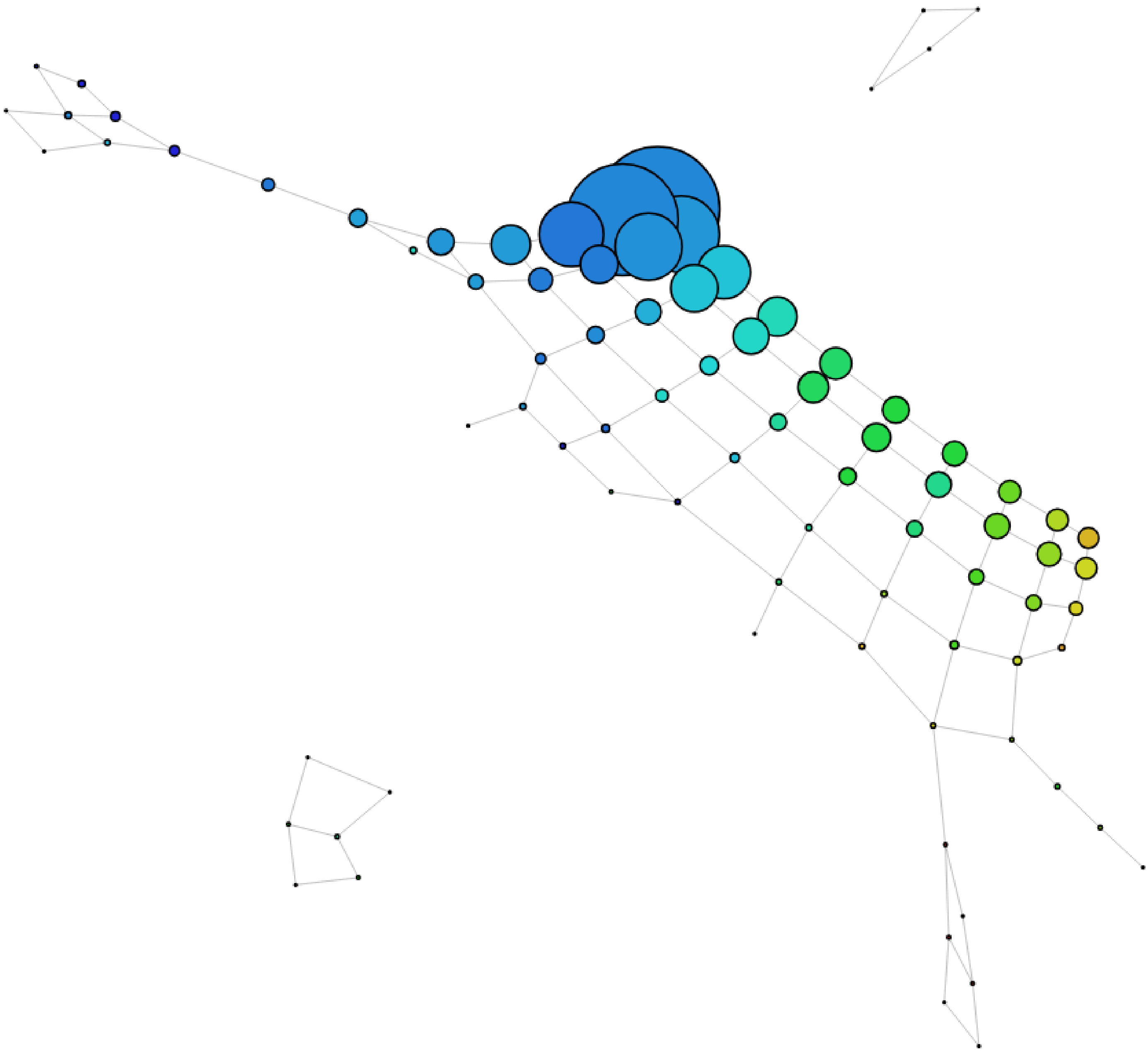
We get a much more interesting structure over eccentricity:



The tails have slightly higher attention scores. The top tail has higher total wave playtime, the very end of the bottom tail doesn't, the very end of the tails have low wellbeing (wemwbs). The top flare is primarily because they have higher binge sessions, although have little proportions of ADHD participants. The isolated components have low total wave playtime

10 Intervals, 30% overlap, DBSCAN(0.05, 5)

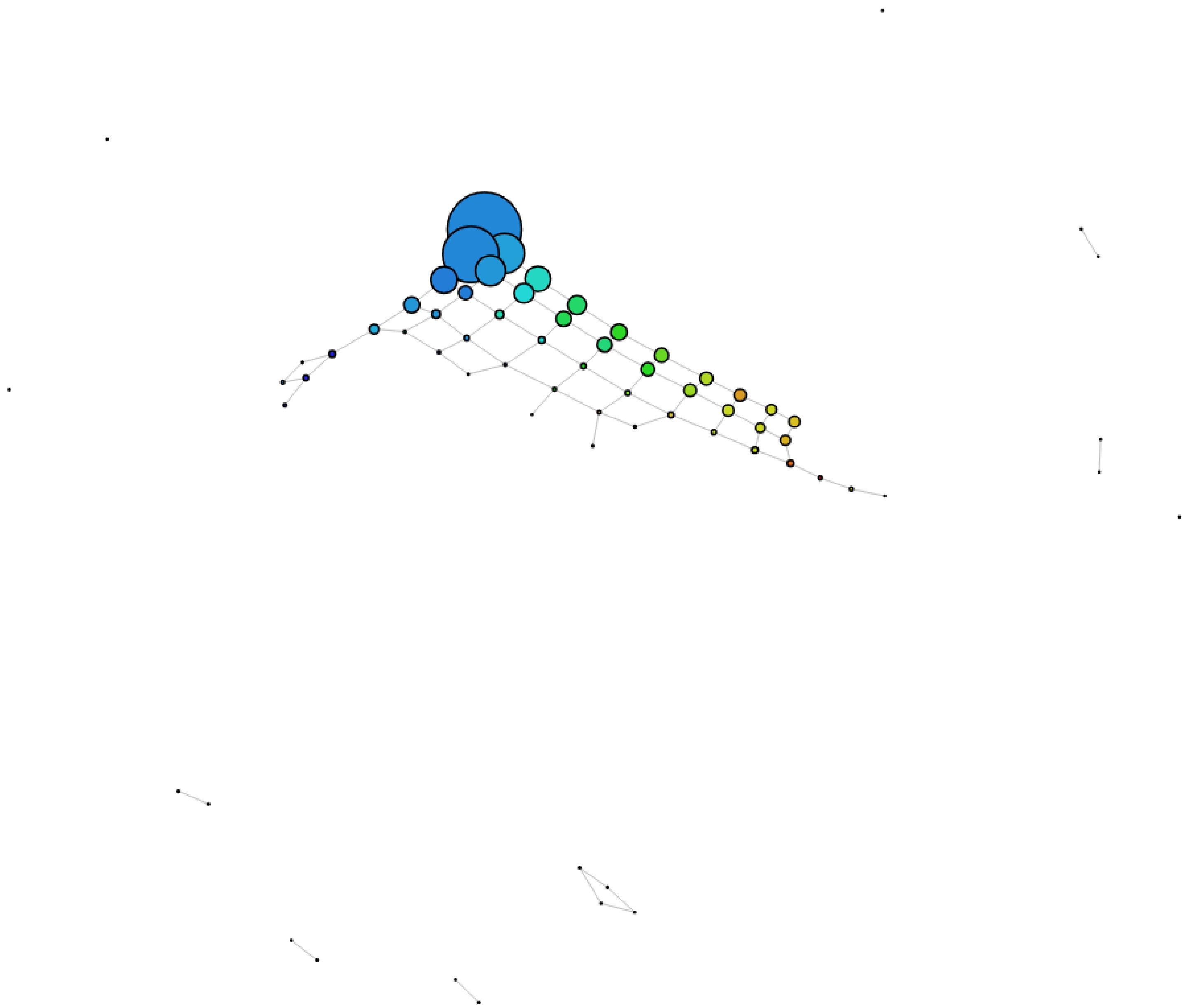
A more relaxed density requirement manages to join the some of the isolated components back:



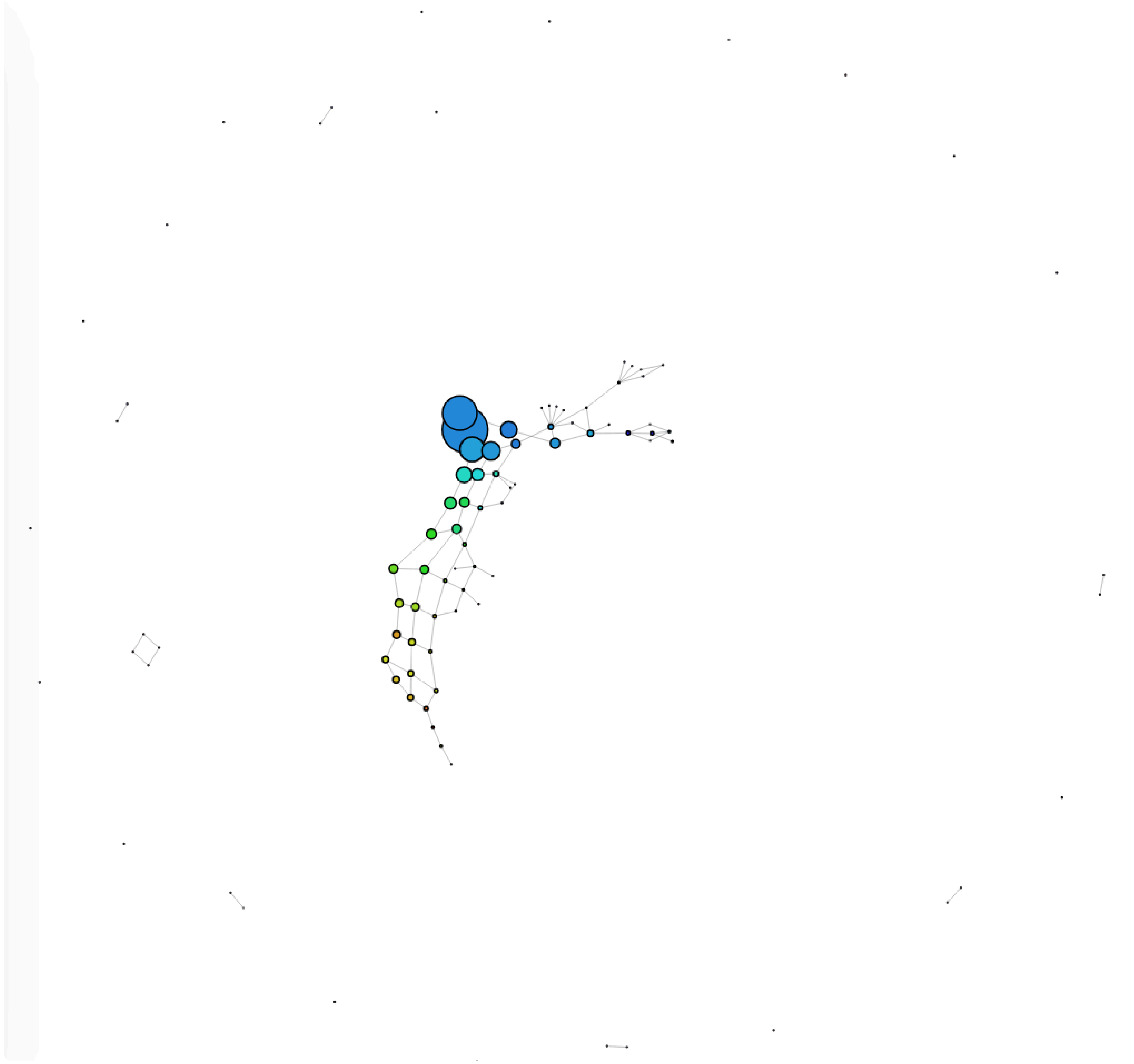
The two flares are still there, but now the top flare is mostly non ADHD participants (with two end nodes however being a self-identified participant). The lower flare is only 3 participants with some level of ADHD, it appears to be a little bit more of a capture of noise. The top flare has longer binge sessions, but the two ADHD nodes don't (slightly lower comp frustration, other factors don't seem to line up). This is a potential issue arising from the XGBoost decision tree with false negatives. They are self-identified ADHD, but the model decided they fit the neurotypical more. With the investigation of the node following the patterns from the adhd participants, I think this is an XGBoost error.

10 Intervals, 20% overlap DBSCAN(0.05, 5)

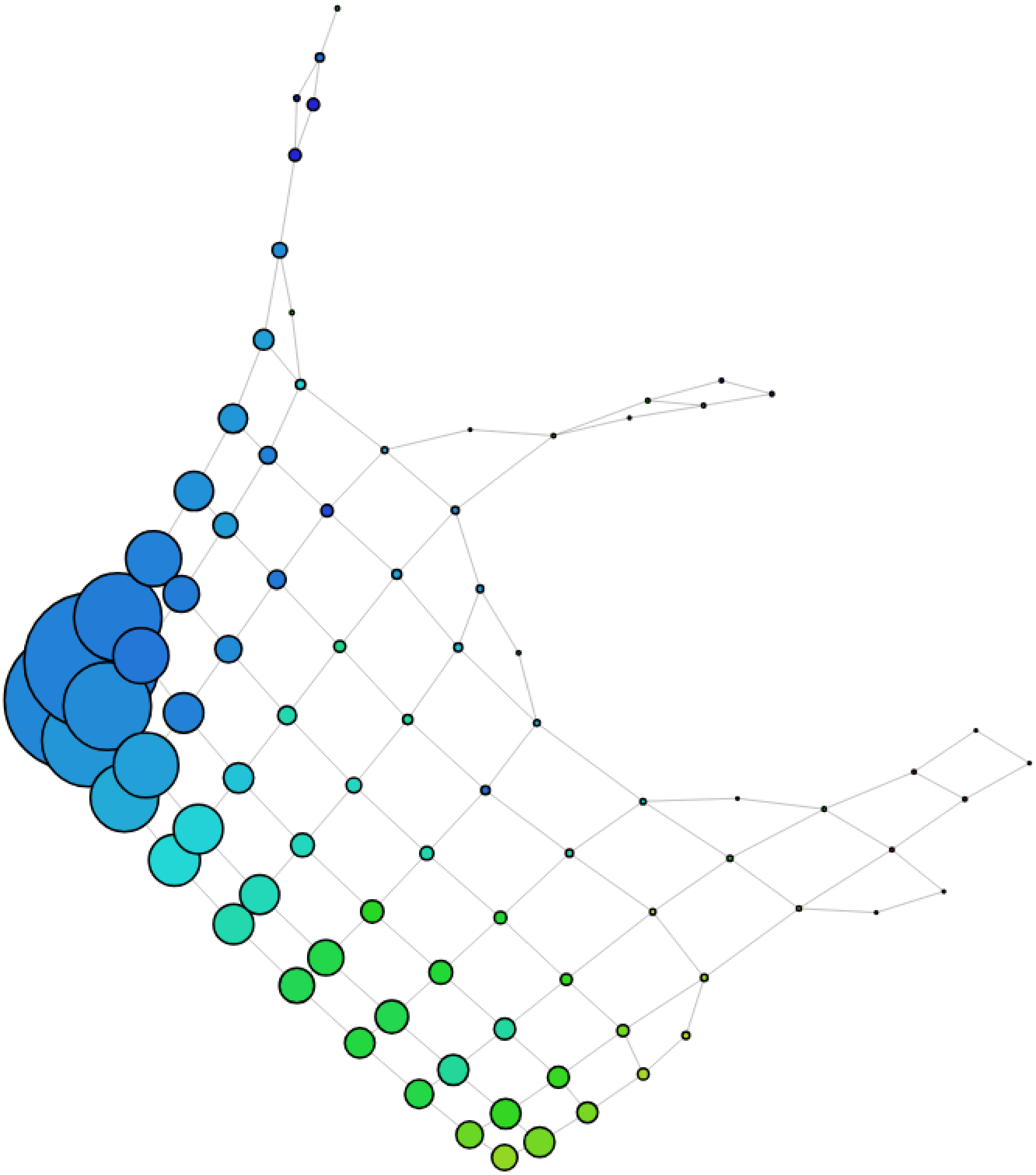
Structure is more generalised, and we have more isolated components:



10 Intervals, 30% overlap HACA(single, 5)



10 Intervals, 50% overlap, DBSCAN(0.05, 5)



The structure moves upward as the total wave playtime increases. The branch/flare in the middle has mixed participants. Low wemwbs counts are located in the middle of the structure

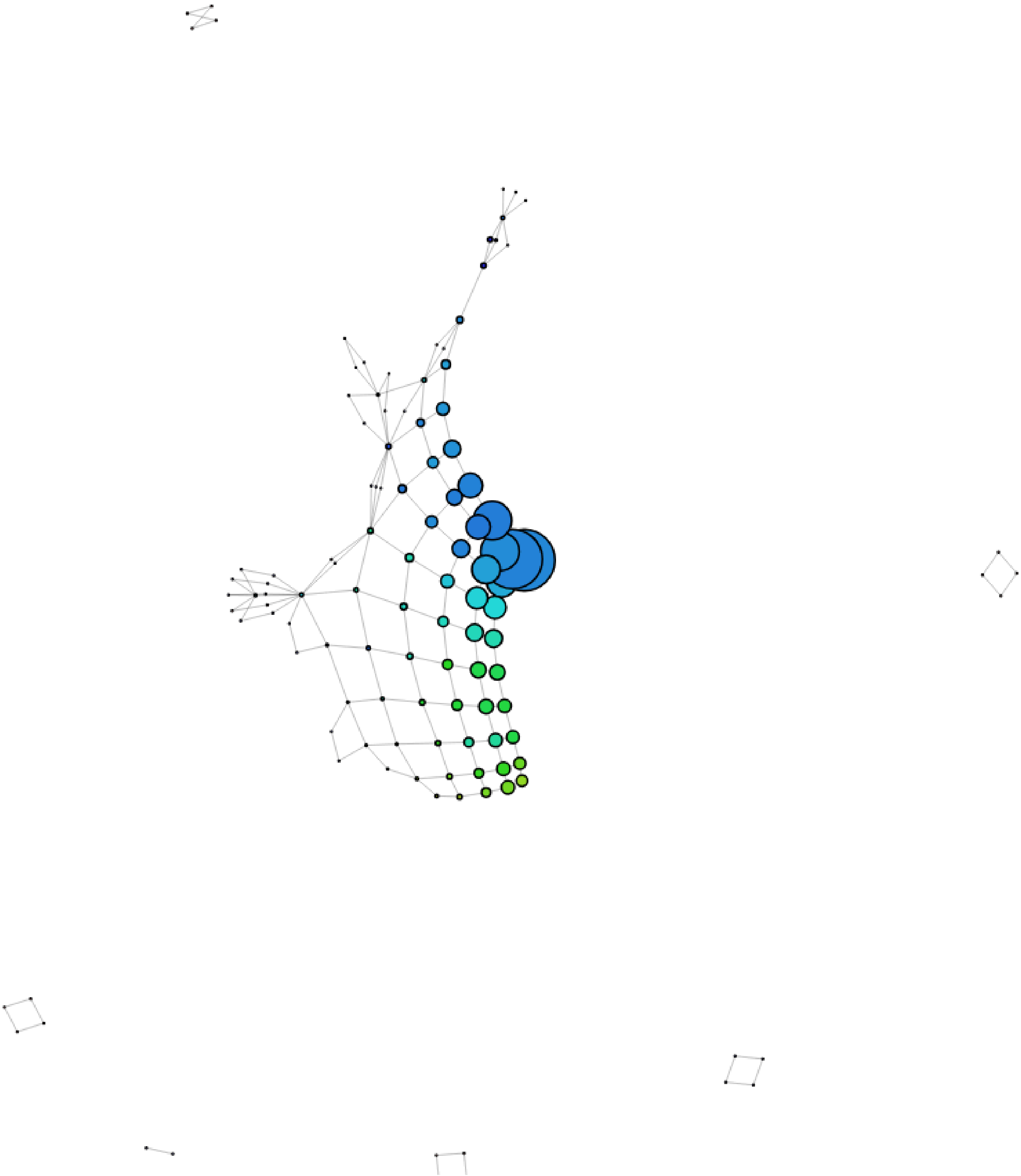
10 Intervals, 50% overlap, DBSCAN(0.05, 10)

No change

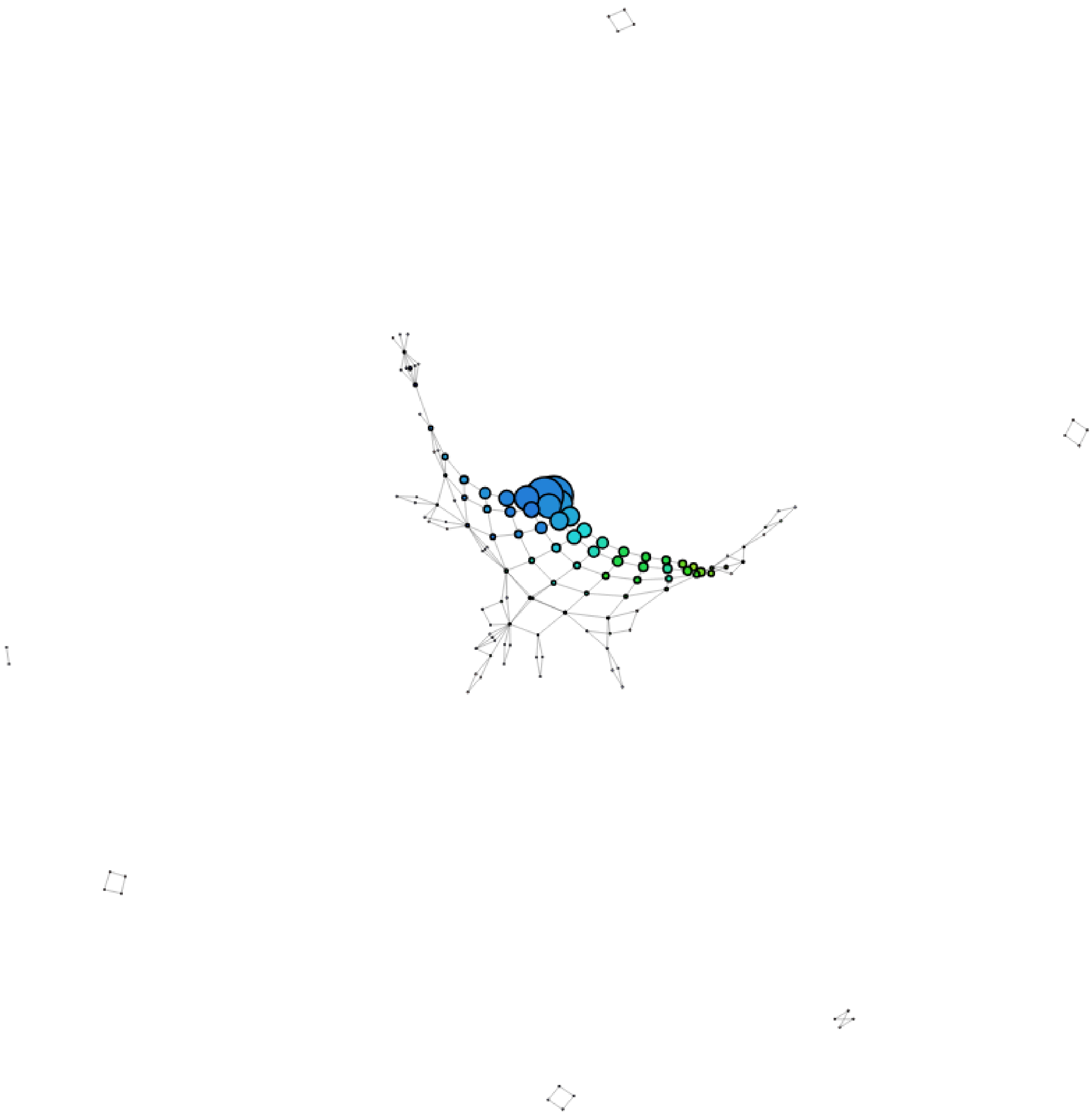
10 Intervals, 50% overlap, DBSCAN(0.1, 5)

No change

10 Intervals, 50% overlap, HACA(single, 5)



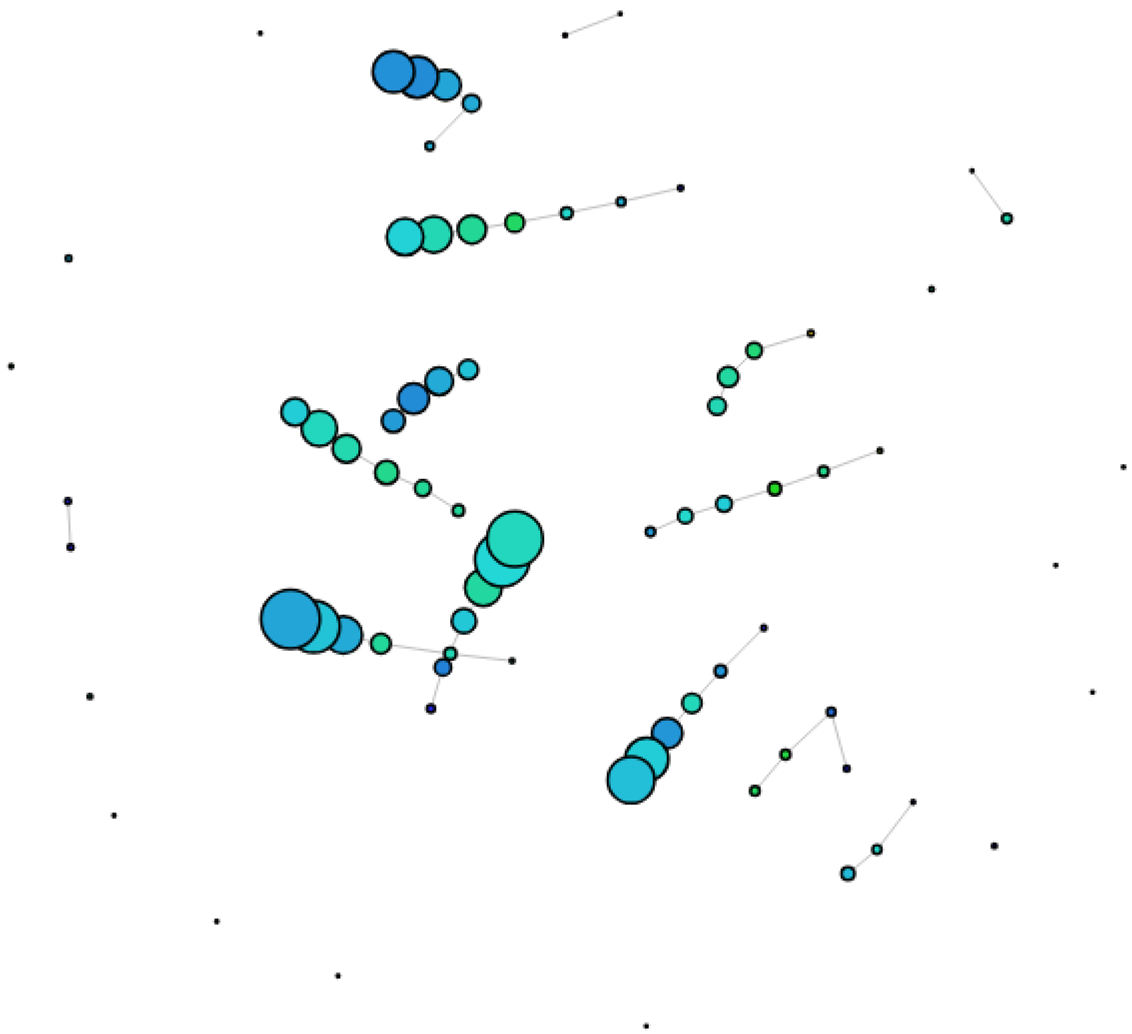
10 Intervals, 50% overlap, HACA(single, 10)



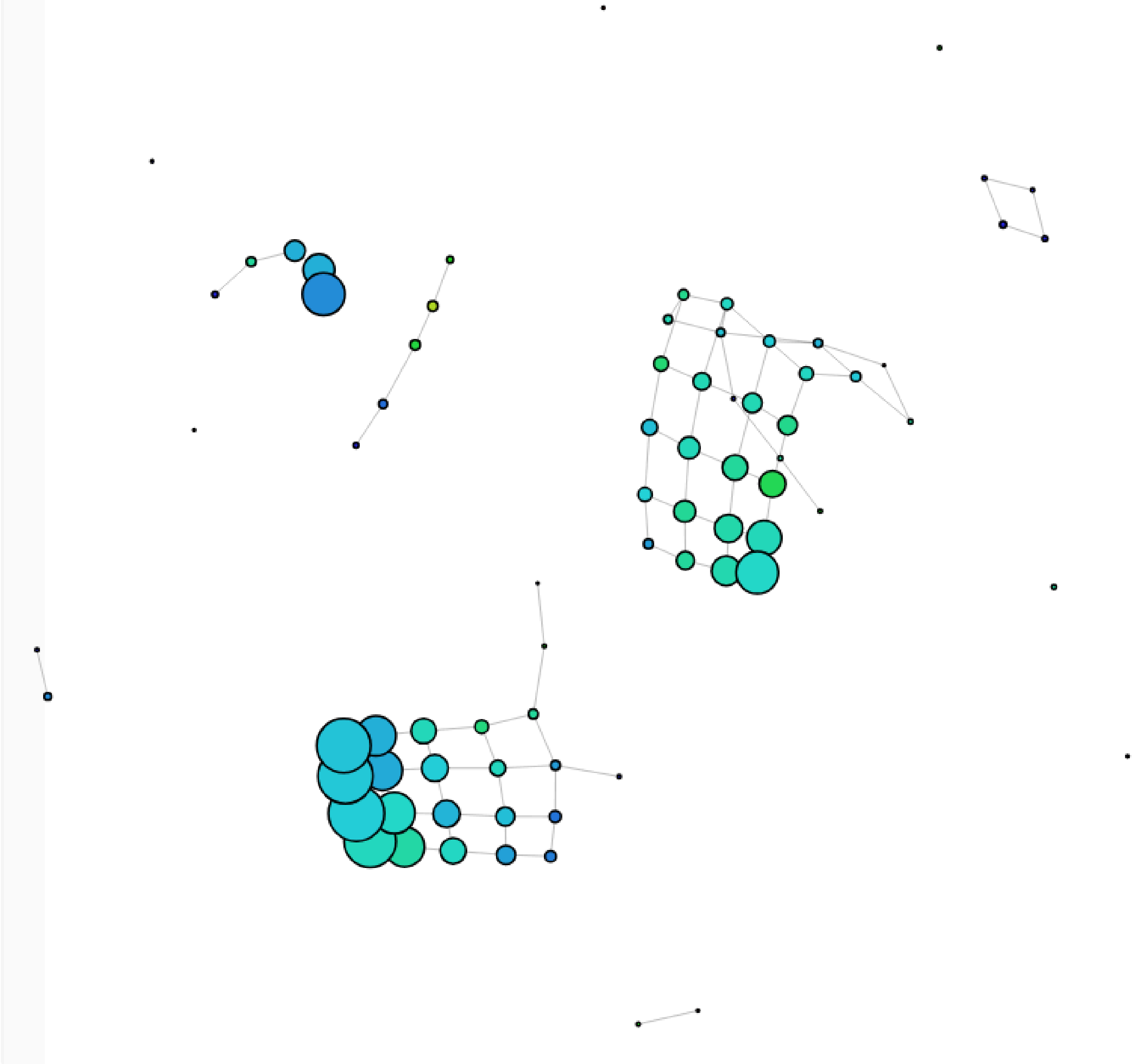
Density Estimation and the WEMWBS Total

This is to explore the wellbeing of the participants.

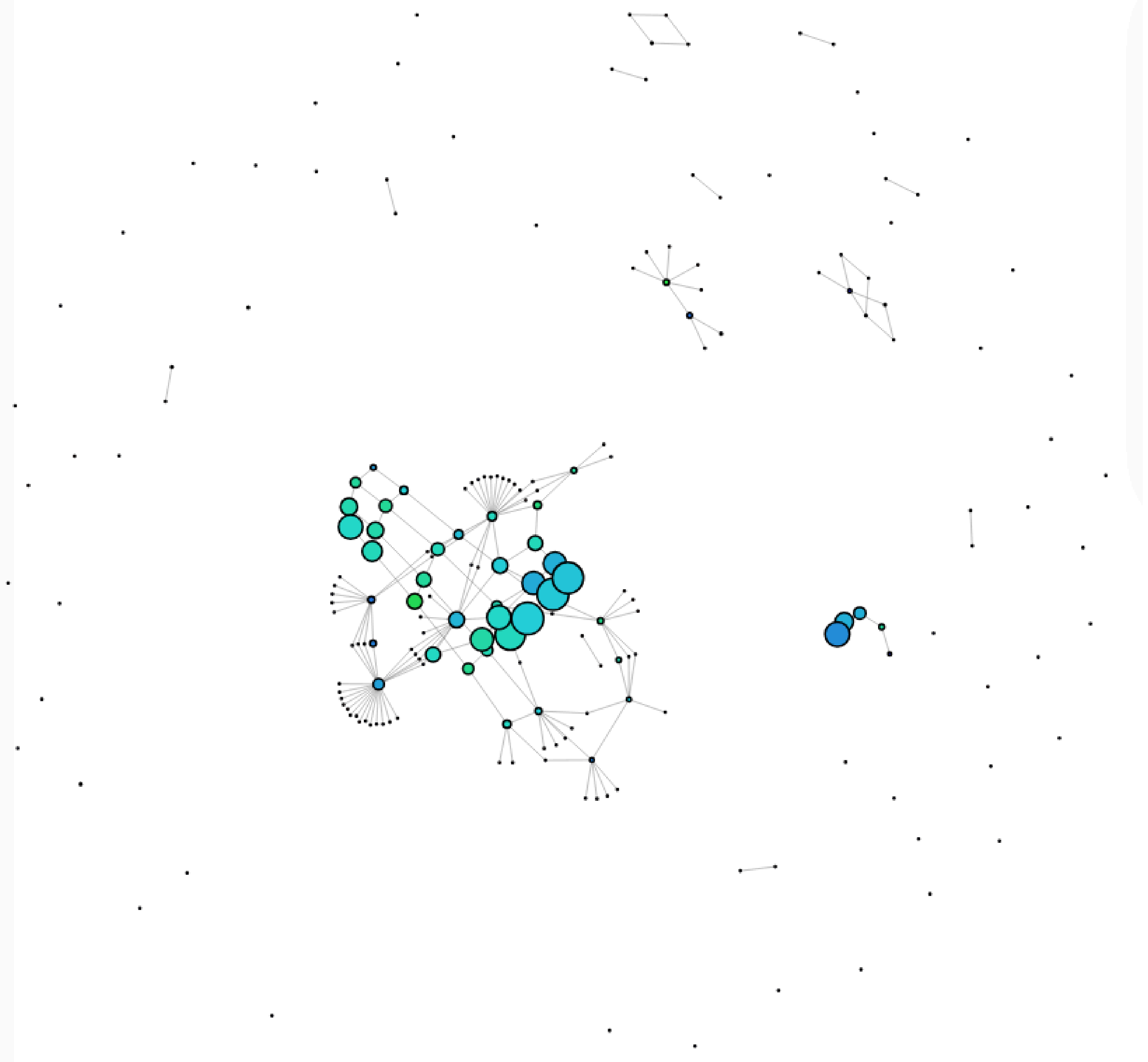
10 Intervals, 10% overlap, DBSCAN(0.05, 10)



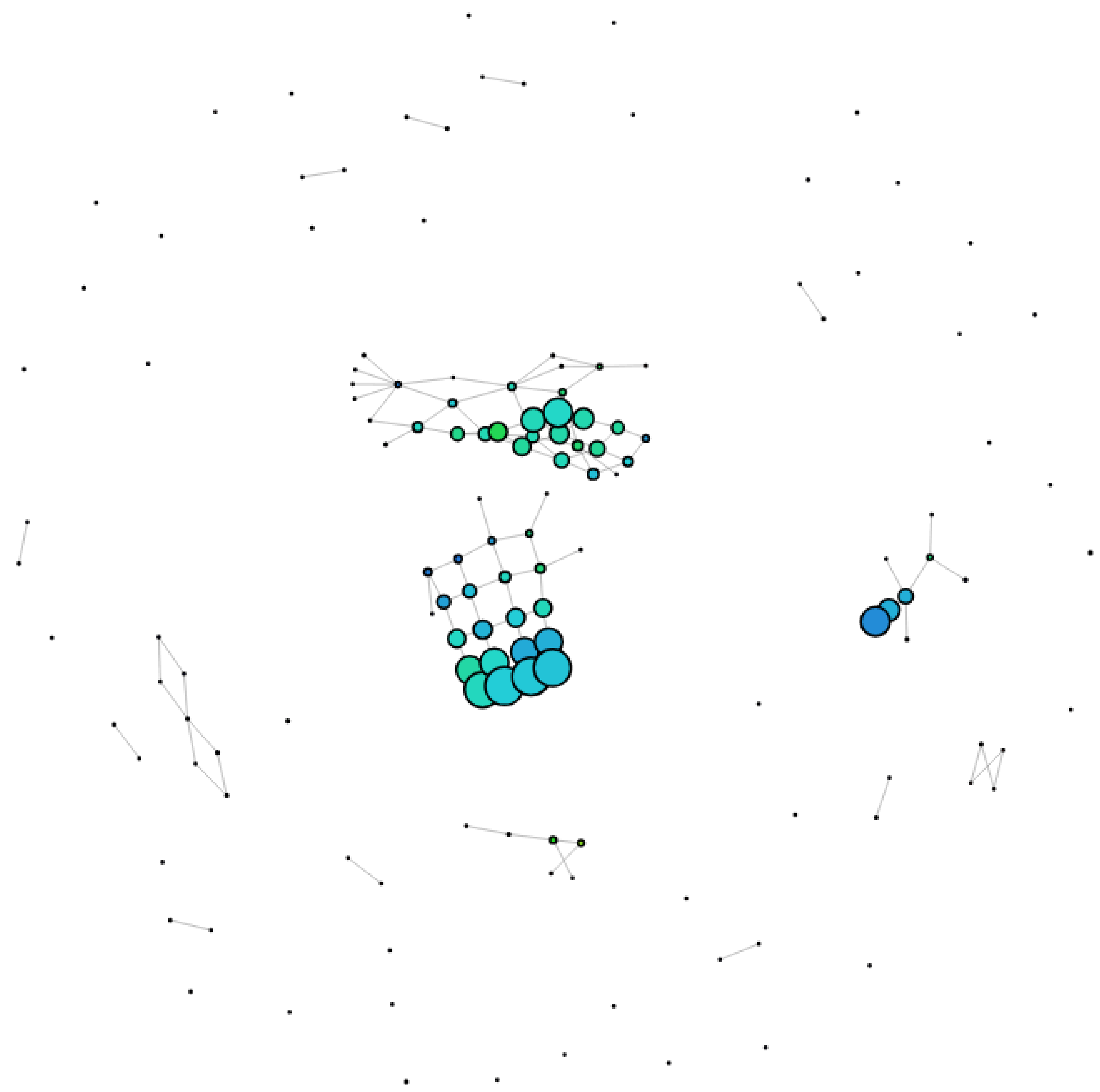
10 Intervals, 20% overlap, DBSCAN(0.05, 10)



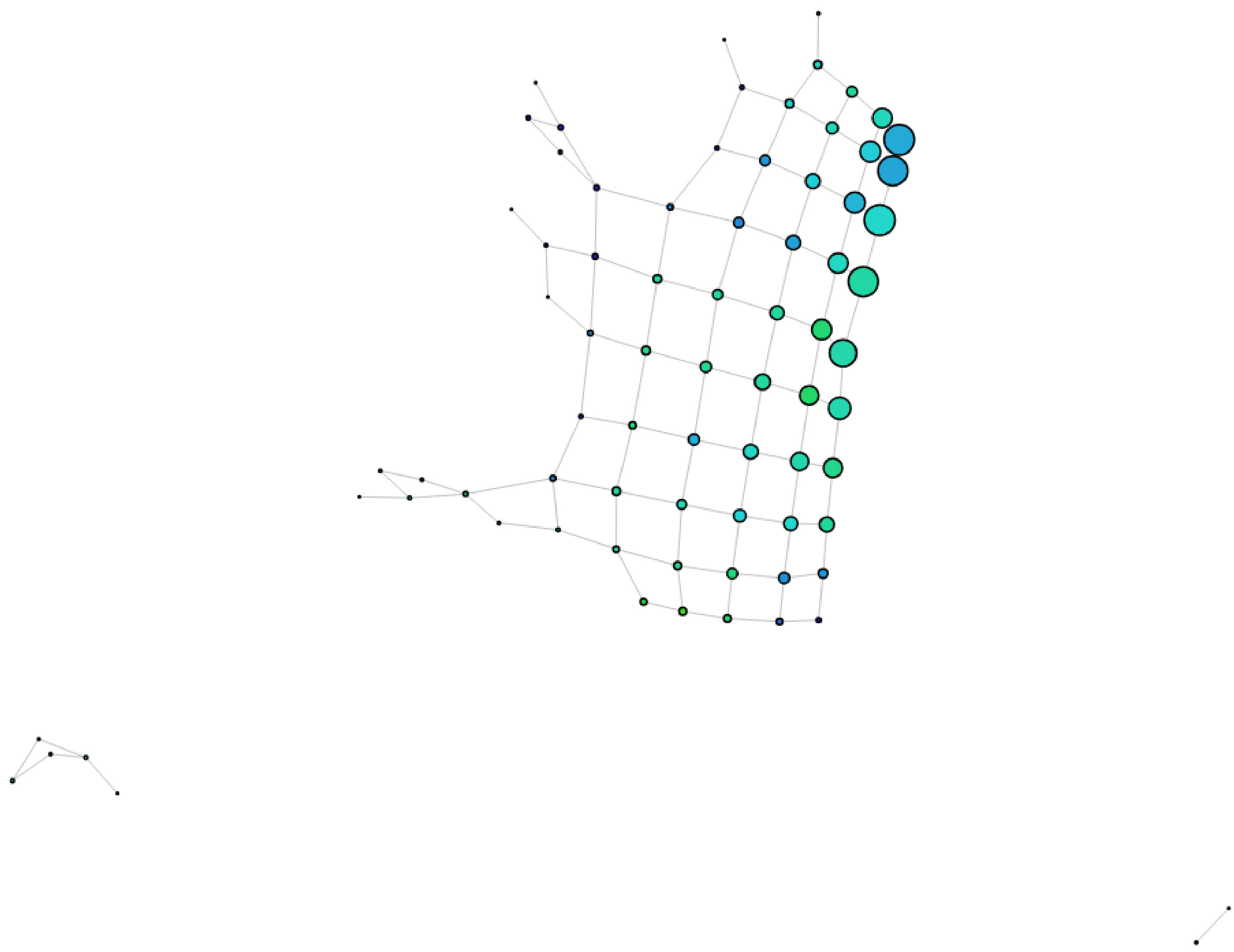
10 Intervals, 20% overlap, HACA(single, 5)



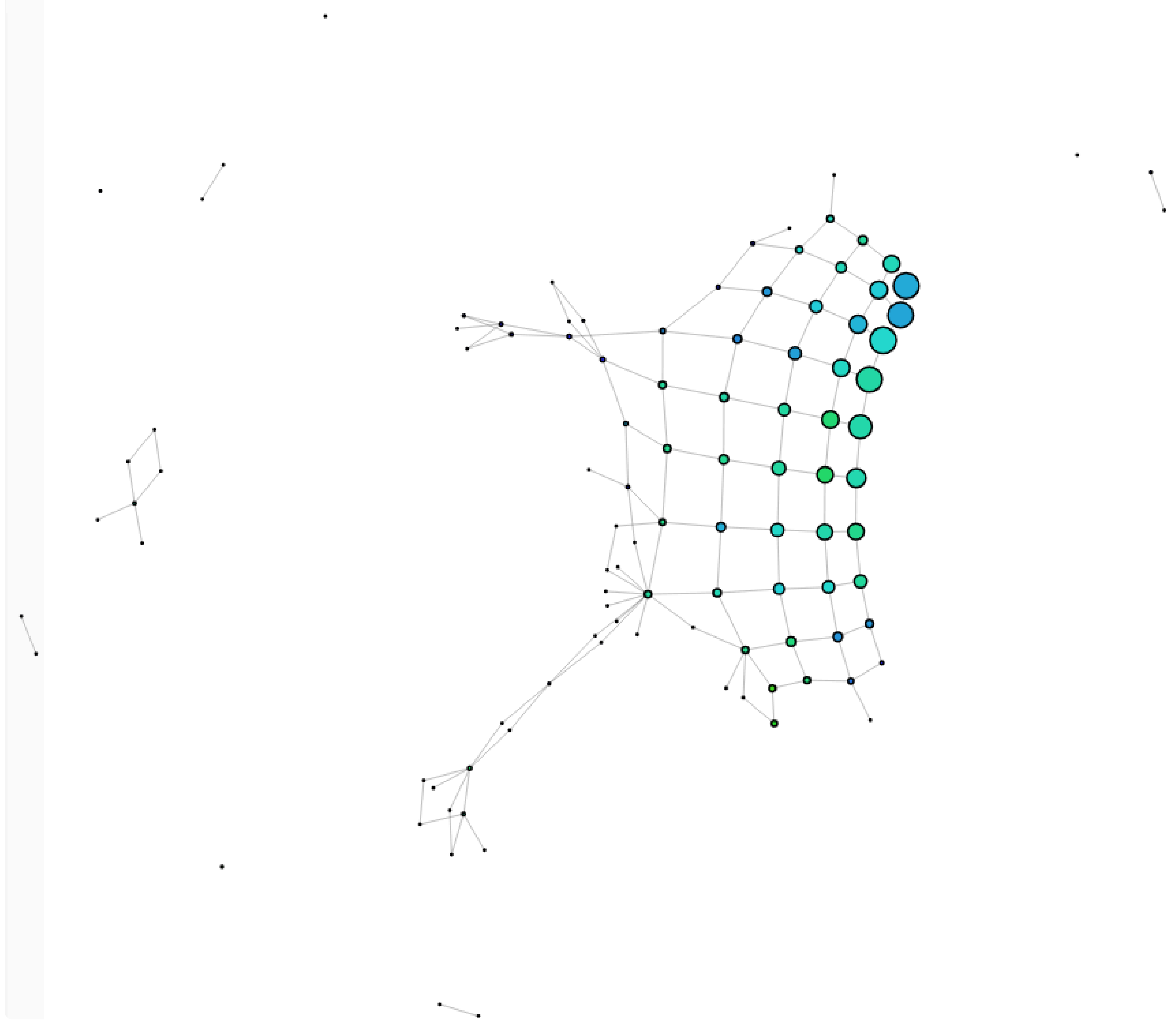
10 Intervals, 20% overlap, HACA(single, 10)



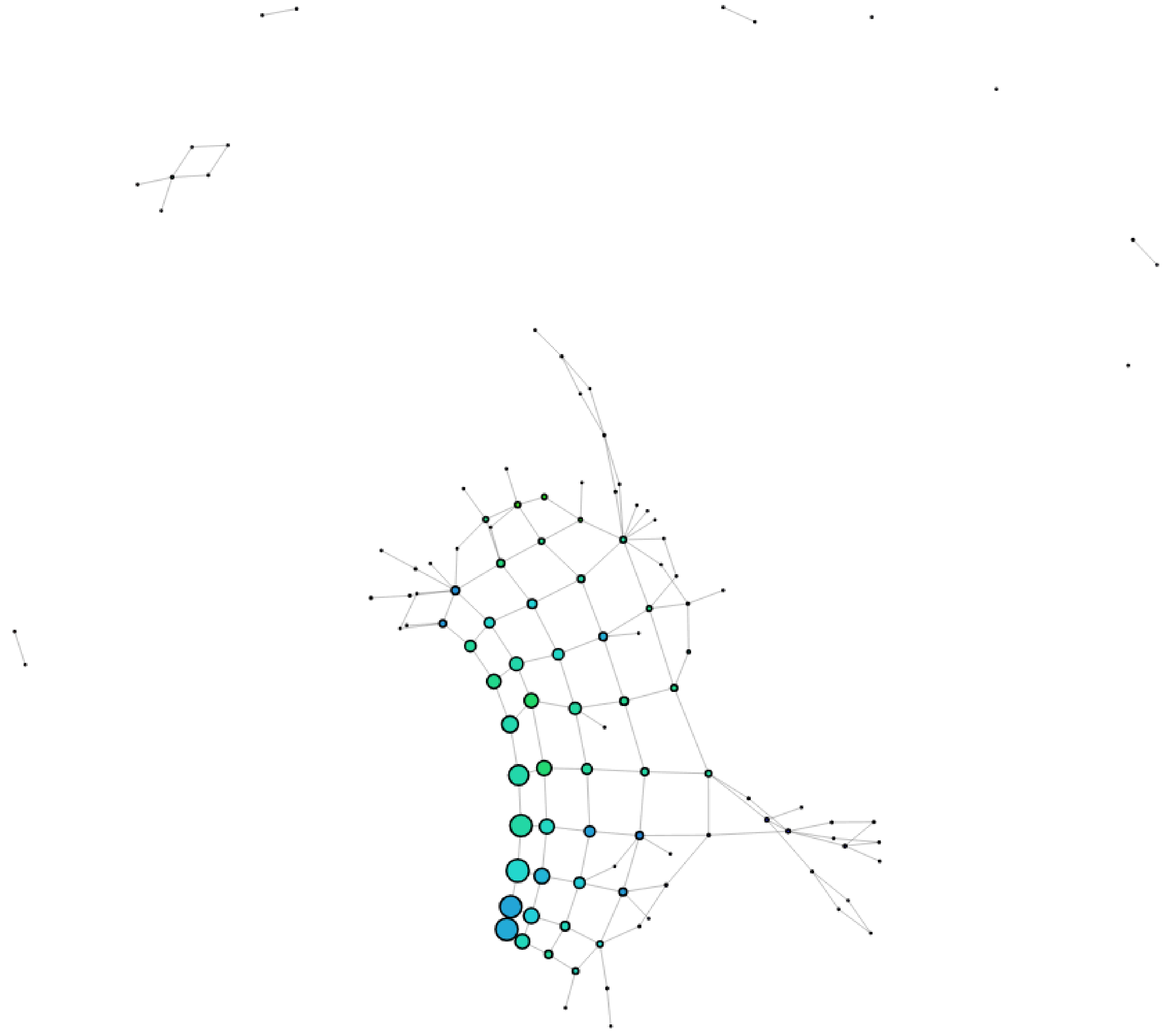
10 Intervals, 30% overlap, DBSCAN(0.05, 10)



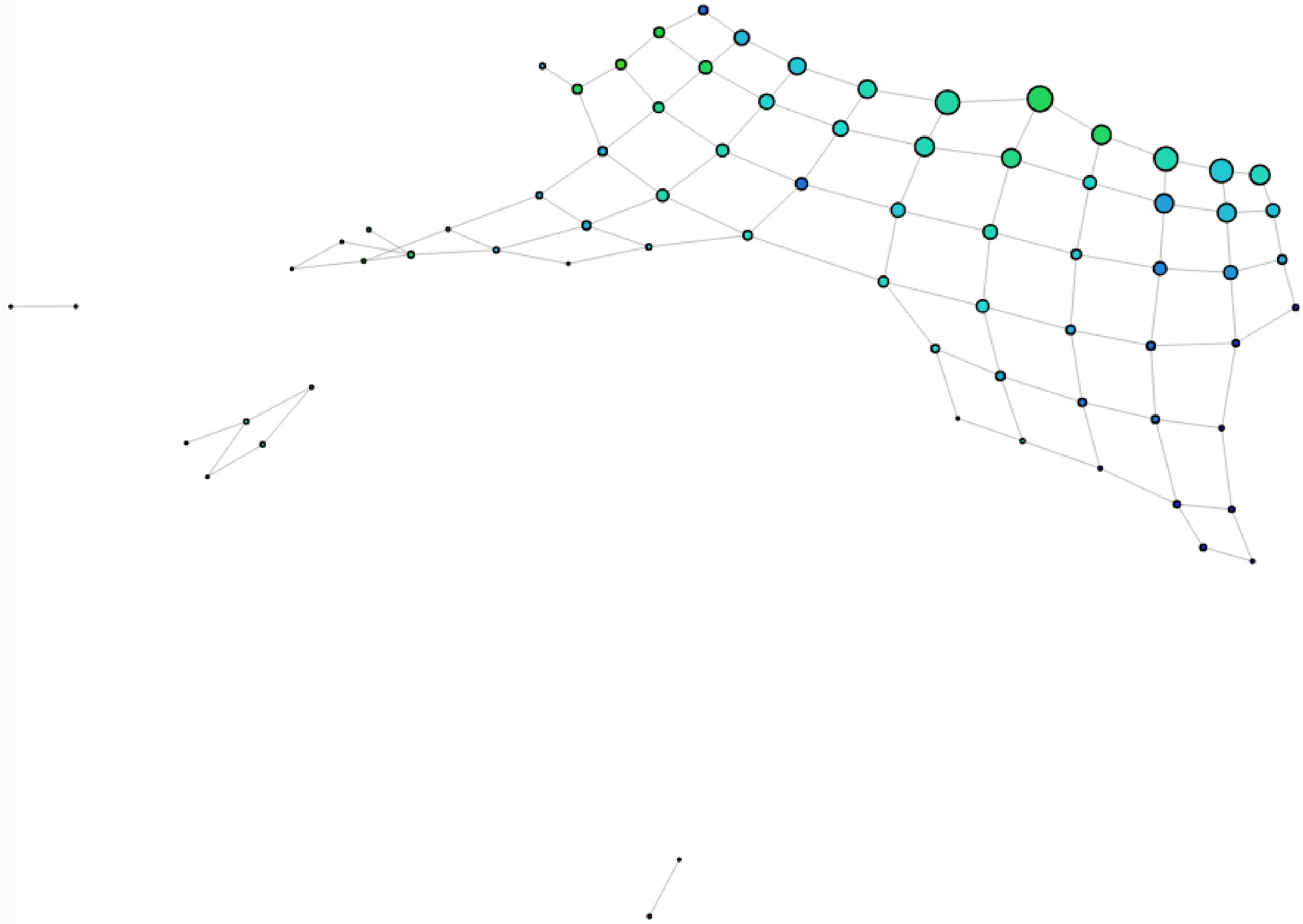
10 Intervals, 30% overlap, HACA(single, 5)



10 Intervals, 30% overlap, HACA(single, 10)



10 Intervals, 40% overlap, DBSCAN(0.05, 10)

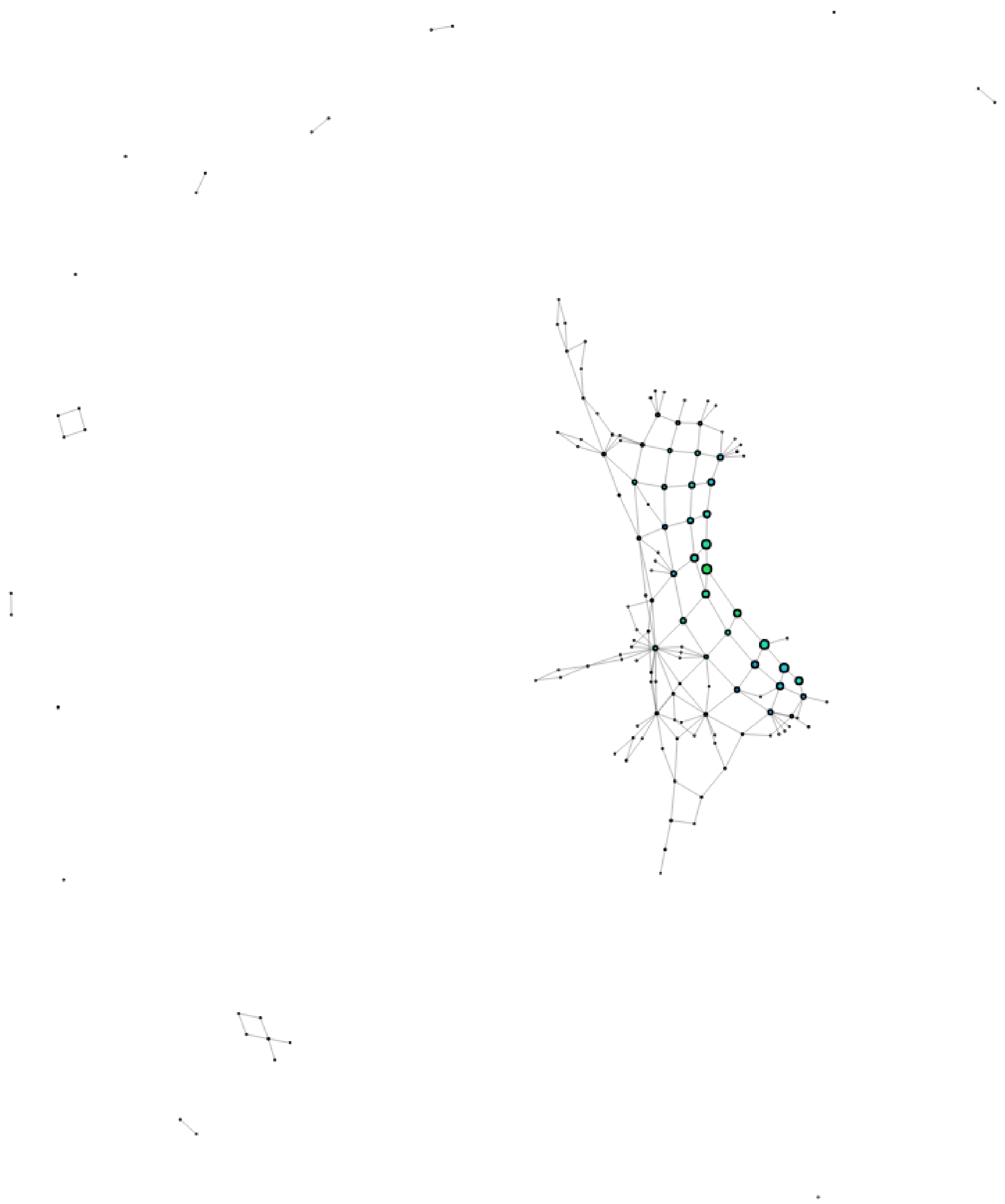


10 Intervals, 40% overlap, DBSCAN(0.05, 5)

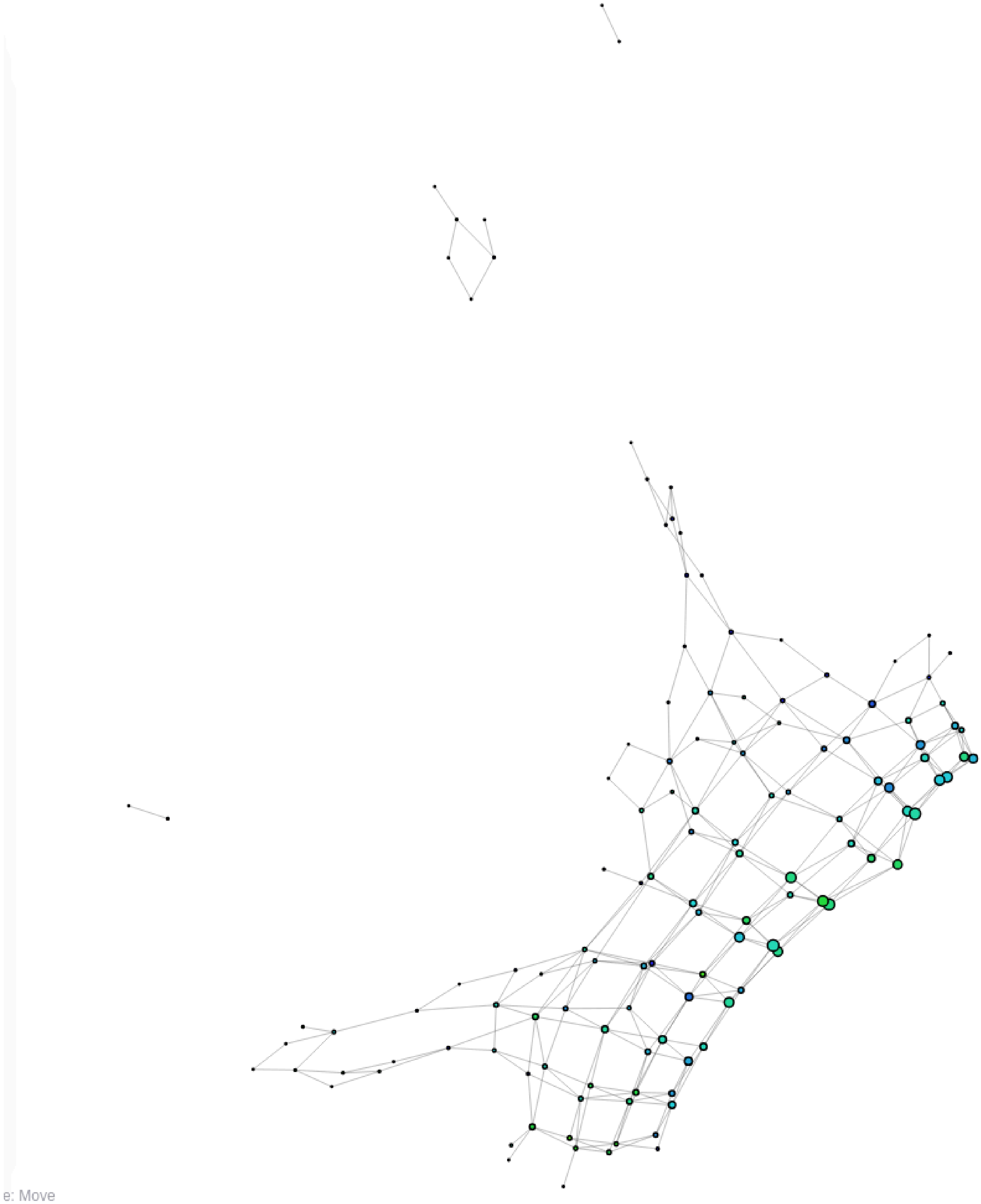
No change

10 Intervals, 40% overlap, HACA(single, 5)

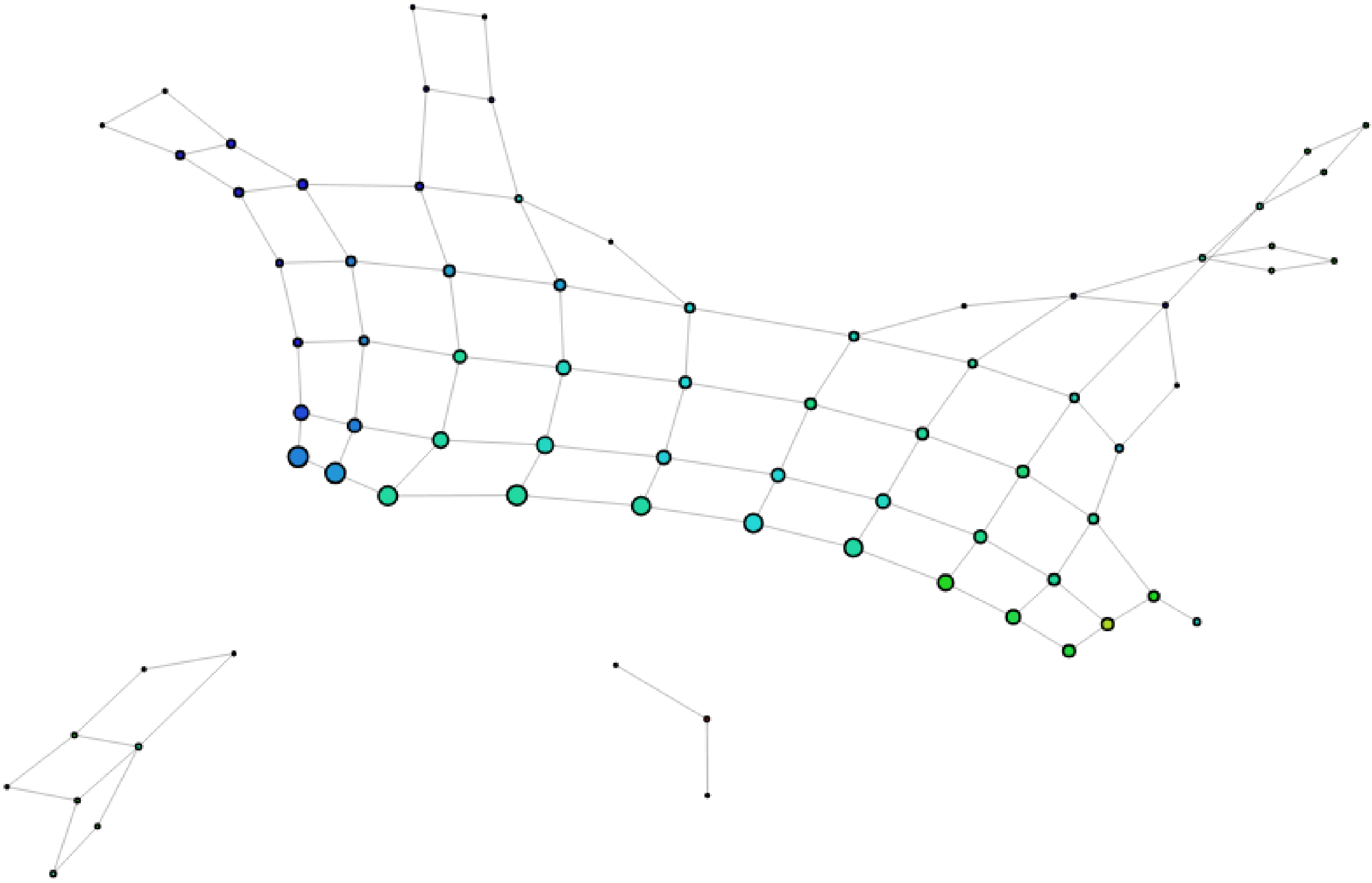
10 Intervals, 40% overlap, HACA(single, 10)



10 Intervals, 40% overlap, kmeans(2)

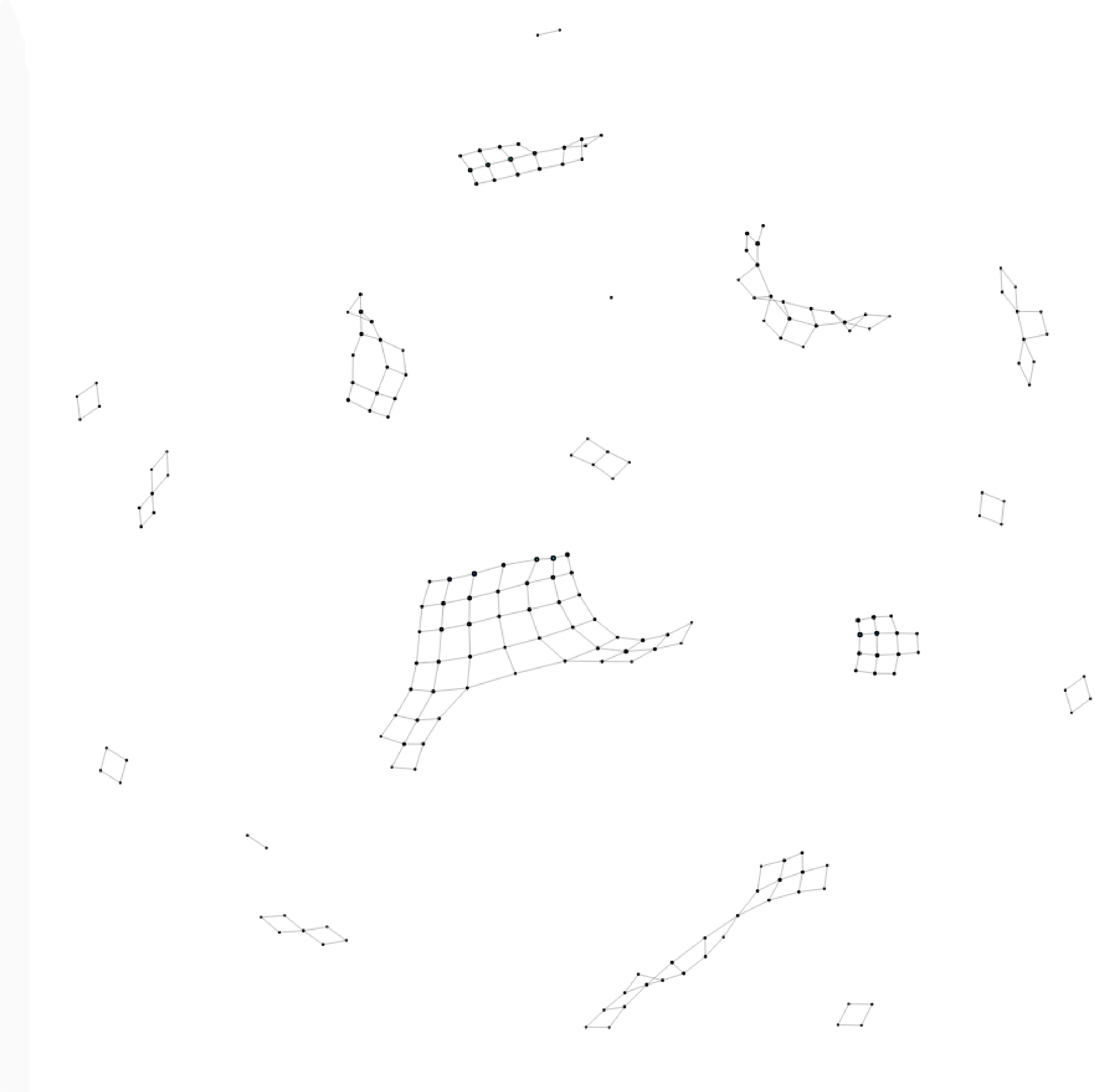


10 Intervals, 50% overlap, DBSCAN(0.05, 10)



10 Intervals, 50% overlap, DBSCAN(0.05, 5)

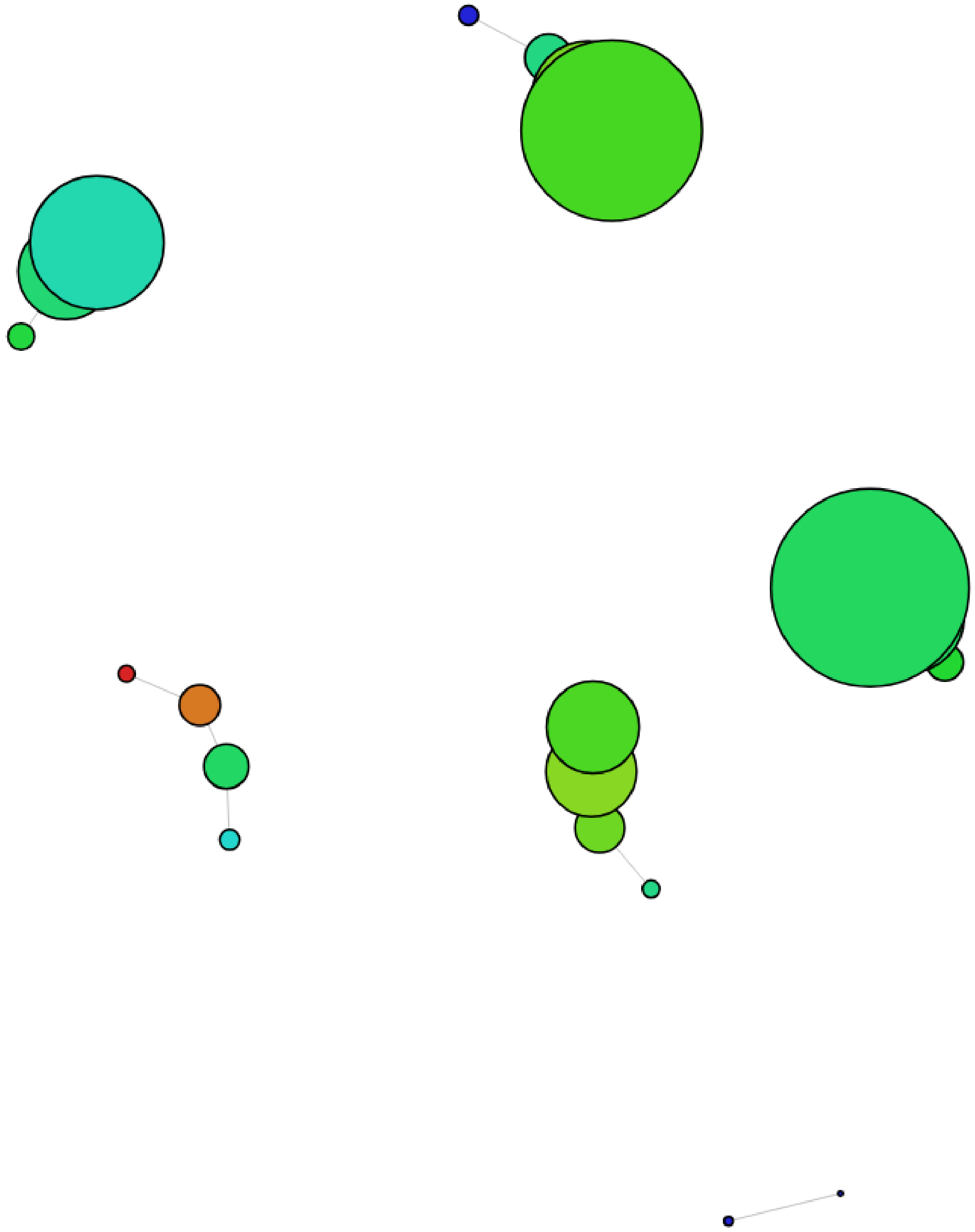
Wow! Lots of isolated components, even with an increased overlap!



No clear signs of ADHD isolated components, but lots without any. The components are mainly split by wembs total (understandable) and some by waves. Maybe the number of intervals are affecting it. I will try to reduce the intervals but keep the overlap.

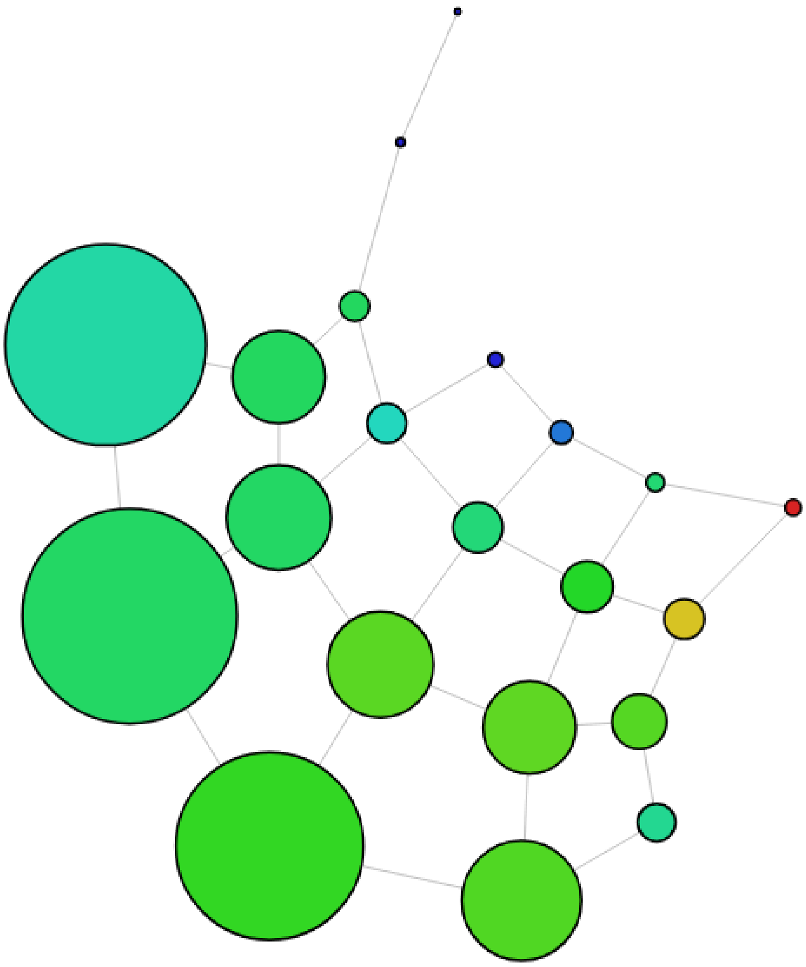
5 Intervals, 10% overlap, DBSCAN(0.05, 10)

Too vague

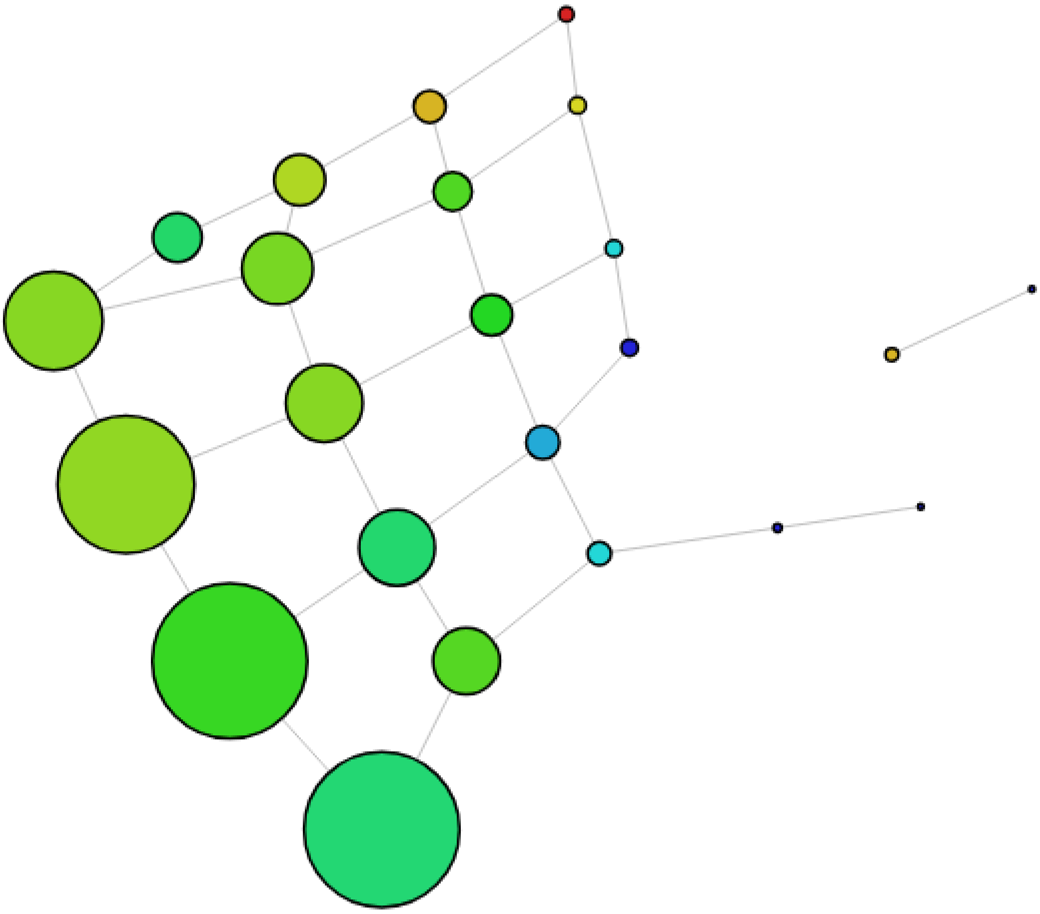


A lot of ADHD participants in the lower section.

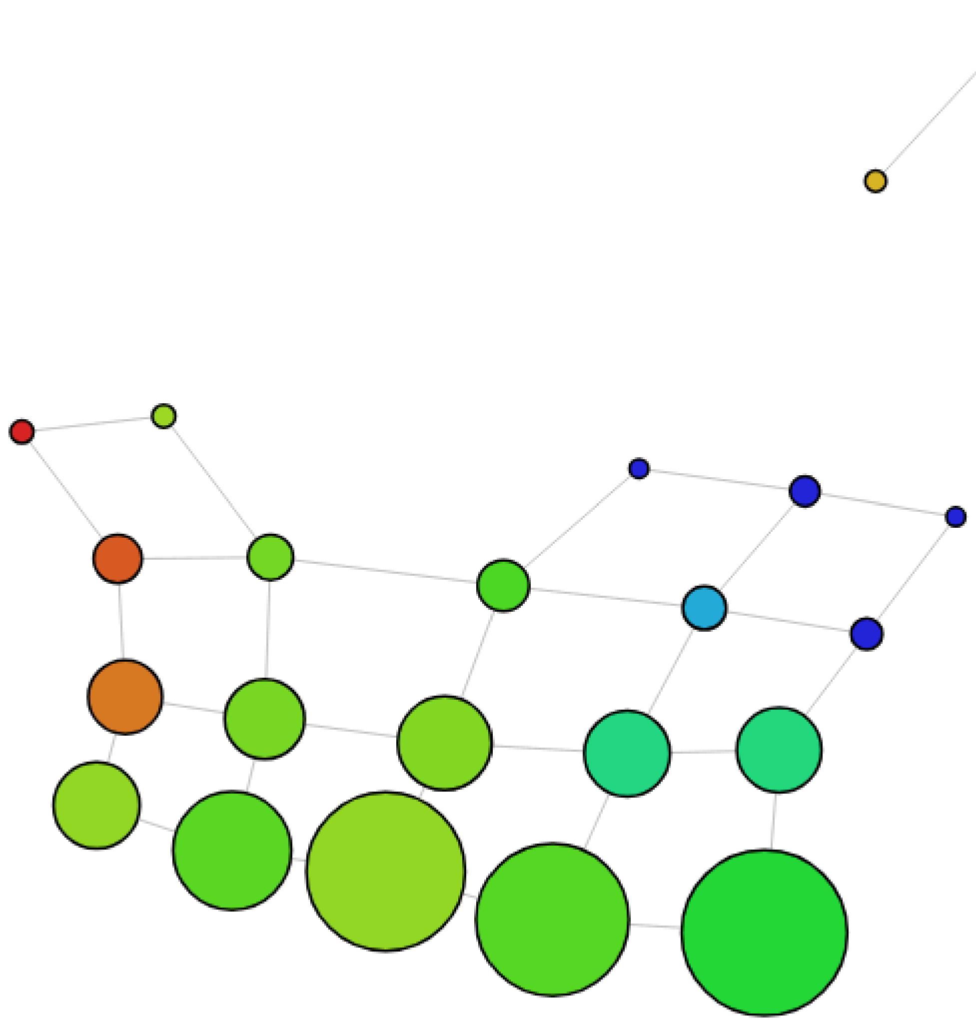
5 Intervals, 20% overlap, DBSCAN(0.05, 10)



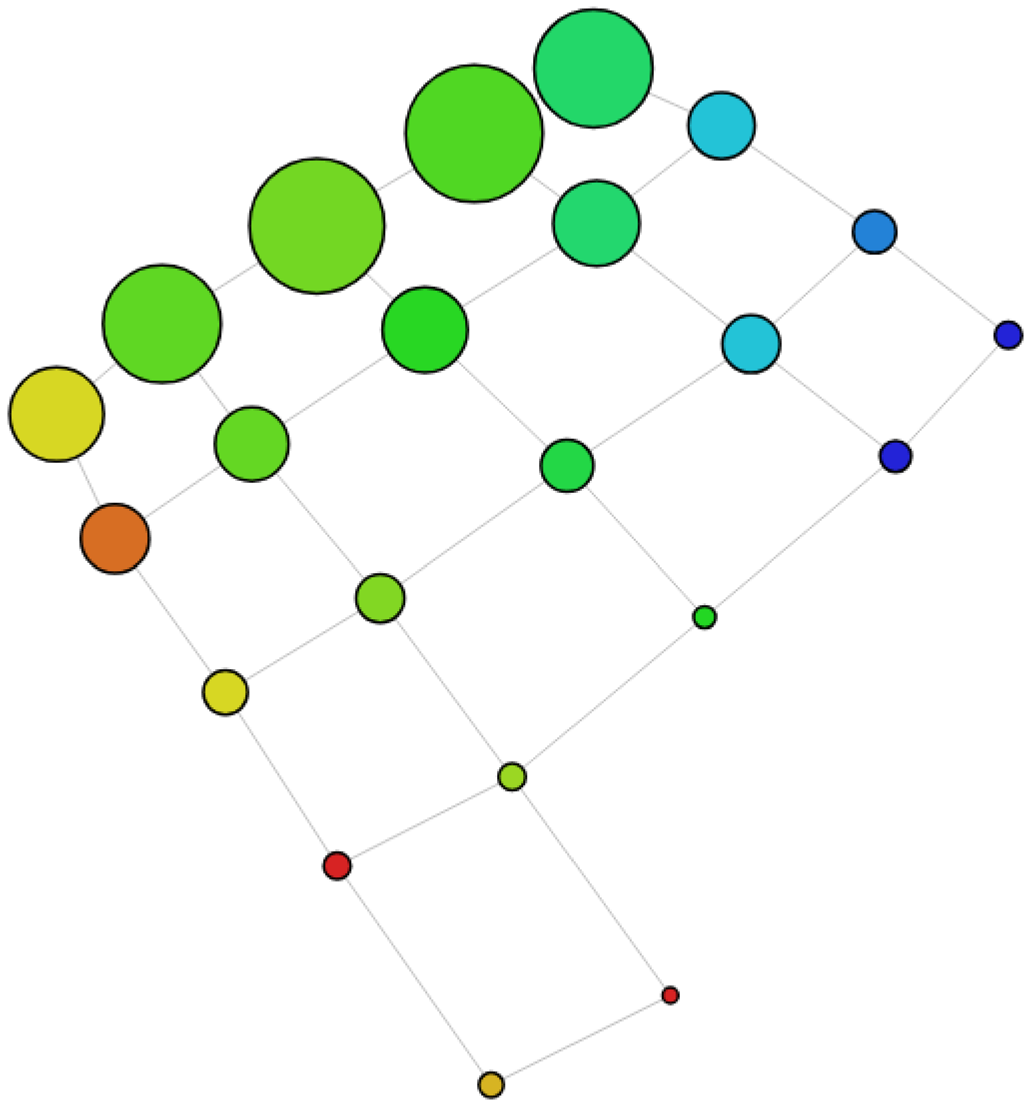
5 Intervals, 30% overlap, DBSCAN(0.05, 10)



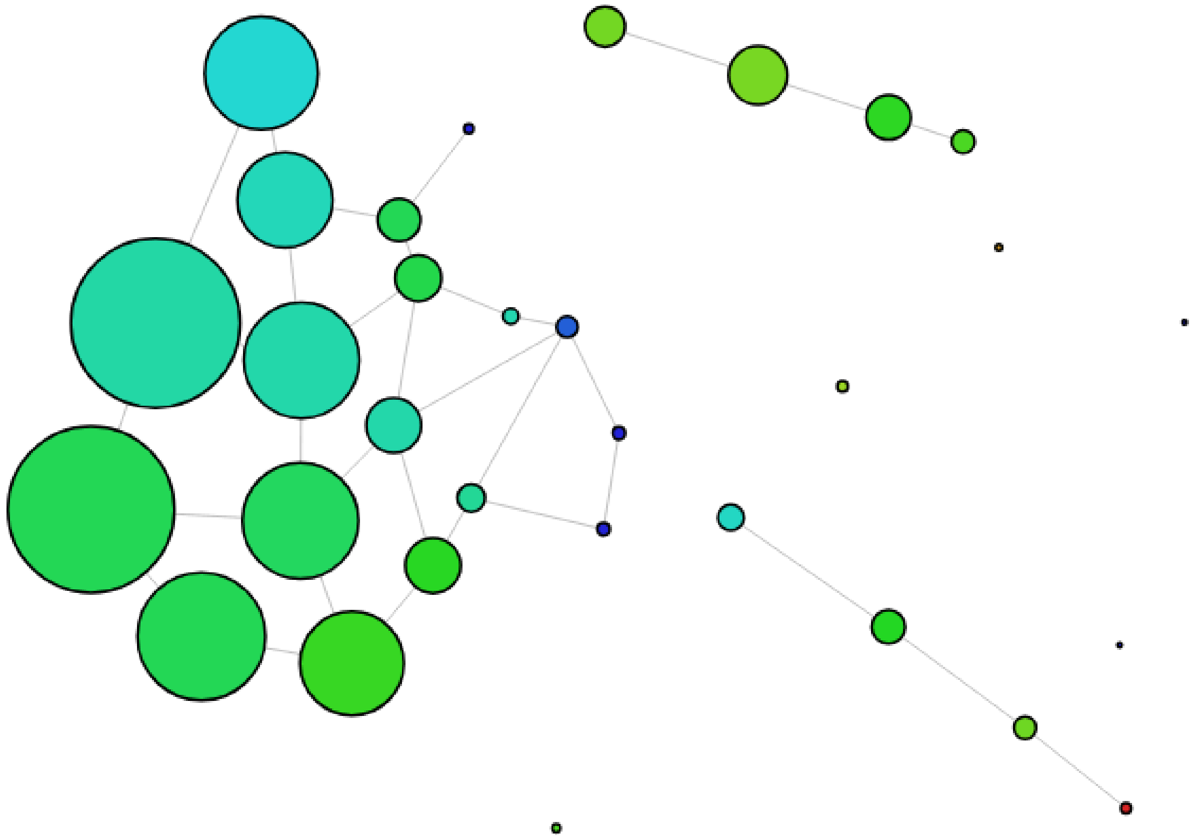
5 Intervals, 40% overlap, DBSCAN(0.05, 10)



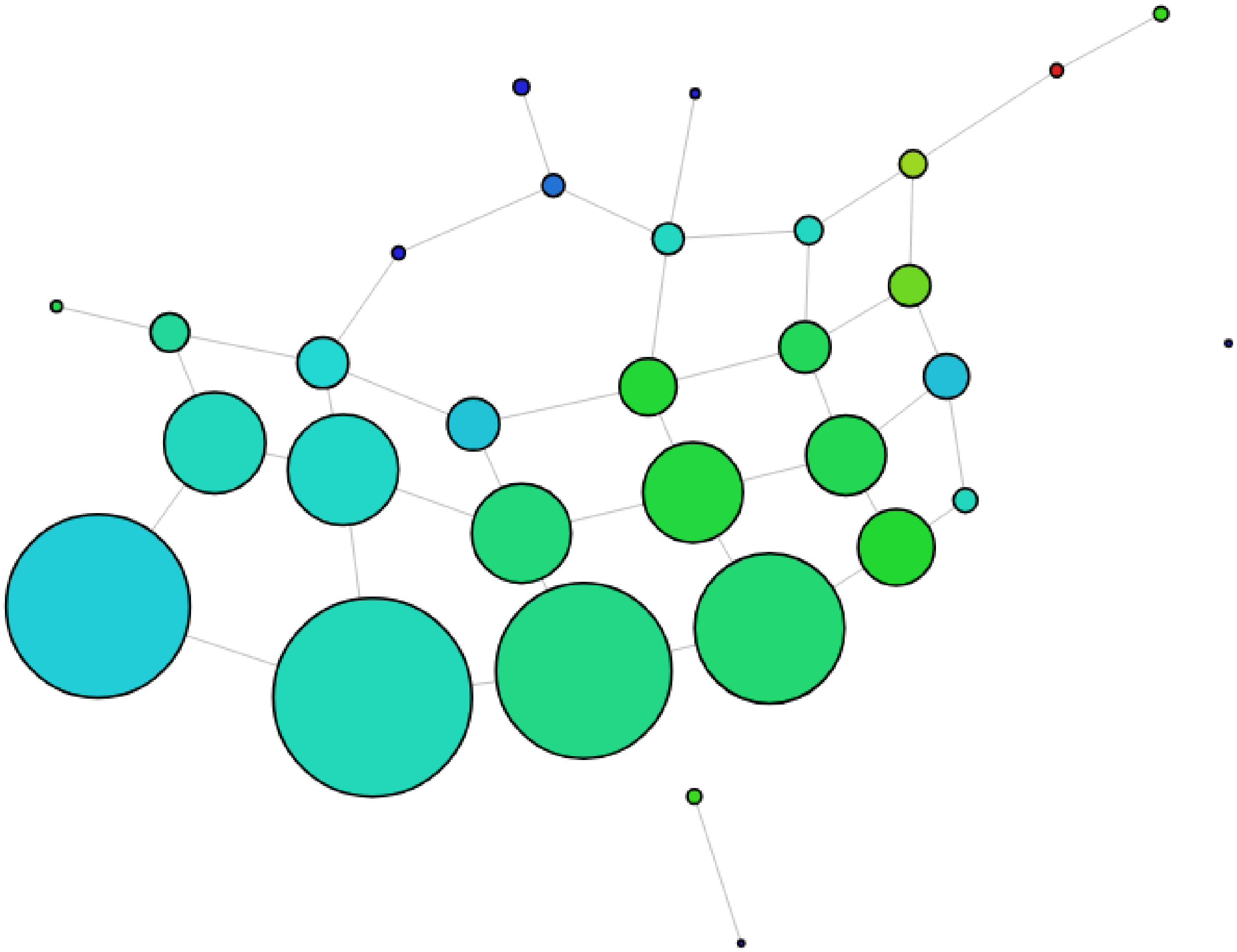
5 Intervals, 50% overlap, DBSCAN(0.05, 10)



6 Intervals, 10% overlap, DBSCAN(0.05, 10)

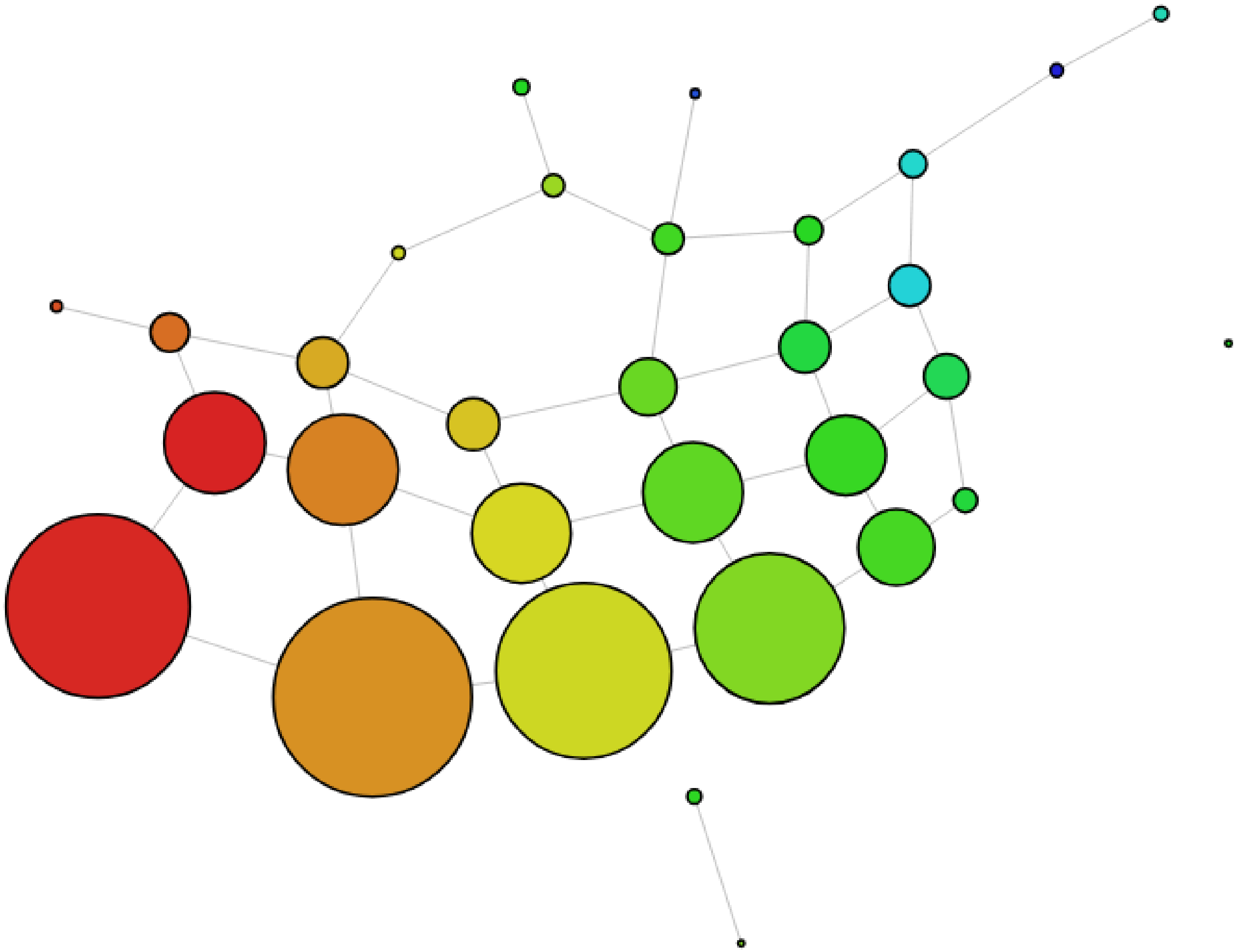


6 Intervals, 20% overlap, DBSCAN(0.05, 10)



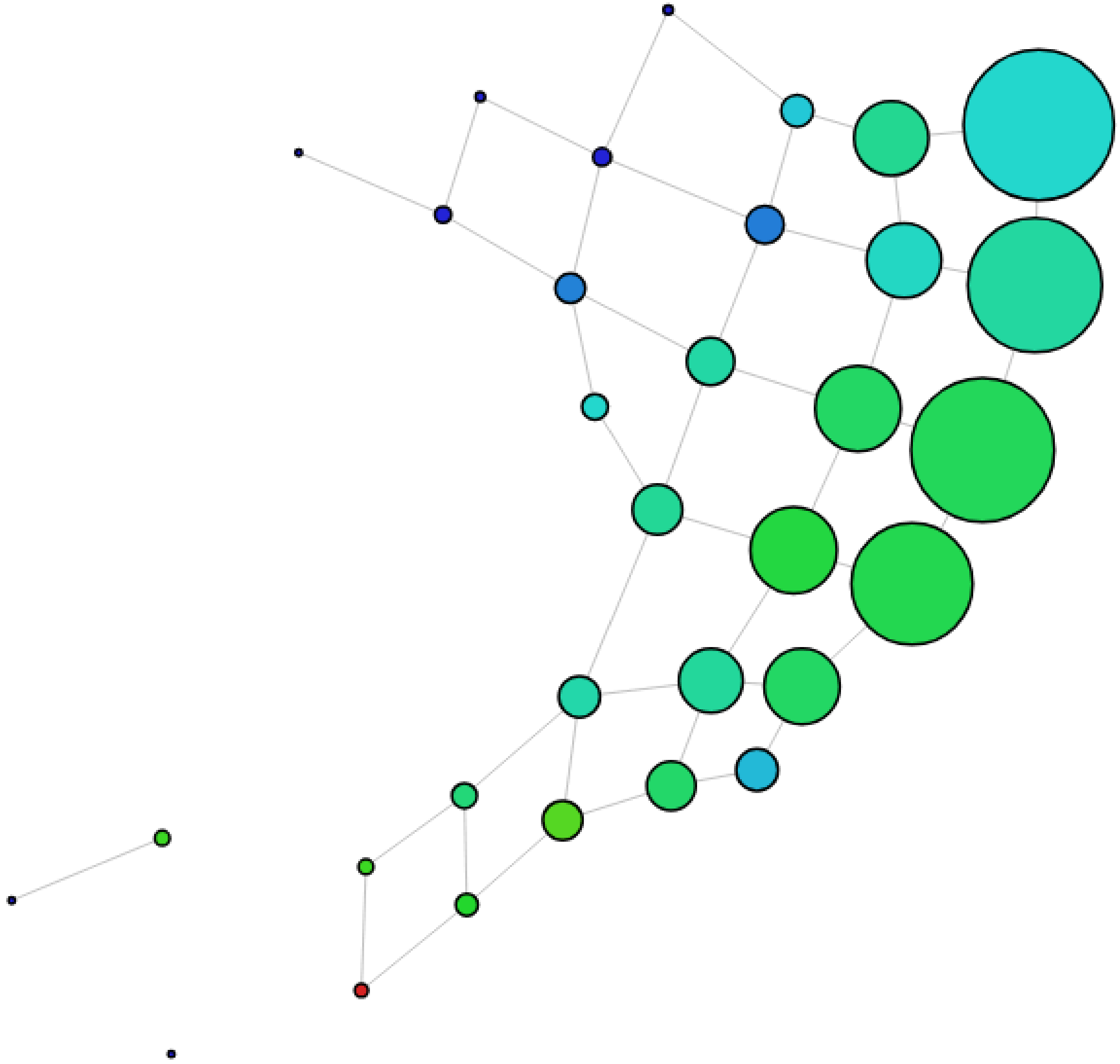
We do have some grouping of ADHD, green is 50%.

By wellbeing (red is higher):

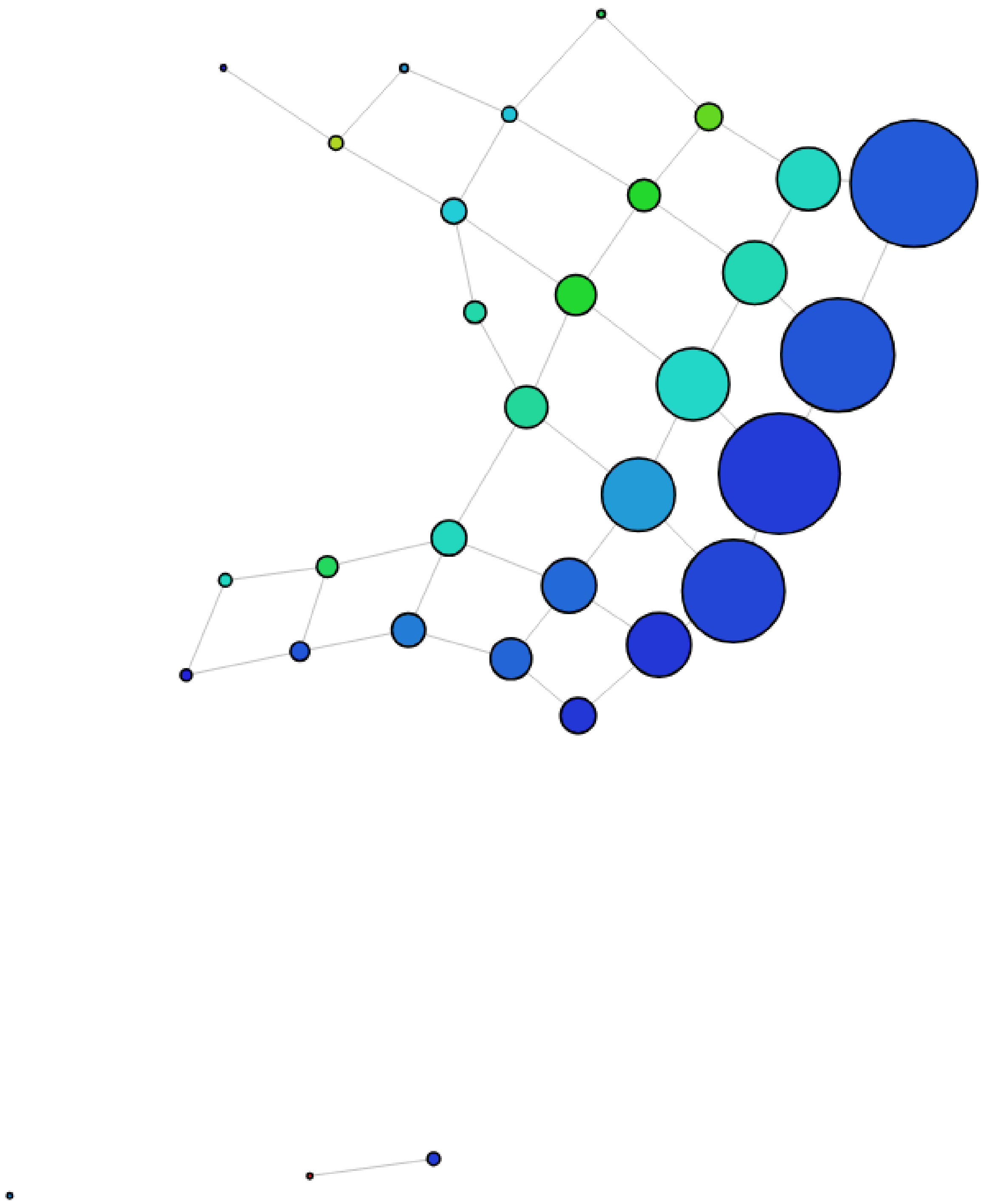


We see that as wemwbs total decreases, the proportion of ADHD participants increase. However the node sizes are massive and with only 15% of the participants being diagnosed with ADHD, we are probably not finding true patterns.
The large binge sessions and playtime consist of the loop found at the top, which also has less of the ADHD participants.

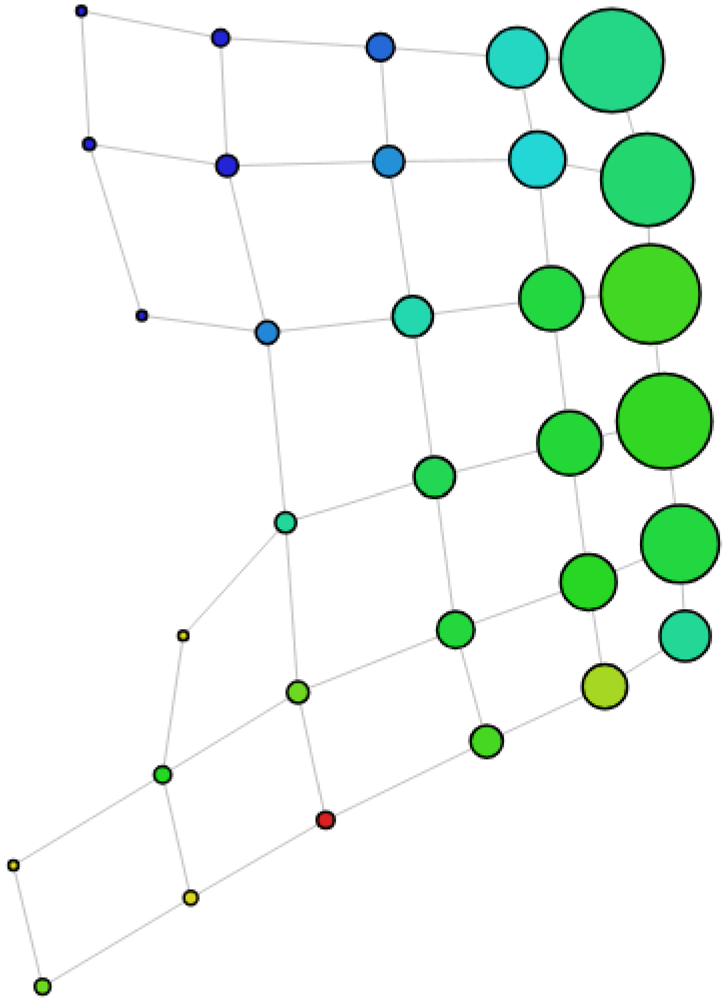
6 Intervals, 30% overlap, DBSCAN(0.05, 10)



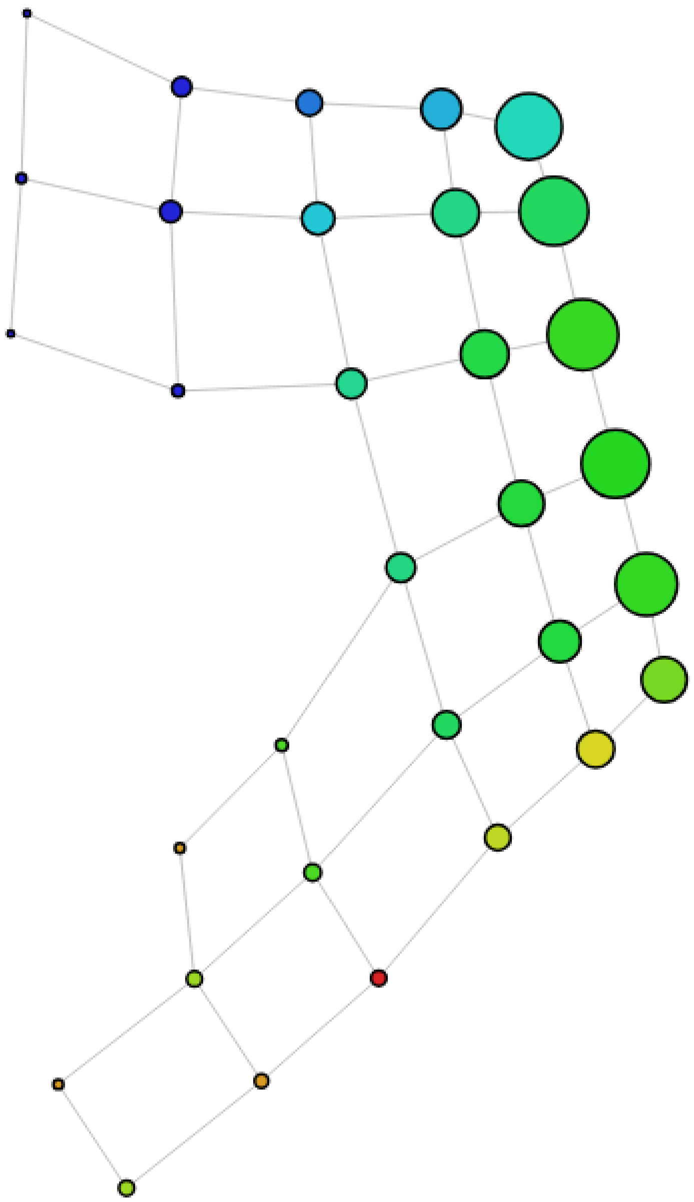
The higher attention scores are located around the high ADHD. But playtime is a bit of a banana shape:



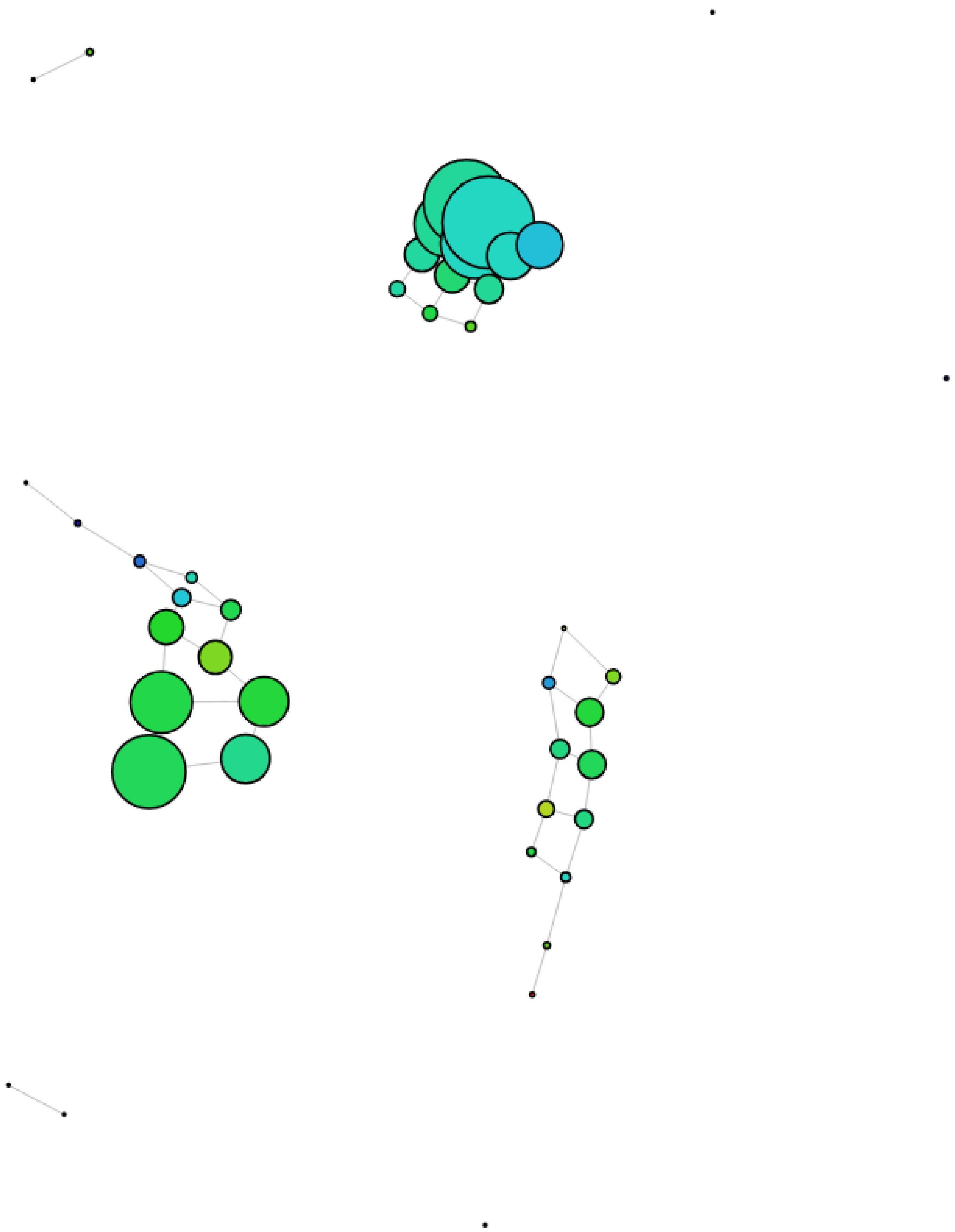
6 Intervals, 40% overlap, DBSCAN(0.05, 10)



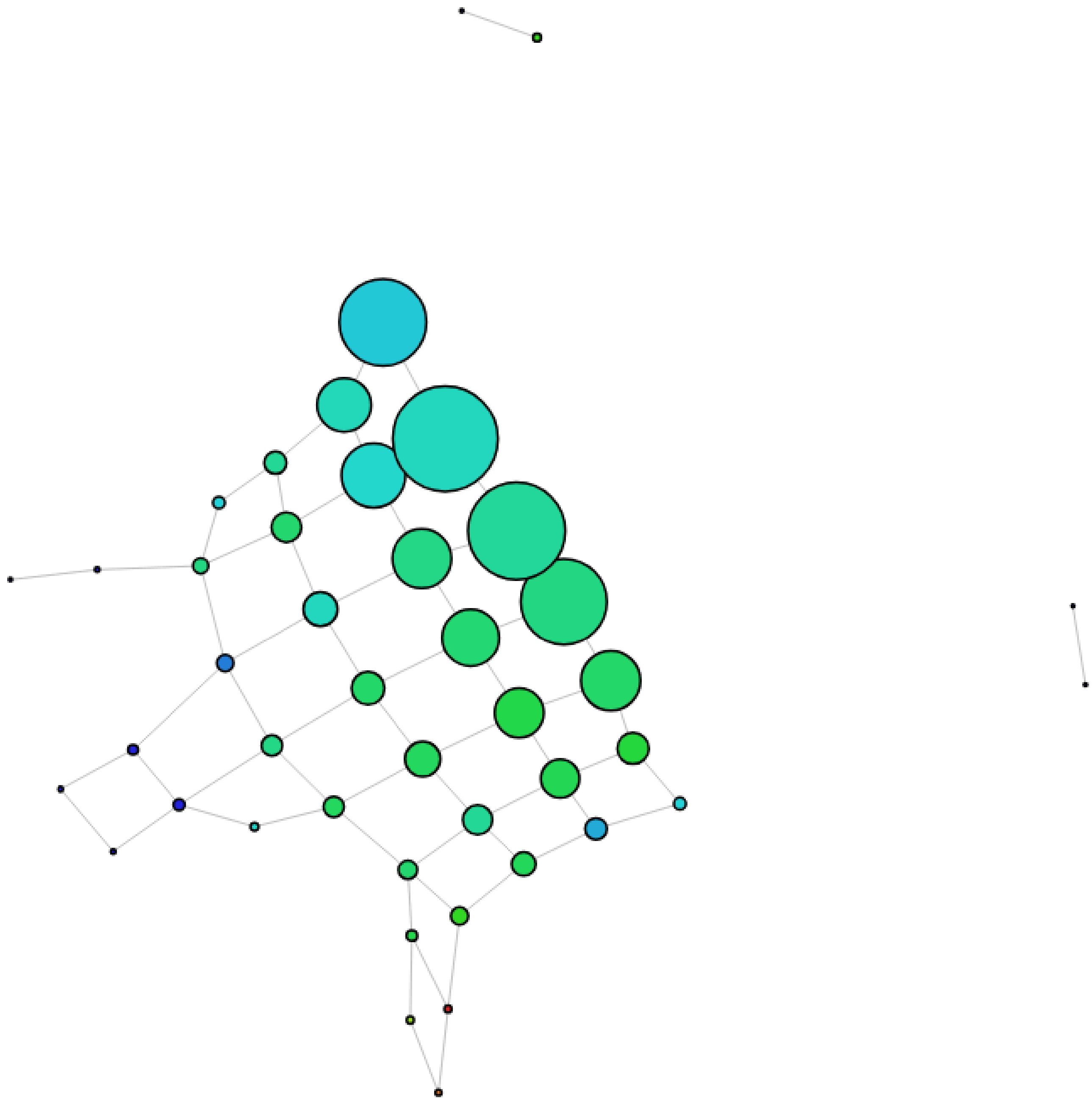
6 Intervals, 50% overlap, DBSCAN(0.05, 10)



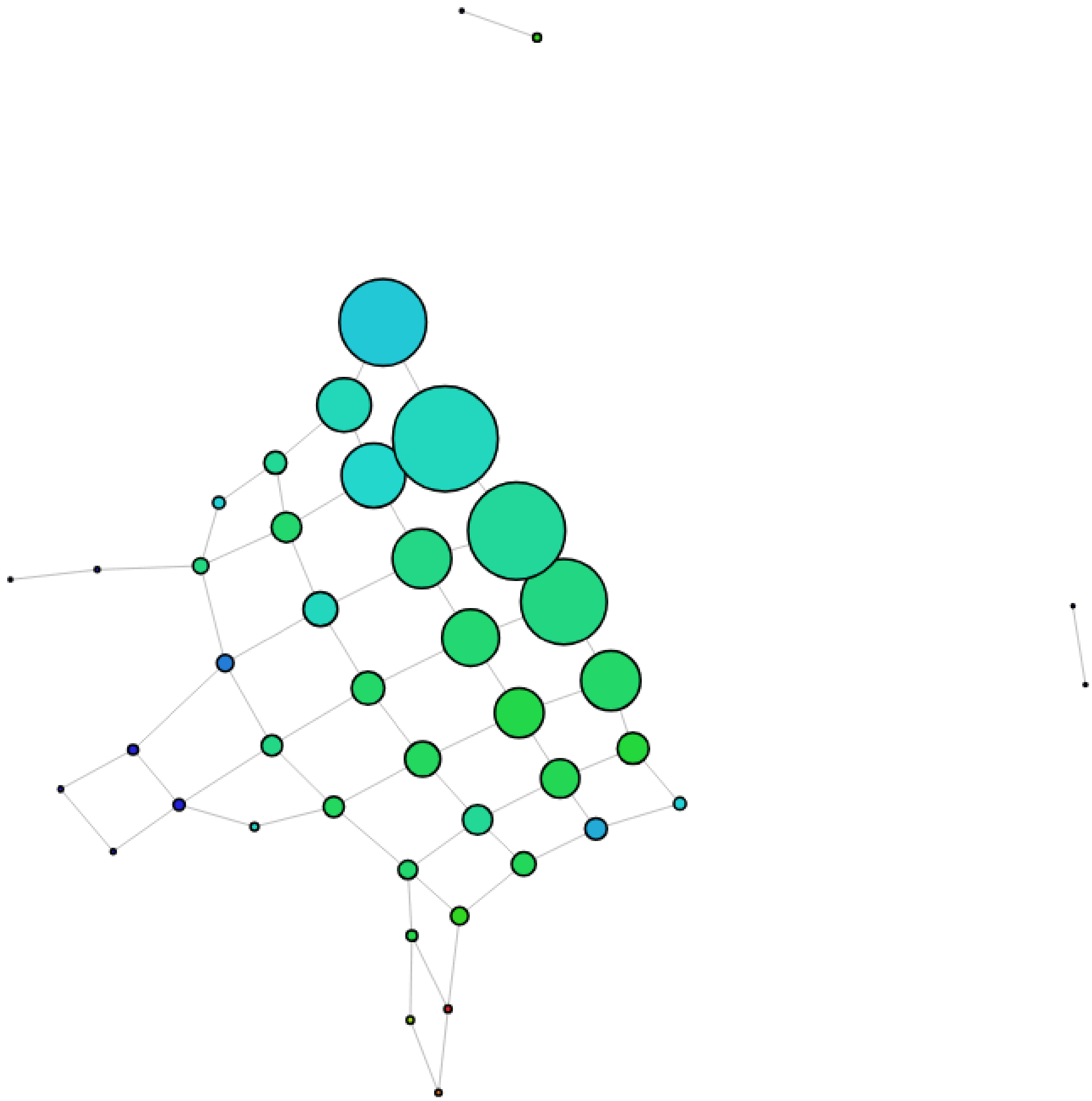
7 Intervals, 10% overlap, DBSCAN(0.05, 10)



7 Intervals, 20% overlap, DBSCAN(0.05, 10)



7 Intervals, 30% overlap, DBSCAN(0.05, 10)



7 Intervals, 40% overlap, DBSCAN(0.05, 10)